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Part A



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Australian Government
Bureau of Meteorology

National performance report 2023–24:
urban water utilities

Part A

Contents

A guide to this report 9
Key terms and abbreviations 9
Bulk water authorities 9
Utility groups 9
Reporting year 10
Missing or unavailable data 10
Interpreting 'overview of results' tables 10
Interpreting box-and-whisker plots 11
Interpreting data and commentary 11
Executive summary 13
Increased volume of water supply from major sources with extended warm weather condition in 2023–24 13
Return to surface water use in Sydney and increased reliance on recycled water in Melbourne 13
Slowed decrease in greenhouse gas emissions for water and wastewater services in major urban areas 13
Increased typical residential bills for water and wastewater services 14
Significant increase in total capital expenditure for water and wastewater services 14
1. Introduction 17
1.1. Context and overview 17
1.2. Reporting 17
1.3. Locations of utilities 18
1.4. Key drivers 19
1.4.1. Rainfall 19
1.4.2. Temperature 21
1.4.3. Utility size 23
1.4.4. Sources of water 23
2. Major urban centres 27
2.1. Water resources 27
2.1.1. Volume of water sources – W1, W2, W3.1, W26 27
2.1.2. Average volume of residential water supplied per property – W12 29
2.1.3. Total volume of recycled water supplied – W26 29
2.2. Pricing 30
2.2.1. Total typical residential bill – P8 30
2.3. Environment 31
2.3.1. Total net greenhouse gas emissions per 1,000 properties – E12 31
2.4. Finance 32
2.4.1. Total capital expenditure: water supply and wastewater – F16 32
2.5. Customers 33
2.5.1. Average duration of an unplanned interruption: water supply – C15 33
3. Water resources 35
3.1. Average annual residential water supplied – W12 35
3.1.1. Key findings 35
3.1.2. Results and analysis – Major utility group 36
3.2. Total recycled water supplied – W26 38
3.2.1. Key findings 38
3.2.2. Results and analysis – Major utility group 39
4. Pricing 40

4.1. Typical residential bill: water supply and wastewater – P8 40
4.1.1. Key findings 40
4.1.2. Results and analysis – Major utility group 42
4.2. Annual residential bill based on 200 kL per annum: water supply and wastewater – P7 44
4.2.1. Key findings 44
4.2.2. Results and analysis – Major utility group 45
5. Finance 47
5.1. Total capital expenditure: water supply and wastewater – F16 47
5.1.1. Key findings 47
5.1.2. Results and analysis – Major utility group 48
5.2. Capital expenditure per property: water supply (F28) and wastewater (F29) 48
5.2.1. Key findings 49
5.2.2. Results and analysis – Major utility group 50
5.3. Community service obligations ratio – F8 52
5.3.1. Key findings 52
5.3.2. Results and analysis – Major utility group 52
6. Customer 54
6.1. Average duration of an unplanned interruption: water – C15 54
6.1.1. Key findings 54
6.1.2. Results and analysis – Major utility group 55
7. Asset 57
7.1. Number of water main breaks, bursts and leaks per 100 km of water mains – A8 57
7.1.1. Key findings 57
7.1.2. Results and analysis – Major utility group 58
7.2. Number of sewer mains breaks and chokes per 100 km – A14 and number of property connection sewer breaks and chokes per 1,000 properties – A15 60
7.2.1. Key findings 60
7.2.2. Results and analysis – Major utility group 62
7.3. Real losses: service connections – A10 65
7.3.1. Key findings 65
7.3.2. Results and analysis – Major utility group 66
8. Environment 68
8.1. Total net greenhouse gas emissions per 1,000 properties – E12 68
8.1.1. Key findings 68
8.1.2. Results and analysis – Major utility group 69
9. Health 70
9.1. Percentage of population for whom microbiological compliance was achieved – H3 70
9.1.1. Key findings 70
9.1.2. Results and analysis – Major utility group 71
Appendix A Individual utility group tables 72
Appendix B Audit framework 109
Appendix C Utilities reporting 111
Appendix D Urban performance indicators 114
Appendix E CPI indexation 120
Appendix F Jurisdictional summaries 121
F1 Australian Capital Territory 121
F2 New South Wales 122
F3 Northern Territory 126
F4 South Australia 128

F5 Tasmania 131
F6 Queensland 133
F7 Victoria 135
F8 Western Australia 138

Figures

Example Figure 1 How to interpret an 'Overview of results' table 10
Example Figure 2 How to interpret a box-and-whisker plot 11
Figure 1 Typical residential bill: water supply and wastewater ($), 2013–14 to 2023–24 14
Figure 2 Total capital expenditure: water supply and sewerage ($ billion) for utilities that reported all 9 years (excluding bulk water utilities) 15
Figure 1.1 The administrative boundaries of all utilities reporting data for 2023–24 19
Figure 1.2 Rainfall decile map for 2023–24 (based on all years of data since 1900) 21
Figure 1.3 Mean daily temperature deciles for 2023–24 (based on all years of data since 1910) 22
Figure 1.4 Water source breakdown in each state and territory, 2019–20 to 2023–24 25
Figure 3.1 Average annual residential water supplied (kL/property) 36
Figure 3.2 Average annual residential water supplied (kL/property) – Major utility group 37
Figure 4.1 Typical residential bill: water supply and wastewater ($), 2013–14 to 2023–24 42
Figure 4.2 Typical residential bill: water supply and wastewater ($) – Major utility group 43
Figure 4.3 Annual bill based on 200 kL: water supply and wastewater ($) – Major utility group. 46
Figure 5.1 Total capital expenditure: water supply and wastewater ($ billion) for active utilities that reported all 9 years (excluding bulk water utilities) 48
Figure 5.2 Capital expenditure: water supply and wastewater ($/property) – Major utility group 51
Figure 6.1 Average duration of an unplanned interruption: water (minutes) – Major utility group 56
Figure 7.1 Number of water main breaks, bursts and leaks per 100 km of water mains 57
Figure 7.2 Water main breaks, bursts and leaks per 100 km of water mains – Major utility group 59
Figure 7.3 Sewer mains breaks and chokes per 100 km of sewer main – Major utility group 63
Figure 7.4 Property connection sewer breaks and chokes per 1,000 properties – Major utility group 64
Figure 7.5 Real losses (L/service connection/day) 66
Figure 7.6 Real losses: service connections (L/service connection/day) – Major utility group 67

Tables

Table 1.1 Utilities reporting in the 2024 Urban NPR by size group and jurisdiction 18
Table 2.1 Data sources for capital city analyses 27
Table 2.2 Volume of water sourced in each urban centre (ML) 28
Table 2.3 Average volume of residential water supplied per property (kL/property) 29
Table 2.4 Total volume of recycled water supplied (ML) 30
Table 2.5 Total typical residential bill ($) 31
Table 2.6 Total net greenhouse gas emissions per 1,000 properties (t CO_2 equivalent/1,000 properties) 32
Table 2.7 Total capital expenditure: water supply and wastewater ($000s) 33
Table 2.8 Average duration of an unplanned interruption: water supply (minutes) 34
Table 3.1 Overview of results: Average annual residential water supplied (kL/property) 36
Table 3.2 Overview of results: Total recycled water supplied (ML) 39
Table 4.1 Overview of results: Typical residential bill: water supply and wastewater ($) 41

Table 4.2 Overview of results: Annual residential bill based on 200 kL per annum: water supply and wastewater ($) 44
Table 5.1 Overview of results: Total capital expenditure: water and wastewater ($000s) 47
Table 5.2 Overview of results: Capital expenditure per property: water supply ($/property) 49
Table 5.3 Overview of results: Capital expenditure: wastewater ($/property) 49
Table 5.4 Overview of results: Community service obligations ratio 52
Table 6.1 Overview of results: Average duration of an unplanned interruption: water (minutes) 54
Table 7.1 Overview of results: Number of water main breaks, burst and leaks per 100 km of water mains (mains breaks/100 km) 58
Table 7.2 Overview of results: Number of sewer mains breaks and chokes per 100 km of sewer main (breaks and chokes/100 km) 61
Table 7.3 Overview of results: Number of property connection sewer breaks and chokes per 1,000 properties (breaks and chokes/1,000 properties) 61
Table 7.4 Overview of results: Real losses: service connections (L/service connection/day) 65
Table 8.1 Overview of results: Total net greenhouse gas emissions per 1,000 properties (t CO_2 equivalent/1,000 properties) 69
Table 9.1 Overview of results: Percentage of population for which microbiological compliance was achieved 70



A guide to this report

The following explanatory notes are provided to assist you to read and interpret the report.

Key terms and abbreviations

Utilities that form part of a city, shire or regional council, or a similar local government entity, are reported under the town or city name within the tables and charts in the report. For example, Dubbo Regional Council is referred to as 'Dubbo' in tables and charts throughout the report.

In addition, several utilities are represented by shorter names to aid presentation in charts and tables:

- Aqwest–Bunbury Water Corporation (W) = Aqwest–Bunbury (W)
- Busselton Water (W) = Busselton (W)
- City of Kalgoorlie–Boulder (S) = Kal–Boulder (S)
- Goldenfields Water County Council = Goldenfields Water (W)
- Queanbeyan–Palerang Regional Council = Queanbeyan
- Queensland Bulk Water Supply Authority = Seqwater

Abbreviations used in utility names include:

- P&W = Power and Water
- WC = Water Corporation

Most utilities in this report provide both water and sewerage services. Where a utility provides only a single service (for example, only water supply), it is denoted using a code in brackets after the utility's name. The codes are:

- W = water supply only
- S = sewerage only

Bulk water authorities

Bulk water authorities operate in several jurisdictions across Australia and provide wholesale water and wastewater services. These authorities do not have direct relationships with retail customers. For example, Melbourne Water supplies bulk water and wastewater services to the 7 retail utilities in and surrounding the Melbourne metropolitan region.[1]

Utility groups

In this report, the contributing utilities are grouped by the number of connected properties. The utility groups used are:

- Major – 100,000+ connected properties
- Large – 50,000 to 100,000 connected properties
- Medium – 20,000 to 50,000 connected properties
- Small – 10,000 to 20,000 connected properties.

[1] South East Water Corporation, Yarra Valley Water Corporation, Greater Western Water, Central Gippsland Water, Barwon Water, South Gippsland Water, and Westernport Water

Bulk water authorities are grouped separately and are not included in the analysis unless explicitly stated.

Reporting year

References to the ‘reporting year’ or ‘2023–24’ refer to the reporting year between 1 July 2023 and 30 June 2024 inclusive.

Missing or unavailable data

Missing or unavailable data are denoted in a table by a blank cell.

Interpreting ‘overview of results’ tables

Example Figure 1 shows how to interpret the ‘Overview of results’ table provided for each indicator.

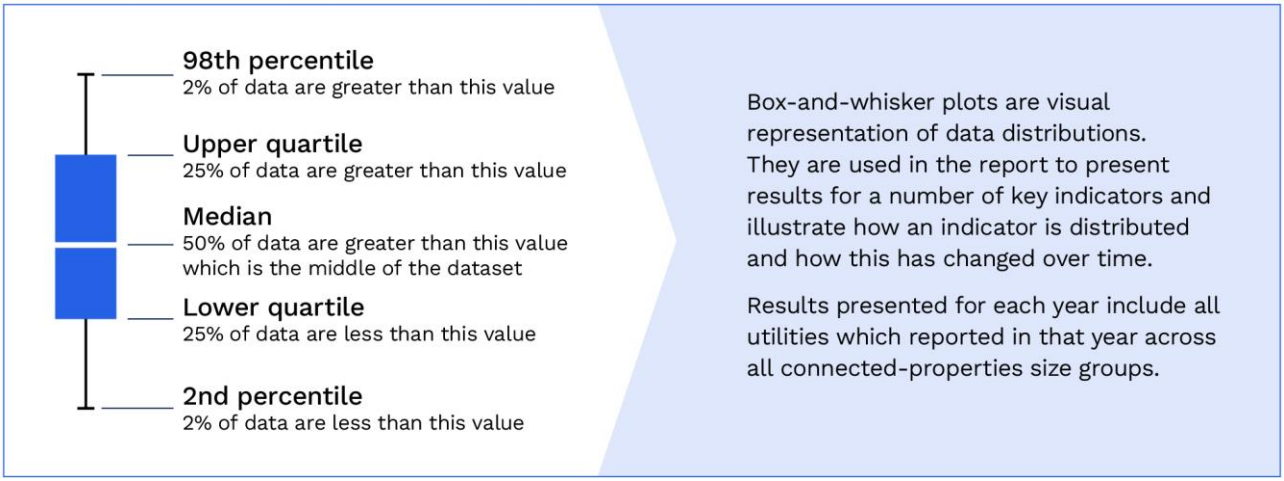
Size group	Range		Number of utilities with increase/decrease from previous year		Median or Total		Change in median/ total from previous year (%)
	High	Low	Increase	Decrease	Previous year	Current year	
Major	249	140	10	1	158	162	3
	WC (Perth)	Logan					
Large	454	130	9	1	164	181	10
	P&W (Darwin)	Toowoomba					
Medium	479	143	17	1	175	201	15
	Lower Murray Water	MidCoast Water					
Small	450	80	19	5	174	179	3
	Multiply utilities (5)	Ballina					
All size groups	**479**	**80**	**55**	**8**	**166**	**177**	**7**
	Lower Murray Water	**Ballina**					

1. The range shows the utilities with the highest and lowest result in the current reporting year for each utility group.
2. These columns show the number of utilities reporting an increase or decrease from the previous year's results for each utility group. Utilities that did not report in both years are included in this column. The figures do not include utilities that reported no change from the previous year.
3. The median value is the middle number in the range of results. For example, if five utilities reported for this indicator and their results are 190, 195, 206, 207, and 210, the median is 206 as it is the middle number. The median value is calculated using data from all utilities reporting against this indicator in that year. For indicators that are not represented as an 'average' for the utility (for example, average duration of water interruptions), or are divided by the number of properties (per property), the summary tables present the sum (total) of the results. The total value for a year is the sum of the results for that year of all the utilities that reported in both years.
4. This column shows the percentage change between the current and previous years and is rounded to the nearest integer, except in cases where additional precision is required.
5. 'Multiple utilities' is used when more than one utility recorded the same value.

Example Figure 1 How to interpret an ‘Overview of results’ table

Interpreting box-and-whisker plots

The report uses box-and-whisker plots to show trends in the annual distribution of key indicators as shown in Example Figure 2.



Example Figure 2 How to interpret a box-and-whisker plot

Interpreting data and commentary

When interpreting data and commentary in the report, it is important to consider the following:

- The indicator codes are noted in the titles of each section (for example, W12, F3, P7) and can be cross-referenced with the National urban water utility performance reporting framework: indicators and definitions handbook, January 2018. The median is the preferred metric for the reporting dataset, as outlier results can affect the average (mean), which can skew results towards the outliers.
- The median is the value at which 50% of utilities fall above and 50% fall below that point. Where average results are presented in addition to the median, they are to be interpreted together with the full dataset.
- Many factors can influence the performance of a utility, and individual performance indicators need to be interpreted in context. A low ranking for one indicator may not accurately reflect the overall performance of a utility. For example, a utility might have a low operating cost per property, but also poor drinking water quality and environmental performance, and a high level of complaints.
- In discussions of indicators, the ‘normaliser’ is often omitted to improve the flow of the commentary about indicators. For example, in the discussion of results for water main breaks per 100 km of water main, the commentary refers to a utility’s ‘water main breaks’. In this case, it is not the absolute number of water main breaks, rather, the number of breaks per 100 km of the water main.
- Single-service utilities are only included in the analysis of an indicator when a comparison can be made on a like-for-like basis with utilities that provide both water and sewerage services. For example, the overview tables for water and sewerage operating expenditure per connected property and for typical residential bills do not include single-service providers, but the overview tables for sewer overflows per 100 km of sewer main include all utilities that provide sewerage services.

- Financial time series information is given in real 2023–24 dollars ($); the impact of inflation is removed to ensure that years can be compared on a like-for-like basis. Consumer price index (CPI) figures can be found in Appendix E (CPI indexation).
- The percentage (%) change is calculated from the 2022–23 reporting year to 2023–24 with figures rounded to the nearest integer, except in cases where additional precision is required.

Executive summary

The National performance report 2023–24: urban water utilities (2024 Urban NPR) compares the performance of 81 utilities and councils (utilities) and 5 bulk water authorities providing urban water services to over 26 million people across Australia. The 2024 Urban NPR is published by the Bureau of Meteorology (the Bureau) with information provided by utilities across Australia's states and territories. The report is the 19th in the series, and the 11th to be produced by the Bureau.

Part A of the report provides commentary on and analysis of key indicators. Part B of the report contains data for the full set of 155 indicators reported by utilities and bulk water authorities for all reporting years.

Increased volume of water supply from major sources with extended warm weather condition in 2023–24

2023–24 was the second warmest financial year on record, with above to very much above average temperatures across most of the country, except for parts of the central inlands of the Northern Territory. The total urban water sourced nationally increased by 3.9%, likely due to the extended warm conditions, with some variations across different states and territories. South Australia and Western Australia reported the highest increase at 13.6%, while New South Wales saw a decrease of 3.5%. Surface water remained the dominant source in all states and territories except Western Australia, which relied mainly on groundwater. Nationally, groundwater use increased by 9.5% due to limited surface water availability in parts of South Australia and New South Wales, as well as groundwater recovery and chlorine disinfection in the Northern Territory, which allowed for larger groundwater extractions.

Desalinated water use decreased by 5.2% nationally, mainly due to not being required in Victoria and a return to regular lower levels in New South Wales compared to the previous year when flooding affected surface water quality and availability.

Return to surface water use in Sydney and increased reliance on recycled water in Melbourne

In the 2023–24 financial year, Sydney returned to using surface water after its quality improved following major flooding in 2022, and with regionally above average rainfall during the reporting year. This returned the reliance on desalinated water to its regular levels for meeting urban water demand in 2023–24. In Melbourne, high surface storage levels (over 85% capacity) from wet conditions in 2022–23 meant that desalinated water was not required from the Victorian Desalination Plant for 2023–24. However, hot and dry conditions in this region led to increased use of recycled water, mainly for non-residential customers.

Slowed decrease in greenhouse gas emissions for water and wastewater services in major urban areas

Despite a 2.0% decrease in total national net greenhouse gas emissions for water and wastewater services in 2023–24 compared to the previous year, there were variations across states and territories. Canberra, Melbourne, and Sydney continued to reduce emissions for water and

wastewater services in 2023–24. Perth reported the highest net greenhouse gas emissions, mainly due to its high reliance on desalinated water, and it saw the highest percentage increase (19.0%) in emissions compared to the previous year. This increase was driven by dry conditions in the region affecting surface water storage and resulting in increased use of desalinated water from 2022–23. Emissions also increased in South East Queensland and Adelaide, likely due to the increased volume of water supplied in response to extended hot conditions and higher water demands.

Increased typical residential bills for water and wastewater services

In 2023–24, the typical residential bill increased nationally by 2.0% compared to the previous year, marking the first rise since 2019–20. All major urban areas experienced an increase in their water and wastewater service bills, except Melbourne where the declining trend continued mainly driven by decreased wastewater service bills in 2023–24. Despite the increases, typical residential bills remained below 2019–20 levels. Adelaide reported the highest increase of 5.2% while South East Queensland had the lowest increase at 1.8% from 2022–23.

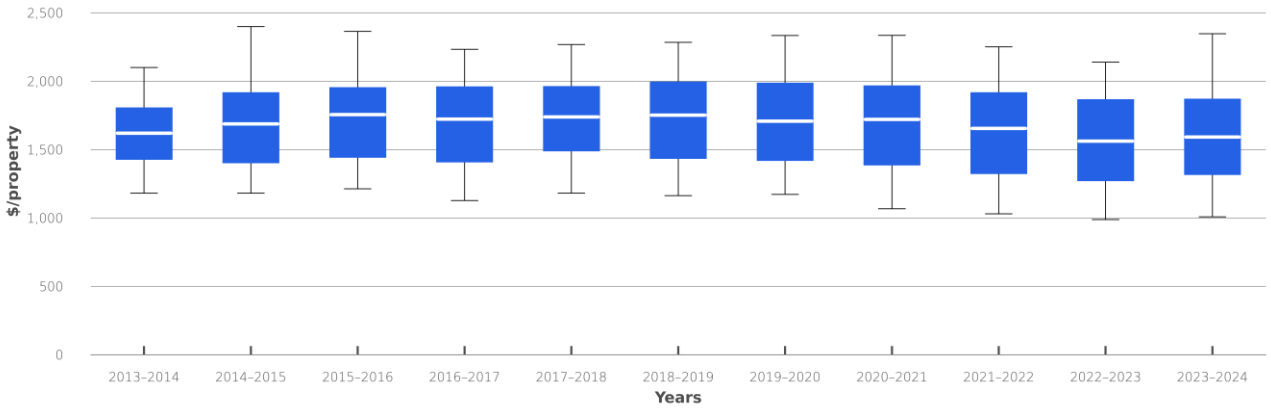


Figure 1 Typical residential bill: water supply and wastewater ($), 2013–14 to 2023–24

Significant increase in total capital expenditure for water and wastewater services

In 2023–24, the total capital expenditure on water and wastewater services significantly increased due to the progression and expansion of capital programs across different states. The increase in expenditure for water supply was slightly larger than that for wastewater services. Among major urban areas, Darwin reported the highest increase (similar to last year) while representing the lowest total expenditure in 2023–24. Canberra was the only major urban area to report a decline, driven by decreased capital expenditure on wastewater services.

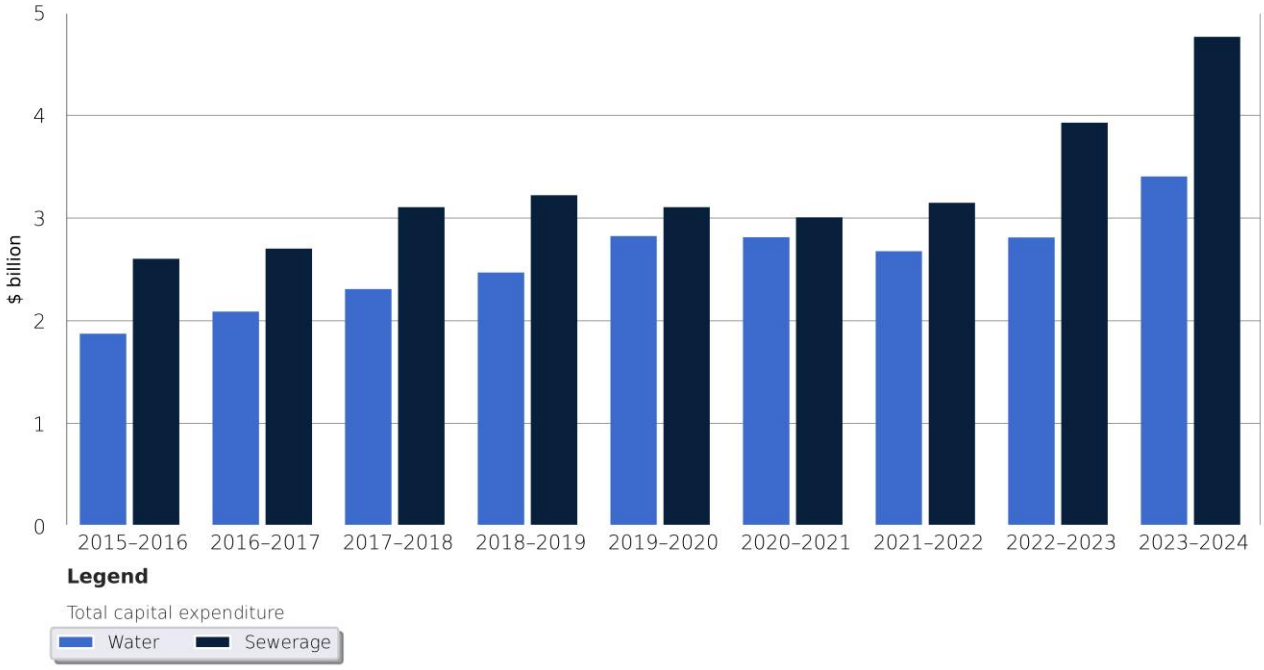


Figure 2 Total capital expenditure: water supply and sewerage ($ billion) for utilities that reported all 9 years (excluding bulk water utilities)



1. Introduction

1.1. Context and overview

This National performance report 2023–24: urban water utilities (2024 Urban NPR) supports the commitments made by states and territories under the National Water Initiative (NWI) to report publicly and independently on the performance of water utilities[2].

The 2024 Urban NPR compares the performance of 81 utilities and councils (utilities) and 5 bulk water authorities that provide urban water and sewerage services to over 26 million people across Australia. It is produced by the Bureau of Meteorology (the Bureau) in conjunction with state and territory governments and the Water Services Association of Australia.

Part A of this report provides commentary on, and analysis of, key indicators that apply to retail and distribution utilities. The Major urban centre analysis in Chapter 2 includes performance data for bulk water suppliers. Part B of this report contains data for the full set of indicators that are reported on by urban water utilities and bulk water authorities for all reporting years.

The 2020 NPR indicator review[3] recommended the retirement of 11 indicators out of the 166 indicators historically defined in the Urban NPR Framework. The retired indicators are C13, C14, C19, F11.1, F12.1, F13, F13.1, F28.1, F29.1, H4a, and W14. The review found no specific use case for these indicators within the NPR Framework, indicating that their removal would not result in any material loss of information, as individuals can still retrieve and assess this information if required. As a result, the 2024 Urban NPR includes a total of 155 indicators.

The analysis and commentary provide context for each indicator, discuss changes in reporting methods, and highlight trends within and/or between different utility groups. The utilities are grouped according to the number of properties they are connected to, as explained in ‘A guide to this report’.

The commentary and analysis in the 2024 Urban NPR are not intended to be a comprehensive explanation of every reported indicator. They present some of the significant trends or differences between years and between utilities. Most of the information is sourced from publicly available sources such as annual reports, regulatory decisions and utility websites. This information is also validated by the Technical Reference Group, which includes state and territory representatives.

1.2. Reporting

The 86 utilities (including 5 bulk water authorities) contributing data to the 2024 Urban NPR are listed in Appendix C. Table 1.1 summarises the utility size groups by jurisdiction.

Seventy-two of the 86 utilities included in this report provide both reticulated water supply and wastewater (sewerage) services. The remaining utilities provide either water supply or sewerage services. In summary, the report includes data for:

- 72 utilities providing water supply and sewerage services
- 5 utilities providing only water supply services
- 4 utilities providing only sewerage services

[2] National Water Initiative clauses 75–76

[3] http://www.bom.gov.au/water/npr/framework-review/

- 5 bulk water authorities.

There were no changes in the number of utilities reporting to the NPR framework in 2023–24 compared to 2022–23. However, the financial data for 2 utilities in regional New South Wales (Dubbo Regional Council and Orange City Council) were unavailable at the time of report preparation. Consequently, this data is not included in the analysis.

Table 1.1 Utilities reporting in the 2024 Urban NPR by size group and jurisdiction

Jurisdiction	Bulk	Major	Large	Medium	Small	Total
Australian Capital Territory		1				**1**
New South Wales	2	3	1	13	12	**31**
Northern Territory			1		1	**2**
Queensland	2	4	5	4	7	**22**
South Australia		1			1	**2**
Tasmania		1				**1**
Victoria	1	4	5	5	1	**16**
Western Australia		1	1		9	**11**
Total	**5**	**15**	**13**	**22**	**31**	**86**

1.3. Locations of utilities

Figure 1.1 shows the administrative boundaries of all utilities reporting data for the 2024 Urban NPR. Further details about the utilities are available from the relevant utility websites. While the SA Water Corporation provides services across South Australia, it does not provide water and wastewater services to all communities, which are also serviced by councils and private entities.[4]

[4] Maps of cities and towns serviced by SA Water are available in SA Water's 2023–24 annual report p.12. 2023-24 South Australian Water Corporation Annual Report

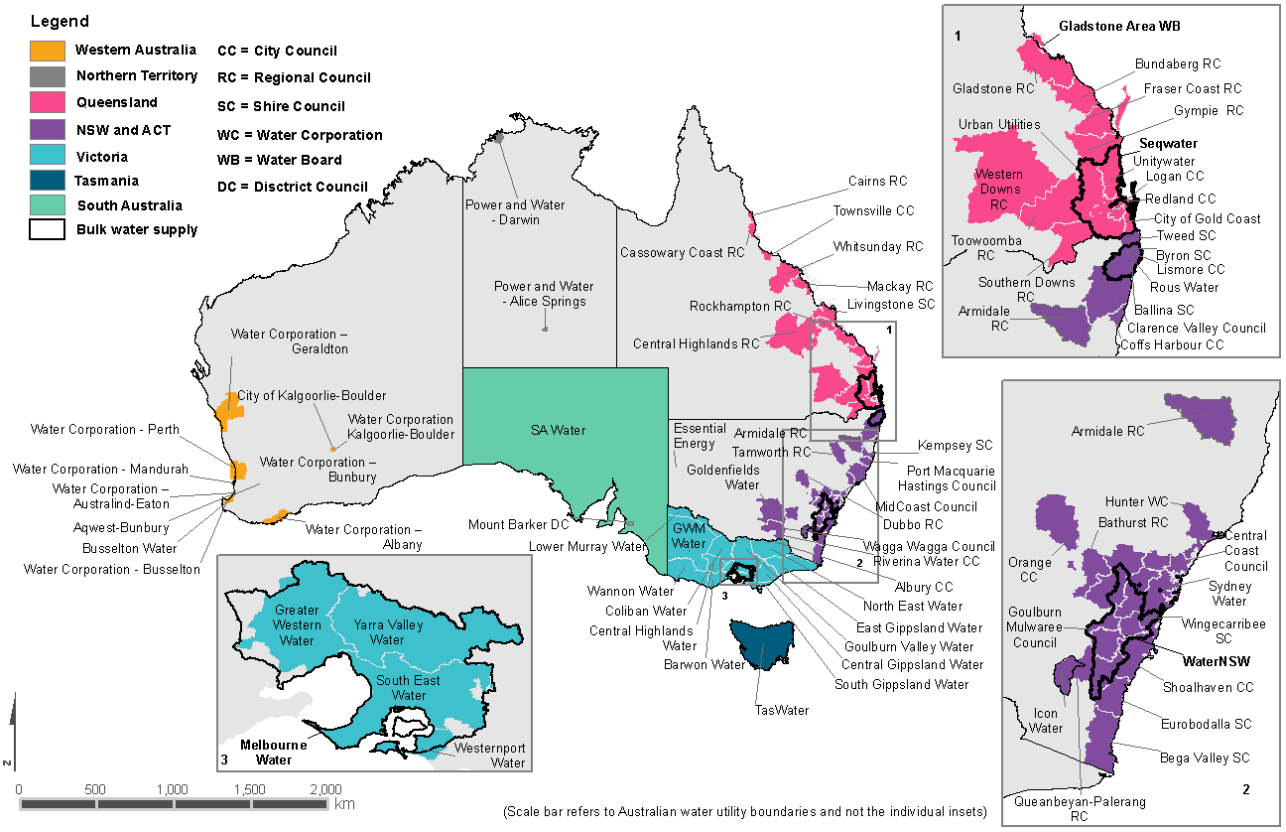


Figure 1.1 The administrative boundaries of all utilities reporting data for 2023–24

1.4. Key drivers

Key drivers of water utility performance presented in the 2024 Urban NPR include rainfall, temperature, utility size and sources of water.

Other factors also affect performance but are not discussed in detail. These include:

- network density
- soil types
- the age and condition of infrastructure
- impacts of the COVID-19 pandemic
- geographic location and remoteness
- government policy and regulation.

1.4.1. Rainfall

Rainfall can affect utility performance in many ways.

- Significant droughts with prolonged periods of low rainfall can stress urban water supply systems. Depending on the severity of the drought, security of the system and availability of climate-resilient water sources (for example, desalinated or recycled water), the utility may impose water restrictions to conserve water and ensure continuity of the water supply.
- Wet or dry conditions can affect demand for outdoor watering, resulting in a change in the volume of urban water and recycled water supplied to residents, councils, and parklands to be used for outdoor leisure activities such as golf courses (Water resource indicators W12, W26).

Changes in water consumption affect the revenue collected by utilities, their profitability, and the strength of their water-usage pricing signal.

- Wet or dry conditions can affect decisions about the water sources used (Water resource indicators W1 to W7). Persistent dry conditions can trigger thresholds for production from desalination plants and the use of groundwater and recycled water sources, which affect the operating costs of utilities (Finance indicators F11 and F12).To mitigate risks associated with the variable quality of raw water due to prolonged severe wet weather and flood conditions, utilities might decide to increase use of desalinated water (increasing W3.1).
- Increased rainfall can result in infiltration of water into sewer systems, which can increase the volume of sewage to be pumped and treated, increasing the operating costs of utilities (Finance indicator F12) and greenhouse gas emissions from sewage (Environment indicators E10, E12). Additional rainfall and sewer infiltration can also result in additional sewer overflows, especially during heavy rainfall.
- Extreme wet or dry conditions can cause expansion and shrinking of reactive clay soils in some parts of Australia. This can result in ground movement causing an increase in water or sewer main breaks (Asset indicators A8, A14), especially when conditions fluctuate rapidly from wet to dry or dry to wet. In periods of more consistent rainfall, the soils maintain more even moisture levels, resulting in less ground movement.

In 2023–24, Australia's total rainfall was 14% above the 1961–90 average (at 532.2 mm). Rainfall for the financial year (Figure 1.2) was above to very much above average for most of the Northern Territory, northern and western Queensland, inland and coastal areas of New South Wales, some parts of north-eastern South Australia, and an area extending from north-east to south-east Western Australia. It was the sixth-wettest financial year on record in the Northern Territory, with highest on record rainfall for its central regions. However, rainfall was below to very much below average for Tasmania, much of Victoria, coastal and some inland areas of South Australia, and the west coast and southern areas of Western Australia. Some areas of south-west Western Australia, south-eastern South Australia, South-western Victoria, and north-western Tasmania experienced the lowest rainfall on record. Tasmania had the third-driest financial year on record.

Winter and spring rainfalls were below average while summer and autumn rainfalls were above average for Australia as a whole. Autumn rainfall was the highest since 2011. There were monthly and regional variations. November 2023 rainfall was 43% above average with showers and locally severe thunderstorms across parts of the north and east. Australia had its ninth-wettest January on record in 2024, with widespread storms, tropical cyclones and heavy rainfall across the north. There were frequent thunderstorms in areas of Queensland, New South Wales and Victoria.

Tropical cyclones Jasper and Kirrily, which formed in December 2023 and January 2024 respectively, caused widespread heavy rainfall on the east and north of Queensland. March 2024 was the third-wettest March on record, with areas of central Northern Territory and inland Western Australia experiencing their highest rainfall on record. Tropical Cyclone Megan brought heavy rainfall to northern and coastal parts of the Northern Territory and northern Queensland. February to June 2024 rainfall was below average for much of Victoria, south-western Western Australia, the South Australia coastline and Tasmania.

More information is available in the Bureau's 2023–24 Climate and Water Statement.

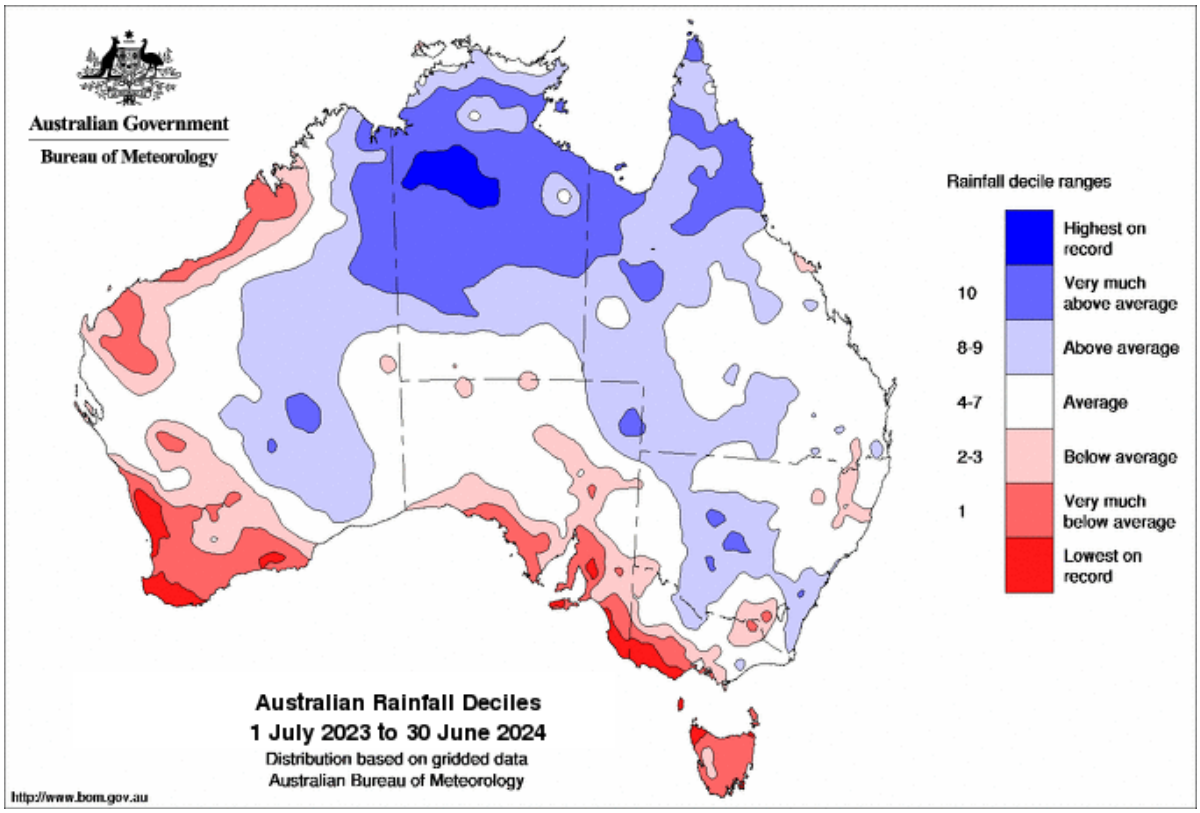


Figure 1.2 Rainfall decile map for 2023–24 (based on all years of data since 1900)

1.4.2. Temperature

Temperature can affect utility performance in many ways.

- Temperature can influence demand for water, particularly for residential and non-residential outdoor use. Prolonged periods of above-average temperatures can result in increased supply of potable and recycled water (Water resource indicators W12, W26, W27) to residents, councils and parklands for outdoor leisure activities such as to golf courses. Changes in water consumption affect the revenue collected by water utilities, their profitability (Finance indicators F3, F24) and the strength of their water-usage pricing signal (Finance indicator F4).
- Hot weather can increase the risk of bushfires, resulting in resources being deployed to protect water supply catchments and control bushfires. Emergency deployments can affect operating expenditure (Finance indicators F11 and F12). When responding to a bushfire, temporary water restrictions may be put in place to ensure the availability of supply and to meet firefighting requirements during extreme fire weather. These restrictions can affect the volume of water supplied by a utility and its operating cost and revenue. Poor water quality in a burnt catchment can affect the availability of water supply and the cost of treatment.
- Extended periods of heat or cold can affect the quality of water sources and supply, and in turn, decisions about water sources used (Water resource indicators W1 to W7) and the level of treatment required. For example, a heatwave can contribute to the decline in dissolved oxygen levels in a waterbody and can trigger the need to supply water from an alternative source, or increase water treatment, which affects the operating costs of utilities (Finance indicators F11 and F12).
- Changes in temperature can affect water quality as biological processes are particularly sensitive to extremes of heat or cold and rapid fluctuations in temperature. This can affect the

quality of water supplied (Health indicators H1 to H5) and treatment requirements, which impact the operational costs of a utility (Finance indicators F11 and F12).

- Extended hot conditions cause dry soil conditions. Trees will seek out moisture and their roots can enter the sewer system, causing blockages and breaks (Asset indicators A14, A15), and increasing the number of water main breaks (Asset indicator A8).

In 2023–24, the national mean temperature was 1.31 °C above the 1961–90 average, the second warmest on record, behind 2015–16 (Figure 1.3). The national average maximum and minimum temperatures were 1.56 °C and 1.05 °C warmer than the long-term average, respectively. Maximum temperatures were the highest on record for the west coast extending to inland of Western Australia, northern and eastern Tasmania, some parts of inland South Australia, and the New South Wales coast. An area of central Northern Territory experienced below average maximum temperatures. Minimum temperatures were the highest on record for the south and central coasts of Western Australia extending to inland, central and northern South Australia and into adjacent western Queensland and southern Northern Territory.

The national mean temperature for all seasons for the financial year was above the 1961–90 average, with the warmest winter, the fifth-warmest spring, and the third-warmest summer on record since national observation began in 1910. Frequent low-intensity to severe heatwaves affected many areas of Australia, particularly large parts of Western Australia. However, the national mean temperature for April was 0.51 °C below average, the lowest since 2015.

More information is available in the Bureau's 2023–24 Climate and Water Statement.

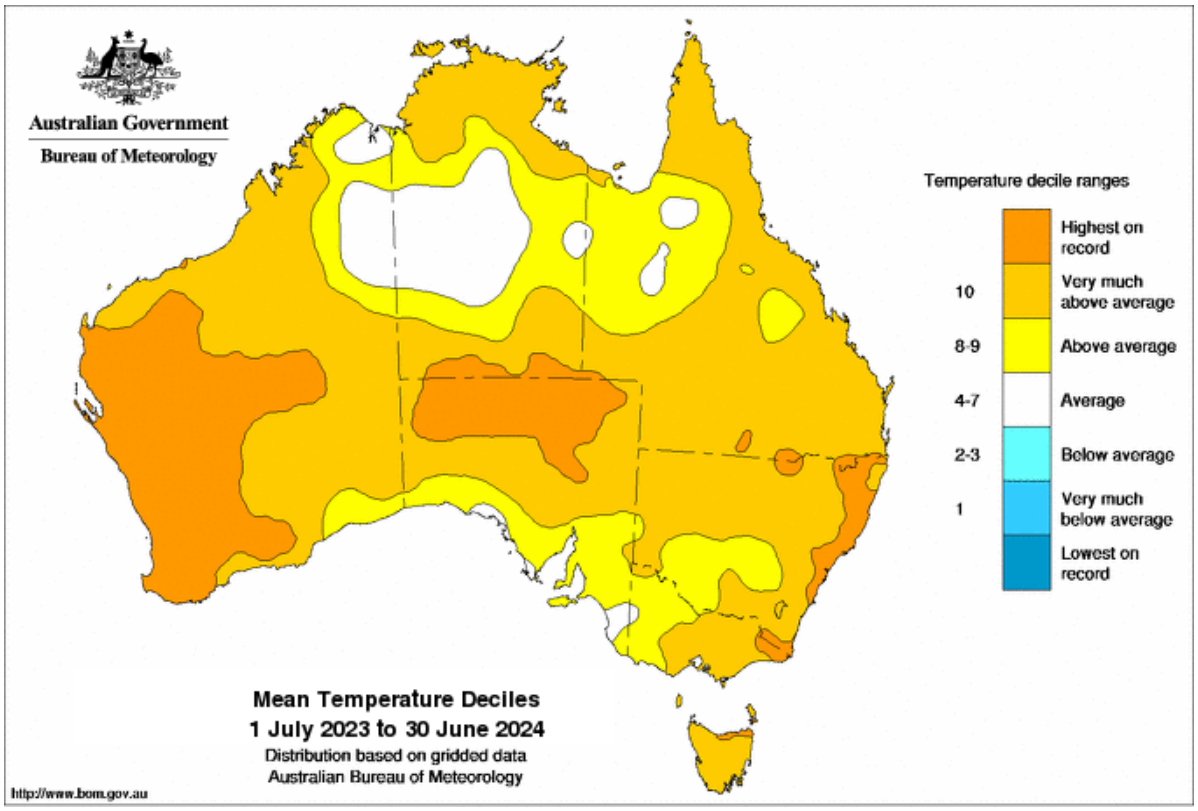


Figure 1.3 Mean daily temperature deciles for 2023–24 (based on all years of data since 1910)

1.4.3. Utility size

The size of a utility's customer base influences its performance on a range of indicators. This relationship may be causal, coincidental, or due to a related matter (for example, larger utilities may be subject to price regulation, unlike smaller utilities).

1.4.4. Sources of water

Two important drivers of performance are the sources of water used by a utility and the geographical relationship between the source and the urban centre it supplies. The combination and interaction of these drivers serve to create wide variations in engineering, operations and social challenges between utilities across the country.

The sources of water available to a utility are an important driver of several key performance indicators. Examples include the cost of treating water to an acceptable standard and supplying it to users affects the revenue collected by water utilities, their profitability (Finance indicators F3 and F24) and the strength of their water-usage pricing signal (Finance indicator F4).

Traditionally, Australians have relied on surface water and, to a lesser extent, groundwater to meet their urban consumption needs. The increased demand for urban water – resulting in a need to further develop and maintain ongoing water supply – is driven by many factors, including population growth and the reliability and security of existing sources (predominantly driven by water quality and climate variability). Financial, environmental and social factors reduce the feasibility of developing further traditional sources of water especially considering that most suitable dam sites have already been developed. In response to this situation, utilities and bulk water authorities across the country are developing non-traditional supply sources – such as desalinated and recycled water – while continuing to explore options for harvesting stormwater and rainwater.

The diversification of water sources affects the performance of utilities by increasing the cost of treating water to acceptable standards (to meet regulatory requirements) and to supply multiple water types to end users. For example, water from a 'protected' or 'closed' storage catchment is usually higher quality than water from an 'open' storage catchment and requires less treatment, which reduces the cost of supply.

The quality of water from groundwater sources varies greatly depending on the type and depth of the aquifer and has a significant impact on the extraction and treatment processes used and subsequent infrastructure and operational costs. Urban water supplied from recycled sources typically requires dual-pipe supply systems to separate recycled water from potable water, incurring greater infrastructure costs.

Figure 1.4 shows the annual supply from major sources of water, and the total supply, for utilities in each state and territory from 2019–20 to 2023–24.

- Water sourced from surface waters (that is, rivers, streams and dams; Water resource indicator W1) is the dominant sources of water for all states and territories (81.4% of the total water sourced from major sources of water) except Western Australia, where most of the water (39.3%) is sourced from groundwater (Water resource indicator W2).
- In 2023–24, total water sourced nationally increased by 3.9%, primarily due to a 13.6% increase in the total water sourced in South Australia and Western Australia. Extended hot weather and heatwave conditions throughout the year increased water demand across Australia, resulting in higher water supply requirements.

- The total surface water sourced on the national scale increased by 3.6%, mainly influenced by a 13.5% increase in the volume of surface water sourced in Western Australia. Nationally above average rainfall during summer and autumn increased the availability of surface water across the country. The increase in surface water use in Western Australia was driven by higher water demand from hot weather conditions. The volume of surface water sourced in New South Wales decreased by 1.6%, mainly due to below average streamflow in parts of north-eastern and south-eastern New South Wales.
- The volume of groundwater sourced increased by 9.5% from 2022–23 across the country. This was mainly driven by a 25.3% increase in the Northern Territory, where groundwater levels showed some improvements since 2021 due to above to very much above average rainfall, resulting in more bores reaching average levels compared to the previous year. These average levels, along with increased chlorine disinfection, allowed for higher groundwater extraction in the Northern Territory compared to the previous year. South Australia and New South Wales followed with 16.7% and 12.3% increases, respectively. Below average rainfall in southern South Australia and parts of north-eastern and south-eastern New South Wales resulted in below average streamflow, impacting surface water availability and leading to higher groundwater extraction.
- The volume of water sourced from desalinated water (Water resource indicator W3.1) in 2023–24 decreased (5.2%) for the fourth time since 2019–20 (18.3% decrease in 2020–21, 9.9% decrease in 2021–22, and 36.6% decrease in 2022–23). The decrease was mainly due to no need for desalinated water in Victoria[5] as well as a 48.2% decrease in desalinated water sourced in New South Wales. In 2022–23, major flooding affected the quality of surface water resources in New South Wales, significantly increasing reliance on desalinated water. With no such adverse impact in 2023–24, the use of desalinated water decreased from the previous year and returned to regular levels.
- In 2023–24, the use of desalinated water increased in Queensland by 31.3%, in Western Australia by 26.5%, and in South Australia by 0.8% compared to the previous year. Desalinated water use in Queensland returned to its regular levels in 2023–24 after an unusually low usage in the previous year, resulting in the highest increase percentage compared to the preceding year (from 7,834 ML in 2022–23 to 10,283 ML in 2023–24). The increase in desalinated water use in Western Australia was attributed to the integrated operational strategy, where surface water storages were not recovering due to below to very much below average streamflow (Perth's surface water storages experienced 10.3% decline from the previous year[6]). In the current year, as in the previous years, Tasmania, the Northern Territory and the Australian Capital Territory did not source any desalinated water.
- The volume of water sourced from recycled water (Water resource indicator W4/W26) in 2023–24 increased by 8.7%. This increase was primarily driven by an 82.4% increase in recycled water sourced in the Australian Capital Territory which was mainly attributed to non-residential customers. Additionally, recycled water uses in South Australia increased by 33.6% compared to the previous year, due to below average rainfall, a relatively dry summer, and an expansion of the customer base. Queensland and Western Australia were the only states to report a slight decrease of 2.4% in recycled water use compared to the previous year.

[5] Annual Report | Melbourne Water

[6] Financial year climate and water report 2023–24

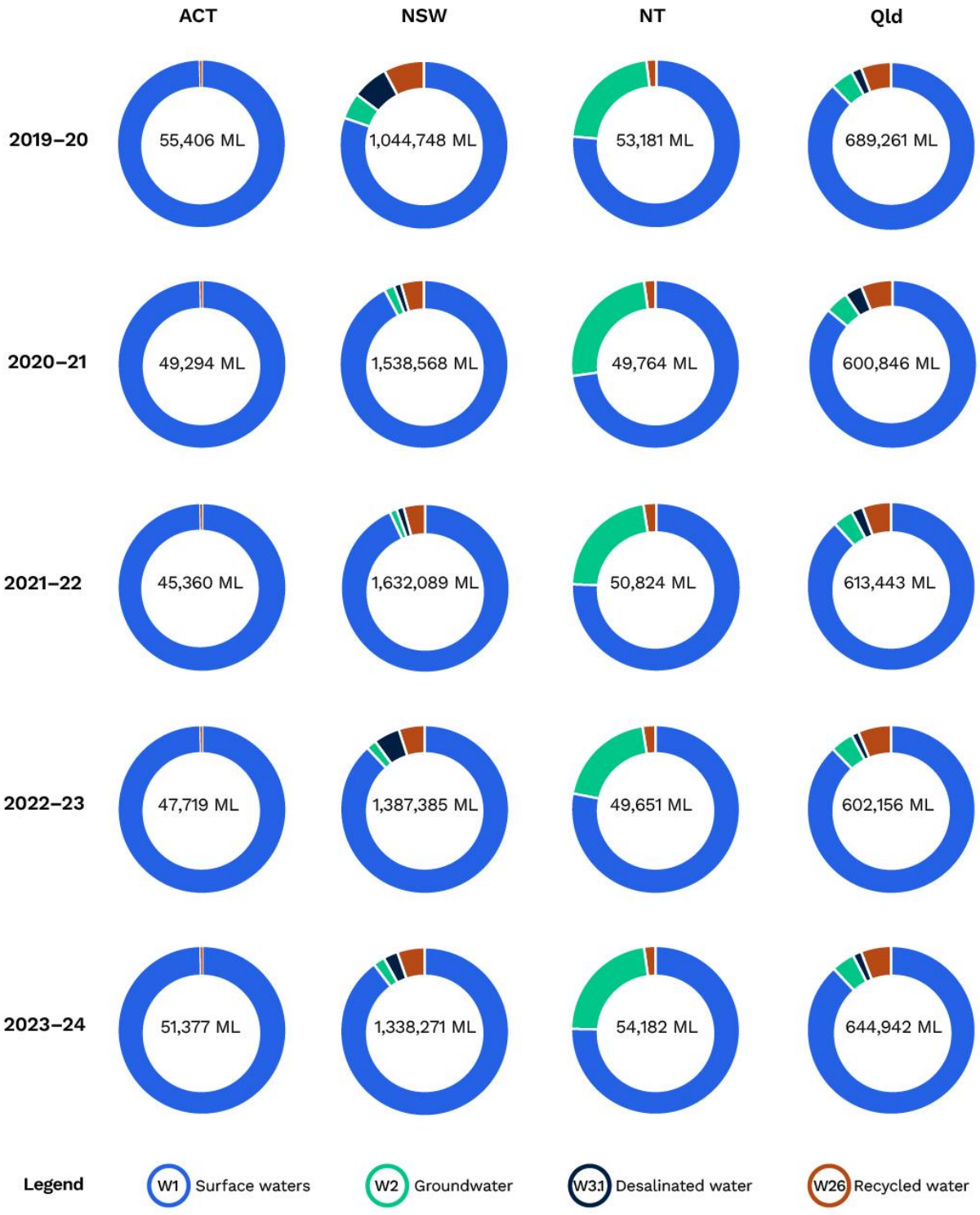


Figure 1.4a Water source breakdown in each state and territory, 2019–20 to 2023–24

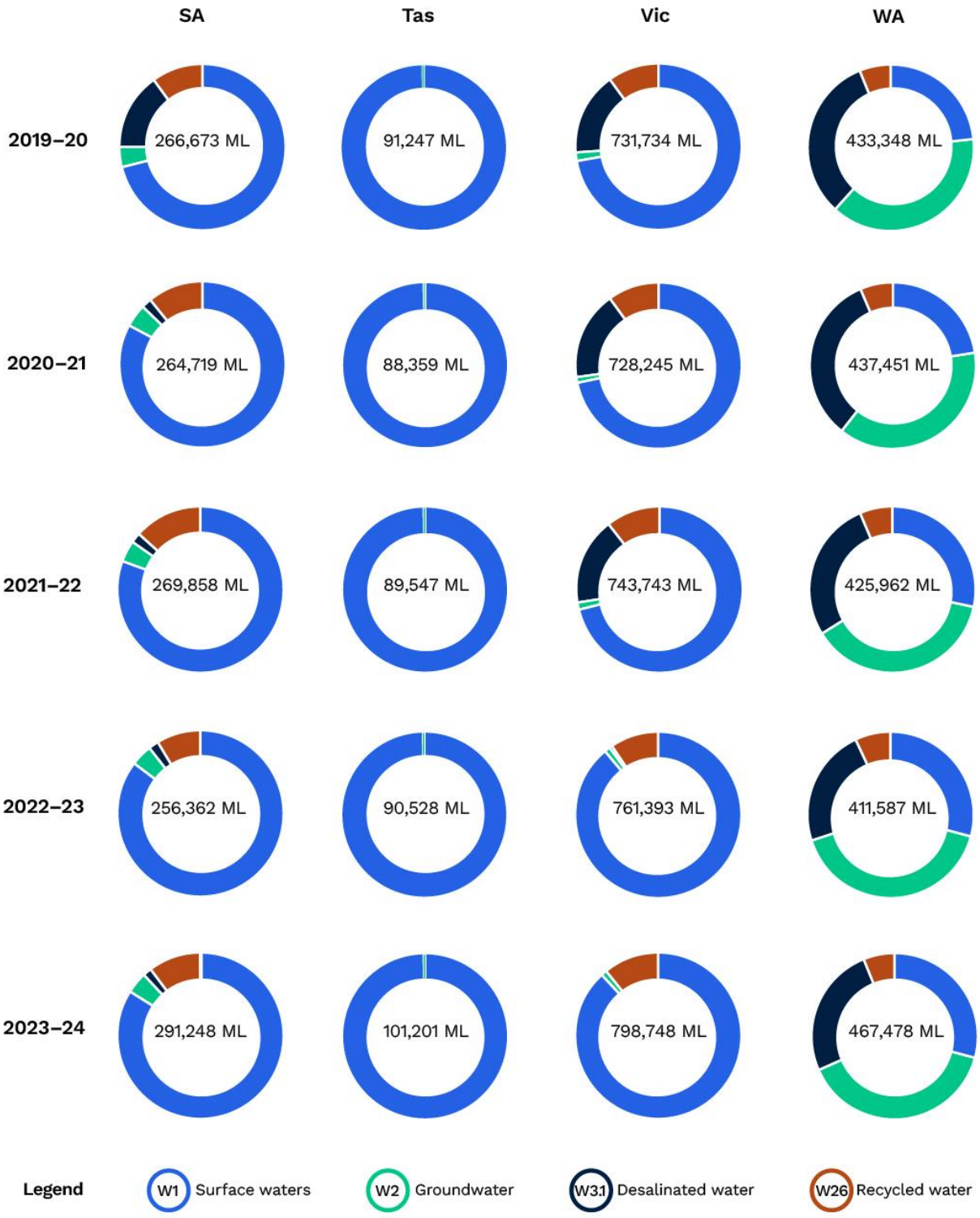


Figure 1.4b Water source breakdown in each state and territory, 2019–20 to 2023–24

2. Major urban centres

This chapter provides comparative tables and figures for a selection of key indicators for major urban centres.

The figures and tables are compiled using data supplied by the utilities detailed in Table 2.1.

The structure of utilities varies, so the figures in this chapter should be treated with some caution and read in conjunction with the notes for each table. For example, to provide figures that represent Sydney and South East Queensland, it may be necessary to aggregate the numbers for both bulk water authorities and utilities servicing those areas. Melbourne (urban centre) data from the 2021–22 reporting year onward is not comparable with pre-2021–22 reporting years due to the creation of Greater Western Water. This resulted in the service area previously managed by Western Water being included in the calculations for Melbourne from the 2021–22 reporting year onward.

The historical values for all financial indicators have been adjusted using consumer price index (CPI) data to facilitate comparisons in real terms.

Table 2.1 Data sources for capital city analyses

Major urban centre	Utility (B denotes bulk supplier)
Perth	Water Corporation – Perth
Adelaide	SA Water Corporation
Canberra	Icon Water Limited
South East Queensland	Queensland Bulk Water Supply Authority (Seqwater) (B), Urban Utilities, Unitywater, City of Gold Coast, Redland City Council, Logan City Council
Sydney	WaterNSW (B), Sydney Water Corporation
Melbourne	Melbourne Water (B), Greater Western Water, South East Water Corporation, Yarra Valley Water Corporation
Hobart	No data – TasWater services this area; performance data are available only on an aggregated basis for the entire state of Tasmania
Darwin	Power and Water – Darwin

2.1. Water resources

2.1.1. Volume of water sources – W1, W2, W3.1, W26

Table 2.2 presents the volume (ML) of water sourced from surface water (W1), groundwater (W2), desalinated marine water (W3.1) and recycled water (W26) for each city.

In 2023–24, the total water sourced for major urban centres increased by 6.2% compared to 2022–23 on the national scale. All major urban centres reported an increase in the total volume of water sourced, with Sydney reporting the smallest increase at 4.0% and Adelaide the largest at 12.2%. Among all water source types, recycled water contributed the lowest volume (7.3%), and surface water contributed the highest volume (77.2%) of the total water sourced for major urban centres. The use of surface water, groundwater and recycled water for major urban centres increased by 7.0%, 8.8% and 10.0%, respectively, as a result of above average rainfall across the country as

well as hot weather conditions leading to higher water demands. Conversely, the use of desalinated water decreased by 5.3% compared to the previous year.

Sydney was the largest supplier of surface water (528,804 ML). Similar to the previous year, Perth remained the largest supplier of groundwater (148,164 ML) and desalinated water (119,548 ML). Melbourne became the largest supplier of recycled water (53,804 ML) to the urban centre in the current year. This increase was driven by Greater Western Water's larger supply to both residential and non-residential customers, as well as increased supply to non-residential customers by Melbourne Water Corporation and South East Water Corporation. For the third consecutive year since 2021–22, Sydney sourced the highest total volume of water (604,381 ML).

In contrast to the previous year, Melbourne did not source any desalinated water to meet urban demands (a change from 4,180 ML to 0 ML). The total volume of desalinated water sourced in Sydney also returned to its regular levels (from 67,996 ML to 35,216 ML) with the recovery of surface water quality after major flooding impacts in 2022–23. An unusually low usage of desalinated water in Queensland during 2022–23 followed by a return to almost regular levels in 2023–24 resulted in a 33.0% increase in desalinated water use (from 7,240 ML to 9,627 ML) compared to the previous year.

In 2023–24, Canberra reported the largest increase (82.4%) in the use of recycled water (from 17 ML to 31 ML) mainly for non-residential customers. Conversely, South East Queensland reported the largest decrease (19.6%) in recycled water use (from 17,949 ML to 14,439 ML), primarily due to the decrease in supply of recycled water to non-residential customers by Seqwater.

Table 2.2 Volume of water sourced in each urban centre (ML)

Major urban centre	Surface water (W1)		Groundwater (W2)		Desalinated marine water (W3.1)		Recycled water (W26)		Total	
	2022–23	2023–24	2022–23	2023–24	2022–23	2023–24	2022–23	2023–24	2022–23	2023–24
Adelaide	155,507	171,334	-	-	4,804	4,827	20,706	27,021	181,017	203,182
Canberra	47,702	51,346	-	-	-	-	17	31	47,719	51,377
Darwin	38,891	40,905	2,258	4,316	-	-	-	-	41,149	45,221
Melbourne[a]	452,668	471,997	140	101	4,180	-	40,704	53,804	497,692	525,902
Perth[b]	64,463	58,838	136,791	148,164	94,474	119,548	23,337	22,660	319,065	349,210
South East Queensland[c]	327,886	344,769	12,928	12,856	7,240	9,627	17,949	14,439	366,003	381,691
Sydney[d]	472,179	528,804	-	-	67,996	35,216	41,198	40,361	581,373	604,381

Notes:

[a] Melbourne's surface water is sourced from Melbourne Water and Greater Western Water, while its recycled water is sourced from Melbourne Water and the 3 retailers (Yarra Valley Water, Greater Western Water and South East Water).

[b] Perth's surface water (W1) volume reflects Water Corporation transferring water into surface water storages. In 2023–24, it diverted 136,512 ML from surface water (W1) and returned 77,674 ML. In 2022–23, Water Corporation – Perth diverted 120,263 ML from surface water (W1) and returned 55,800 ML.

[c] South East Queensland's surface water, groundwater and desalinated water are sourced from Seqwater. South East Queensland's recycled water is sourced from Seqwater and the retailers (Urban Utilities, Unitywater, City of Gold Coast, Logan City Council and Redland City).

[d] Sydney's surface water (W1) volume includes water supplied by WaterNSW to Greater Sydney and the volume of water sourced directly by Sydney Water Corporation from surface water.

2.1.2. Average volume of residential water supplied per property – W12

Table 2.3 shows the annual average volume (kL/property) of residential water supplied to customers in each major urban centre.

In contrast to the previous year, which saw a decrease in residential water supply for almost all major urban centres, 2023–24 experienced increases across the board. The highest increase was in Darwin (11.5%), driven by hot weather, increased demands, and higher groundwater extraction following chlorine disinfection. Perth followed with a 10.5% increase due to unusual hot weather from September 2023 to March 2024 and the associated increased demands. The lowest increase was in South East Queensland (1.3%).

For Darwin and Perth, the annual average volume of residential water supplied per property was the highest since 2019–20. After experiencing annual declines for 3 years since 2019–20, Sydney, Melbourne, Canberra and Adelaide had increases of 2.8%, 2.9%, 6.3% and 8.4%, respectively, compared to the previous year. However, their supplied volumes remained below the 2019–20 levels. For the second consecutive year, South East Queensland reported an increase (2.7%) in the volume of residential water supplied per property compared to 2021–22.

Table 2.3 Average volume of residential water supplied per property (kL/property)

Major urban centre[a]	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Adelaide	198	196	195	179	194	8.4
Canberra	202	176	163	159	169	6.3
Darwin	373	360	374	349	389	11.5
Melbourne[bc]	148	147	146	140	144	2.9
Perth	227	227	228	219	242	10.5
South East Queensland[b]	162	159	147	149	151	1.3
Sydney	189	186	178	176	181	2.8

Notes:

[a] The figures exclude bulk utilities because they do not supply to customers.

[b] Melbourne and South East Queensland figures are the weighted averages for their respective retailers in each year (i.e. W8 – Total volume of water supplied to residential customers/C2 – Number of connected residential properties: water supply).

[c] Melbourne figures from the 2021–22 reporting year onward are not comparable with pre-2021–22 reporting years due to the merging of City West Water and Western Water to form Greater Western Water on 1 July 2021. Values displayed in this table pre-2021–22 do not include the service area previously managed by Western Water.

2.1.3. Total volume of recycled water supplied – W26

Table 2.4 reports the total volume (ML) of recycled water supplied to customers (W26), aggregated by major urban centre. Unlike W4 (volume of water sourced from recycling plants), W26 includes all recycled water supplied for various uses.

The total recycled water supplied across the major urban centres reached its highest volume in the past 5 years, reflecting a 6.1% increase from 2019–20 and a 10.0% increase from 2022–23 on the national scale. Among all major urban centres, Adelaide, Melbourne and Canberra reported an increase in their total supplied recycled water, with Canberra having the highest increase (82.4%; from 17 ML in 2022–23 to 31 ML in 2023–24) mainly due to higher recycled water supply to non-residential customers than the previous year. South East Queensland, Perth and Sydney reported

a decrease in their total supplied recycled water, with South East Queensland representing the highest decrease of 19.6% from the previous year. In the current year, as in the past 5 years, Darwin did not supply any recycled water to customers.

See Section 3.2 for recycled water supplied by all utilities.

Table 2.4 Total volume of recycled water supplied (ML)

Major urban centre	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Adelaide	23,803	26,627	33,122	20,706	27,021	30.5
Canberra	75	27	24	17	31	82.4
Darwin	0	0	0	0	0	0.0
Melbourne[ab]	42,877	41,716	45,242	40,704	53,804	32.2
Perth	20,681	22,579	21,759	23,337	22,660	-2.9
South East Queensland[a]	14,874	15,468	13,554	17,949	14,439	-19.6
Sydney	46,919	37,669	37,693	41,198	40,361	-2.0

Notes:

[a] Melbourne and South East Queensland figures are the aggregated figures for the bulk utility and the existing retailers in that reporting year.

[b] Melbourne figures from the 2021–22 reporting year onward are not comparable with pre-2021–22 reporting years due to the merging of City West Water and Western Water to form Greater Western Water on 1 July 2021. Values displayed in this table pre-2021–22 do not include the service area previously managed by Western Water.

2.2. Pricing

2.2.1. Total typical residential bill – P8

Table 2.5 shows the total typical residential bill ($) for water supply and wastewater in each major urban centre.

Nationally, the total typical residential bill increased by 2.4% from the previous year. All major urban centres reported increases in their total typical residential bill except Melbourne which reported a decrease of 1.5% from the previous year. Despite the increase in the typical residential bill for water supply in Melbourne, the overall decrease in the total bill was primarily due to reductions in wastewater service bills by South East Water Corporation and Yarra Valley Water Corporation.

Adelaide reported the highest increase (5.2%) and South East Queensland reported the lowest increase (1.8%) from 2022–23. For the past 5 years, Melbourne and Darwin have been reporting the lowest and highest total typical residential bill, respectively, among all major urban centres.

After experiencing annual declines since 2019–20, the total typical residential bill in most major urban centres increased in 2023–24 compared to the previous year. Melbourne was an exception, where the decreasing trend extended to 2023–24. The increased values, however, were still below the 2019–20 levels, with Adelaide reporting the highest decline of 23.5% (from $1,557 in 2019–20 to $1,191 in 2023–24) and Darwin reporting the lowest decline of 7.4% (from $2,198 in 2019–20 to $2,036 in 2023–24) over the past 5 years.

See Section 4.1 for the typical bills charged by all utilities.

Table 2.5 Total typical residential bill ($)

Major urban centre[a]	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Adelaide	1,557	1,279	1,237	1,132	1,191	5.2
Canberra	1,385	1,281	1,215	1,109	1,133	2.2
Darwin	2,198	2,133	2,121	1,982	2,036	2.7
Melbourne[bc]	1,185	1,180	1,088	1,024	1,009	-1.5
Perth	1,901	1,861	1,807	1,708	1,743	2.0
South East Queensland[b]	1,761	1,751	1,653	1,586	1,614	1.8
Sydney	1,329	1,190	1,145	1,115	1,165	4.5

Notes:

[a] The figures exclude bulk utilities as they do not supply to customers.

[b] Melbourne and South East Queensland figures are the weighted average of the retail utilities in that year (i.e. P3 – Typical residential bill: water supply/C2 – Number of connected residential properties: water supply, and P6 – Typical residential bill: wastewater/C6 – Number of connected residential properties: wastewater).

[c] Melbourne figures from the 2021–22 reporting year onward are not comparable with pre-2021–22 reporting years due to the merging of City West Water and Western Water to form Greater Western Water on 1 July 2021. Values displayed in this table pre-2021–22 do not include the service area previously managed by Western Water. The service area managed by Western Water pre-2021–22 makes up approximately 3.5% of total connections in the Melbourne urban centre from the 2021–22 reporting year onward.

2.3. Environment

2.3.1. Total net greenhouse gas emissions per 1,000 properties – E12

Table 2.6 shows the contribution of the utilities' operations to greenhouse gas emissions (t CO_2 equivalent/1,000 properties), aggregated by major urban centre.

Compared to the previous year, emissions per 1,000 properties decreased in Canberra, Melbourne and Sydney to a nearly equal extent (~12.0%), while emissions increased in Adelaide, Darwin, Perth and South East Queensland. Perth had the highest increase of 19.0% from the previous year. Similar to the past 5 years, Perth also had the highest total emissions per 1,000 properties of all major urban centres in 2023–24 (451 t CO_2 equivalent/1,000 properties). This correlates with Perth's high reliance on desalinated water.

Adelaide has been reporting the highest decrease of 57.8% among all major urban centres over the past 5 years (from 332 t CO_2 equivalent/1,000 properties in 2019–20 to 140 t CO_2 equivalent/1,000 properties in 2023–24) despite reporting an increase of 4.5% in 2023–24 from its previous year. The lowest decrease in total emissions per 1,000 properties since 2019–20 was observed in South East Queensland (10.3%; from 204 t CO_2 equivalent/1,000 properties in 2019–20 to 183 t CO_2 equivalent/1,000 properties in 2023–24). Darwin was the only urban centre reporting an increase of 16.4% compared to 2019–20.

Canberra, Melbourne and Sydney followed a downward trend in the total emissions per 1,000 properties over the past 5 years and with notable declines compared to their 2019–20 levels. Darwin has had an increasing trend in total emissions per 1,000 properties since 2020–21 (24.6%).

See Section 8.1 for total net greenhouse gas emissions by all utilities.

Table 2.6 Total net greenhouse gas emissions per 1,000 properties (t CO_2 equivalent/1,000 properties)

Major urban centre	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Adelaide	332	342	143	134	140	4.5
Canberra	331	196	177	175	154	-12.0
Darwin	213	199	223	233	248	6.4
Melbourne[ac]	278	249	245	229	202	-11.8
Perth	701	695	567	379	451	19.0
South East Queensland[b]	204	205	202	164	183	11.6
Sydney	175	169	168	158	139	-12.0

Notes:

[a] Melbourne figures are the weighted average of the 3 retailers (i.e. E12/C4 – Total connected properties) and Melbourne Water. Melbourne Water's emissions are calculated based on the total connected properties of the 3 active retailers in each year.

[b] South East Queensland figures are the weighted average of the retailers (i.e. E12/C4 – Total connected properties).

[c] Melbourne figures from the 2021–22 reporting year onward are not comparable with pre-2021–22 reporting years due to the merging of City West Water and Western Water to form Greater Western Water on 1 July 2021. Values displayed in this table pre-2021–22 do not include the service area previously managed by Western Water.

2.4. Finance

2.4.1. Total capital expenditure: water supply and wastewater – F16

Table 2.7 shows the combined capital expenditure ($000s) related to the utilities' water and sewerage operations, aggregated by major urban centre. Historical values are adjusted using the CPI index.

Overall, the sum of total capital expenditure for water supply and wastewater service increased by 24.4% from the previous year. All major urban centres, except Canberra, experienced an increase compared to the previous year. While Darwin reported the highest increase of 76.7% from 2022–23 – driven by the Manton Dam return to service project – it had the lowest total capital expenditure ($67,251 thousand) of all major urban centres. South East Queensland reported the lowest increase of 18.4% from the previous year. Sydney reported the highest total capital expenditure for the third consecutive year since 2021–22 ($2,237,578 thousand). Canberra was the only major urban centre experiencing a decline (6.5%) in the total capital expenditure compared to the previous year due to a reduction in capital expenditure for wastewater services.

All major urban centres, except Canberra, have followed an increasing trend in total capital expenditure since 2021–22, with Darwin having the highest increase (140.7%; from $27,941 thousand in 2021–22 to $67,251 thousand in 2023–24), and South East Queensland the lowest increase (29.3%; from $974,252 thousand in 2021–22 to $1,259,250 thousand in 2023–24).

See Section 5.1 for combined capital expenditure for all utilities.

Table 2.7 Total capital expenditure: water supply and wastewater ($000s)

Major urban centre	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Adelaide	402,574	332,569	281,405	295,121	364,269	23.4
Canberra	120,935	101,614	74,606	75,450	70,564	-6.5
Darwin	24,231	23,959	27,941	38,063	67,251	76.7
Melbourne[ab]	1,216,434	1,262,030	1,237,091	1,405,221	1,704,883	21.3
Perth	471,994	432,595	385,360	417,631	649,634	55.6
South East Queensland[a]	975,109	964,236	974,252	1,063,611	1,259,250	18.4
Sydney[a]	1,200,962	1,151,570	1,376,062	1,812,901	2,237,578	23.4

Notes:

[a] Melbourne, South East Queensland and Sydney figures are aggregates for the bulk utility and their respective retailers.

[b] Melbourne figures from the 2021–22 reporting year onward are not comparable with pre-2021–22 reporting years due to the merging of City West Water and Western Water to form Greater Western Water on 1 July 2021. Values displayed in this table pre-2021–22 do not include the service area previously managed by Western Water.

2.5. Customers

2.5.1. Average duration of an unplanned interruption: water supply – C15

Table 2.8 shows the average duration (minutes) of unplanned interruptions to water supply in a utility's operation, aggregated by major urban centre.

Compared to the previous year, Adelaide, South East Queensland and Perth reported an increase in their average duration of unplanned water supply interruptions, with Adelaide reporting the largest increase of 64.8%. The 12.9% increase in South East Queensland was due to a single large interruption recorded by Logan City Council in May 2024 that impacted 100 customers for an extended period of time. Melbourne reported no changes compared to the previous year. Canberra, Darwin and Sydney reported a decline in their average duration of unplanned water supply interruptions, with Sydney and Darwin representing the highest and lowest decreases of 22.5% and 1.3%, respectively, while Darwin also reporting the shortest duration of unplanned water supply interruptions (76 minutes) among all major urban centres in 2023–24.

Canberra and Darwin continued decreases in average duration of unplanned interruptions in water supply since 2020–21, with decreases of 45.3% and 19.7%, respectively. After experiencing annual declines since 2019–20, Adelaide reported an increase from 204 minutes in 2019–20 to 272 minutes in 2023–24. In Perth, an change from 111 minutes in 2019–20 to 170 minutes in 2023–24 represented an increase of 53.2%.

See Section 6.1 for unplanned interruptions to water supply for all utilities.

Table 2.8 Average duration of an unplanned interruption: water supply (minutes)

Major urban centre	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Adelaide	204	188	181	165	272	64.8
Canberra	136	147	136	132	118	-10.6
Darwin	-	139	102	77	76	-1.3
Melbourne[ab]	101	98	103	110	110	0.0
Perth	111	140	141	162	170	4.9
South East Queensland[a]	119	121	134	132	149	12.9
Sydney[a]	187	200	192	231	179	-22.5

Notes:

[a] No data is available for Darwin before 2020–21.

[b] Melbourne figures from the 2021–22 reporting year onward are not comparable with pre-2021–22 reporting years due to the merging of City West Water and Western Water to form Greater Western Water on 1 July 2021. Values displayed in this table pre-2021–22 do not include the service area previously managed by Western Water.

3. Water resources

3.1. Average annual residential water supplied – W12

The average annual residential water supplied indicator (W12) reports the average volume (kL/property) of metered and estimated non-metered potable and non-potable water supplied to residential properties during the reporting year. It is derived by dividing the total volume of residential water supplied (W8) by the number of connected residential water properties (C2). The average volume is influenced by a number of factors, including:

- weather events
- rainfall and temperature
- water conservation measures (for example, water restrictions)
- availability of water supply
- housing density
- water prices.

Rainfall is the most influential factor affecting residential consumption. An increase in rainfall is likely to reduce demand and a decrease in rainfall is likely to increase demand. A decrease in rainfall can result in a significant decrease in runoff into storages and trigger demand-management measures such as water restrictions.

Average annual residential water supply (W12) data for all utilities reporting in 2023–24 is given in Table A1, Appendix A.

3.1.1. Key findings

Table 3.1 shows a summary of the median average annual volume of water supplied to residential customers by utility size group.

Nationally, the average annual volume of water supplied in 2023–24 increased by 6.0%, driven by a 5.0% increase in the median for the Major and Medium size groups. The increase in residential water supply across all utility size groups is consistent with the above to very much above average temperatures in most of the country which caused greater demand for water.

More utilities reported an increase in the average annual residential water supplied than those reporting a decrease in all size groups (overall 61 out of 76 reporting utilities recorded an increase). Orange City Council (New South Wales) in the Small size group was the only utility with no change in its average annual residential water supplied compared to the previous year. Dubbo Regional Council (New South Wales) in the Medium size group reported the highest increase (38.4%), and Ballina Shire Council (New South Wales) in the Small size group reported the highest decrease (19.1%) in average annual residential water supplied from 2022–23.

Table 3.1 Overview of results: Average annual residential water supplied (kL/property)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	242	134	14	1	152	159	5
	WC (Perth)	Logan					
Large	389	140	8	4	181	189	4
	P&W (Darwin)	Toowoomba					
Medium	455	111	14	7	171	179	5
	Lower Murray Water	South Gippsland Water					
Small	397	91	25	3	180	187	4
	P&W (Alice Springs)	Westernport Water					
All size groups (national)	**455**	**91**	**61**	**15**	**168**	**178**	**6**
	Lower Murray Water	**Westernport Water**					

Note: The median average annual residential water supplied (kL/property) for each year is calculated using data from all active utilities providing water supply services in that reporting year.

Figure 3.1 shows a box-and-whisker plot of the average annual volume of residential water supplied for all utilities reporting W12. While Australia's total rainfall in 2023–24 was above the 1961–90 average (14.0%), it was lower than the previous year's total rainfall across the country. The relatively drier conditions in 2023–24 compared to the previous year explain the increased median residential water supply across the country.

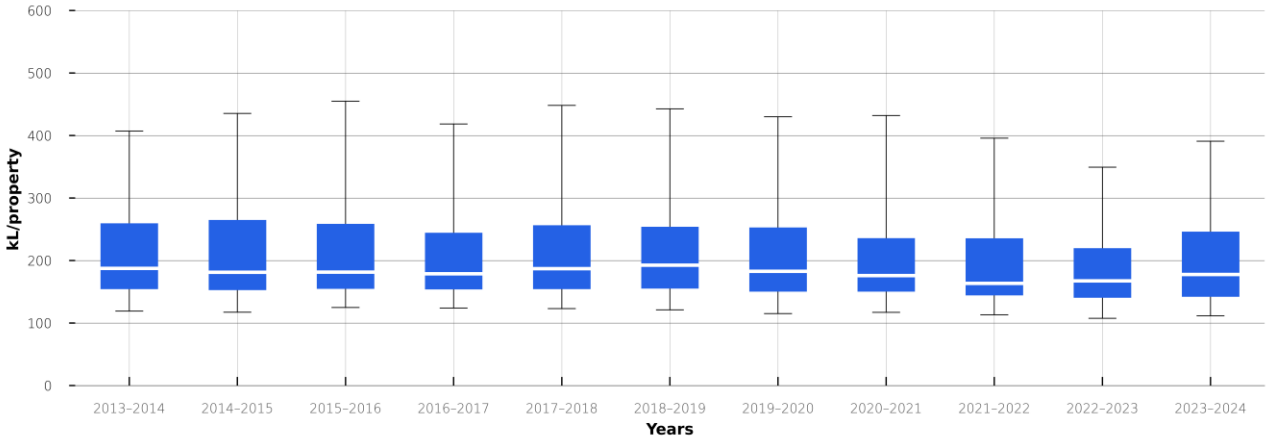


Figure 3.1 Average annual residential water supplied (kL/property)

3.1.2. Results and analysis – Major utility group

Figure 3.2 shows a ranked breakdown of the average volume of residential water supplied for each utility in the Major utility group from 2019–20 to 2023–24.



Figure 3.2 Average annual residential water supplied (kL/property) – Major utility group

For the past 5 years since 2019–20, Water Corporation – Perth (Western Australia) has reported the largest, and Logan City Council (Queensland) has reported the smallest average annual volume of water supplied to residential customers.

Variations from the previous year ranged from a 0.6% decrease reported by Logan City Council (Queensland) (the only decrease among all utilities in the Major size group) to a 10.5% increase reported by SA Water Corporation (South Australia) (Table A1 in Appendix A).

3.2. Total recycled water supplied – W26

Total recycled water supplied (ML) is the sum of all treated sewage effluent used by the utility and its customers. It includes residential, commercial, industrial, agricultural and environmental use as well as on-site use by the utility.

The volume of recycled water supplied is affected by a number of factors, including:

- availability of potable water
- size of the utility
- the utility's proximity to potential customers (for example, agricultural users, major industrial customers and recreational facilities)
- fluctuations in sewage received and effluent available for recycling
- government policy.

Total recycled water supplied (W26) data for all utilities reporting in 2023–24 is shown in Table A2, Appendix A.

3.2.1. Key findings

Table 3.2 shows a summary of the total recycled water supplied by utility size group.

Nationally, the total volume of recycled water supplied increased by 7% in 2023–24, driven by a 10% increase in the Large utility size group. The Small utility size group saw a decrease of 3%, while other groups reported increases compared to the previous year.

More utilities in all size groups reported an increase in the total volume of recycled water supplied than those reporting a decrease (overall 46 out of 73 reporting utilities recorded an increase). The changes varied significantly compared to the previous year, with Wingecarribee Shire Council (New South Wales) in the Medium size group reporting the highest increase (778.0%), and Bathurst Regional Council (New South Wales) in the Small size group reporting the highest decrease (60.1%). Also, Bathurst Regional Council (New South Wales) identified a data consistency issue in reporting the volume of recycled water supply to non-residential customers, which affects the annual change in the total volume of recycled water supplied.

Table 3.2 Overview of results: Total recycled water supplied (ML)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Total		Change in total from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	40,361	31	9	5	120,936	130,518	8
	Sydney Water	Icon Water					
Large	10,626	0	8	3	25,903	28,422	10
	North East Water	P&W (Darwin)					
Medium	5,418	88	11	10	37,289	39,713	7
	Albury	Wingecarribee					
Small	2,290	21	18	8	14,852	14,452	-3
	WC (Albany)	Lismore					
All size groups (national)	**40,361**	**0**	**46**	**26**	**198,980**	**213,105**	**7**
	Sydney Water	**P&W (Darwin)**					

Note: The total recycled water supplied (ML) is calculated using data from all utilities that reported data for W26 in both the 2022–23 and 2023–24 reporting years.

3.2.2. Results and analysis – Major utility group

In 2023–24, the total volume of recycled water supplied by all reporting utilities was 213,105 ML, around 61.2% of which was supplied by the Major utility group. For the past 5 years since 2019–20, Sydney Water Corporation (New South Wales) has reported the largest, and Icon Water Limited (Australian Capital Territory) the smallest total volume of recycled water supplied.

Variations from the previous year ranged from a 13.1% decrease reported by Barwon Region Water Corporation (Victoria) to an 82.4% increase reported by Icon Water Limited (Australian Capital Territory) (Table A2 in Appendix A).

4. Pricing

4.1. Typical residential bill: water supply and wastewater – P8

The typical residential bill ($) for water supply and wastewater (P8) is the sum of fixed charges and volumetric-usage charges for water and sewage billed to a residential customer. The typical bill is based on each utility's average annual volume of residential water supplied (W12) and its pricing structure (P1, P1.2 to P1.7, P4.1 to P4.3). Prices are set by government or, in some jurisdictions, by a regulator, council or utility.

Water bills are influenced by a number of factors, including:

- size of the utility's customer base
- geographical location
- distribution of the customer base
- local topography
- climate
- available sources of water
- government policy and legislation.

The mix of fixed and usage charges, and the level of water consumption, affect the typical residential bill.

When drawing comparisons between utilities, it is important to note that changes in a typical bill may result from both changes to average consumption and changes to the price of water.

Historically, residential water bill pricing models have varied across the nation. Most utilities now have a water supply pricing model based on a 2-part structure: a fixed component and a component based on volumetric usage.

Townsville City Council and Whitsunday Regional Council remain exceptions as ratepayers have a choice between a fixed allocation and a 2-part structure.[7]

Unlike residential water supply pricing, most utilities have a fixed price model for wastewater services. The exceptions are the Melbourne utilities[8], Central Coast Council and Unitywater. These utilities have both a fixed and volumetric component in their wastewater charges.

Billing data is indexed using the consumer price index (CPI) to facilitate comparison in real terms.

Typical residential bill (P8) data for all utilities reporting in 2023–24 is presented in Table A3, Appendix A.

4.1.1. Key findings

Table 4.1 shows a summary of the median typical residential bills by utility size group.

Nationally, median typical residential bills for water and wastewater services increased by 2% from 2022–23. This equates to a $33 increase in the median typical residential bill.

[7] https://www.whitsundayrc.qld.gov.au/our-council/about-council/rates-fees-and-charges/water-billing-options-and-water-tariff-calculator

[8] Greater Western Water, Yarra Valley Water and South East Water

Overall, the water and wastewater utilities in the Major utility group reported the highest increase of 10% from the 2022–23 median bill, compared with 2 to 3% in the other utility size groups. Almost 3 quarters of all utilities reported an increase in their typical residential bill from 2022–23 (52 out of 71 reporting utilities). Among all, 5 utilities reported an increase of 10% or more, with Central Coast Council (New South Wales) in the Major size group being the highest of those at 23.4%. Decreases in the typical residential bill (19 utilities) were all less than 5%, with the largest decrease of 4.5% reported by the Ballina Shire Council (New South Wales) in the Small size group.

Byron Shire Council (New South Wales) and Central Highlands Regional Council (Queensland), both in the Small size group, reported the highest ($2,354) and Goulburn Valley Water (Victoria) in the Large size group reported the lowest typical residential bill ($937).

Table 4.1 Overview of results: Typical residential bill: water supply and wastewater ($)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	1,797	990	12	3	1,123	1,235	10
	Gold Coast	South East Water					
Large	2,036	937	8	4	1,490	1,516	2
	P&W (Darwin)	Goulburn Valley Water					
Medium	2,131	1,096	14	6	1,584	1,635	3
	Gladstone	Lower Murray Water					
Small	2,354	1,101	18	6	1,818	1,857	2
	Multiple utilities	Mount Barker					
All size groups (national)	**2,354**	**937**	**52**	**19**	**1,558**	**1,591**	**2**
	Multiple utilities	**Goulburn Valley Water**					

Note: The typical residential bill in each year is calculated using data from all active utilities supplying both water and wastewater services in that year.

Figure 4.1 shows a box-and-whisker plot of typical residential bills for all utilities reporting data in a given year. Following 2 successive years of decreases, the typical residential bill rose slightly in 2023–24. Despite increasing by 2%, the national median typical residential bill was the second lowest since 2013–14. The difference between the highest and lowest typical residential bill (the distribution in Figure 4.1) in 2023–24 was the largest since 2013–14.

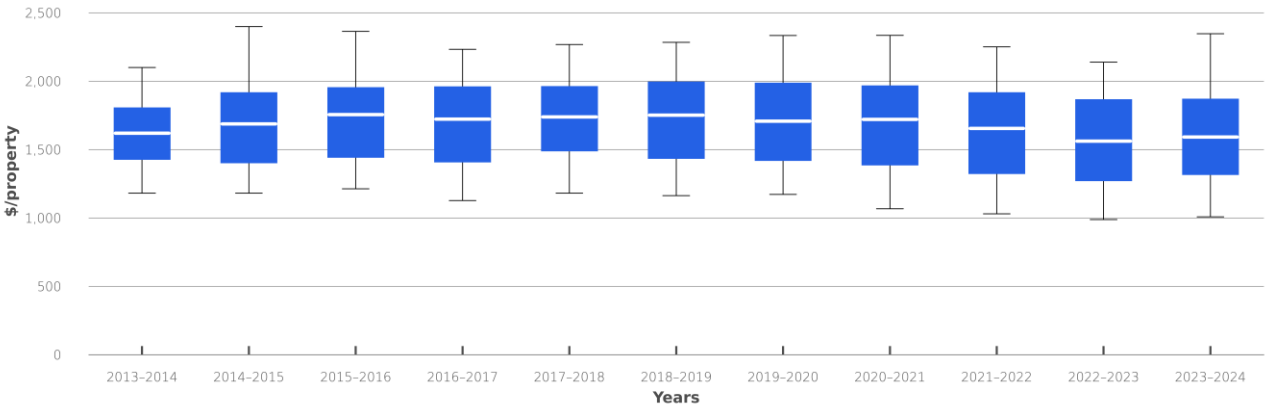


Figure 4.1 Typical residential bill: water supply and wastewater ($), 2013–14 to 2023–24

4.1.2. Results and analysis – Major utility group

Figure 4.2 presents a ranked breakdown of the typical residential bill for the Major utility group. The figure shows the water (P3) and wastewater (P6) components of the bill for active utilities that have reported their information in 2023–24.

The typical residential bill increased from 2022–23 for most of the utilities in the Major size group, with only South East Water Corporation (Victoria), Yarra Valley Water Corporation (Victoria), and Logan City Council (Queensland) reporting decreases of less than 5%. The increase in typical residential bill was largely driven by increases in the typical bills for water supply for all Major utilities, except for Urban Utilities (Queensland), which reported a decrease of less than 1% in its water supply bills. Typical residential bills for wastewater services also increased for 9 of the 15 Major utilities, while South East Water Corporation (Victoria) and Yarra Valley Water Corporation (Victoria) which had significant decreases of close to 23.5% were among the 6 utilities that reported a decrease in their wastewater bills in 2023–24.

The proportional change in typical residential bills from 2022–23 ranged from a decrease of 4.3% (Yarra Valley Water Corporation in Victoria) to an increase of 23.4% (Central Coast Council in New South Wales). City of Gold Coast (Queensland) reported the highest typical residential bill ($1,797/property) in the Major size group and was the highest for water supply ($1,073/property) while Water Corporation – Perth (Western Australia) had the highest typical wastewater bills ($928/property). City of Gold Coast (Queensland), Water Corporation – Perth (Western Australia), and Logan City Council (Queensland) remained the 3 utilities with the highest pricing (Table A3 in Appendix A).

Despite an increase in the pricing of water supply services, a reduction in the typical residential bill for wastewater resulted in South East Water Corporation (Victoria) reporting the lowest typical residential bill this year ($990/property).

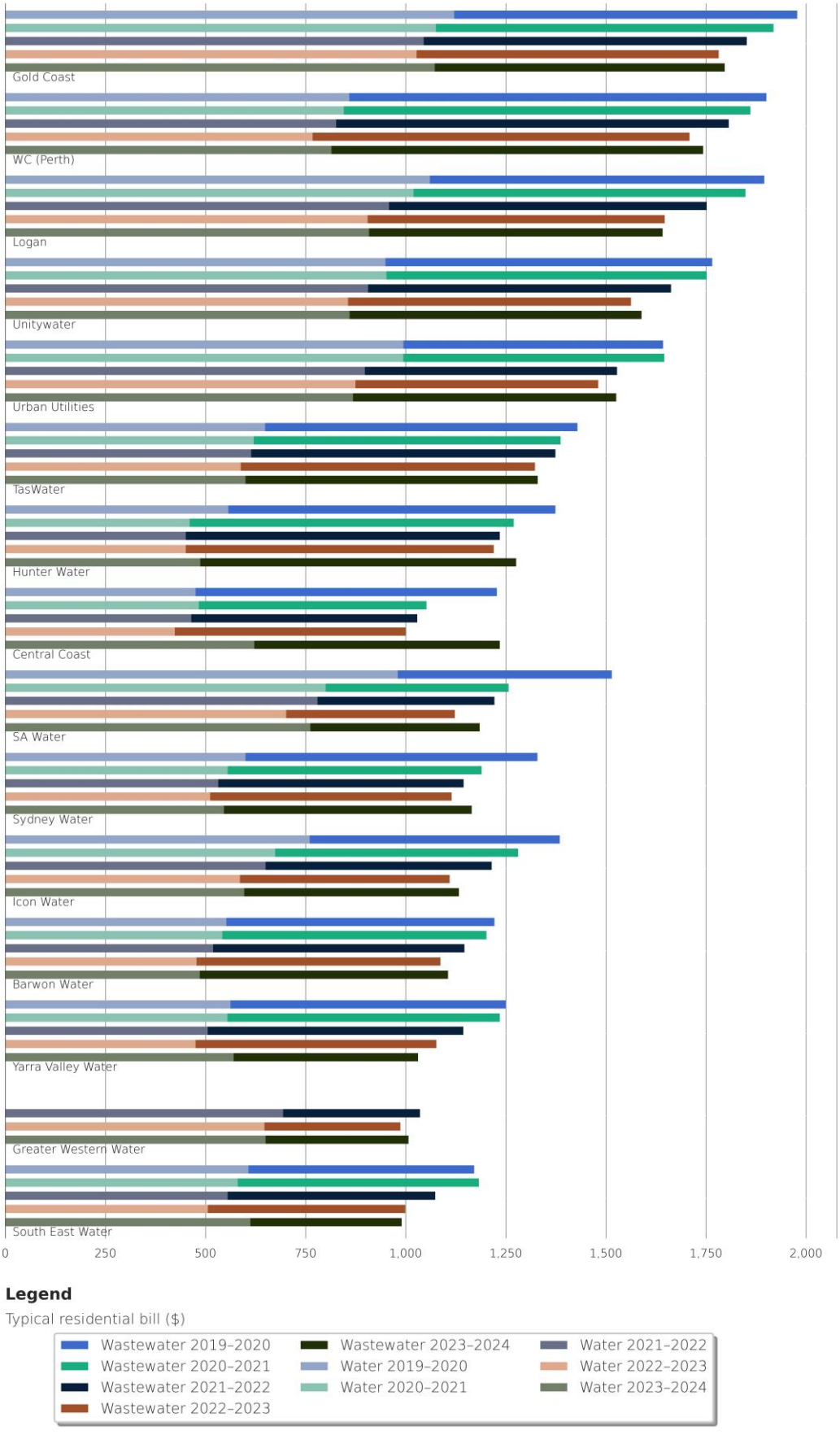


Figure 4.2 Typical residential bill: water supply and wastewater ($) – Major utility group

4.2. Annual residential bill based on 200 kL per annum: water supply and wastewater – P7

The annual bill ($) based on 200 kL for water and wastewater services (P7) is the sum of the annual bill for the supply of 200 kL of water (P2) and the annual bill for the provision of wastewater services for a residential customer using 200 kL of water (P5).

While the typical residential bill (P8) is the best guide to determining the impact of pricing on customers, the annual bill based on 200 kL aids comparisons between utilities. Adopting a consistent basis of 200 kL for the bill partially normalises the data, correcting for differences in the volumes of water supplied and providing insight into price variations.

Billing data is indexed using the consumer price index (CPI) to facilitate comparison in real terms.

The annual bill based on 200 kL (water supply and wastewater) data for related utilities is shown in Table A4, Appendix A.

4.2.1. Key findings

Table 4.2 presents a summary of the median 200 kL/annum residential bill data by utility size group.

Table 4.2 Overview of results: Annual residential bill based on 200 kL per annum: water supply and wastewater ($)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	1,937	1,200	9	6	1,274	1,345	6
	Gold Coast	SA Water					
Large	1,955	885	6	6	1,504	1,513	1
	Toowoomba	Goulburn Valley Water					
Medium	2,460	881	10	9	1,637	1,606	-2
	Tweed	Lower Murray Water					
Small	2,569	1,101	14	10	1,828	1,860	2
	Kempsey	Mount Barker					
All size groups (national)	**2,569**	**881**	**39**	**31**	**1,639**	**1,634**	**0**
	Kempsey	**Lower Murray Water**					

Note: The 200 kL residential bill data for water supply and wastewater for each year are calculated using data from all active utilities reporting against the P2 and P5 indicators in that year.

On a 200 kL/annum basis, the national median bill in 2023–24 was similar to the 2022–23 median bill.

The Major size utility group reported the highest change (6% increase) in the median annual residential bill based on 200 kL/annum, while there was 2% or less difference from 2022–23 in the other size groups. A little over half of all utilities reported an increase in 2023–24 (39 out of 71 reporting utilities, with Coffs Harbour City Council in New South Wales reporting no change).

Kempsey Shire Council (New South Wales) in the Small size group had the highest normalised annual residential bill for the year at $2,569/property. In previous years, Tweed Shire Council (New South Wales) had been significantly higher than all other utilities in this bill category but reported a 25.4% decrease in 2023–24 (the largest decrease in the Medium size group), largely driven by a significant drop (48.6%) in its normalised wastewater bills.

Lower Murray Water (Victoria) in the Medium size utility group reported the lowest annual residential bill based on 200 kL/annum at $881/property and together with Goulburn Valley Water (Victoria), were the only 2 utilities lower than $1000/property in 2023–24.

Orange Shire Council (New South Wales) reported the highest increase in annual residential bills based on 200 kL at 36.7% from 2022–23 which raised it into the top third of utility bills in the Small sized group.

4.2.2. Results and analysis – Major utility group

Figure 4.3 shows a ranked breakdown of the annual residential bill based on 200 kL for the active utilities within the Major size utility group that have reported their information in 2023–24.

The majority of utilities in the Major size group reported less than a 2% change in the combined water and wastewater annual residential bill based on 200 kL in 2023–24. Hunter Water Corporation (New South Wales), Sydney Water Corporation (New South Wales), and Icon Water Limited (Australian Capital Territory) all reported slightly higher increases of between 3% and 4%, while Central Coast Council (New South Wales) had the largest increase of 20.4%.

While South East Water Corporation (Victoria) and Yarra Valley Water Corporation (Victoria) both had increases of more than 20% in the water component of the normalised residential bill, this was offset by an approximately similar decrease in the wastewater services bill resulting in a small overall decrease in the combined bill in 2023–24.

The City of Gold Coast (Queensland) had the highest combined annual residential bill based on 200 kL ($1937/property), with 2 other Queensland utilities, Logan City Council and Unitywater, making up the top 3, largely due to notably higher wastewater bills. Figure 4.3 shows that Water Corporation – Perth (Western Australia) had high normalised wastewater service bills compared with the other Major utilities due to a 2.5% general price increase and an increased average wastewater bill resulting from the revaluation of the Gross Rental Value by Landgate. South Australian Water Corporation (South Australia) had the lowest normalised bill at $1200/property, however, South East Water Corporation (Victoria) and Greater Western Water (Victoria) were just less than $3 higher than that.

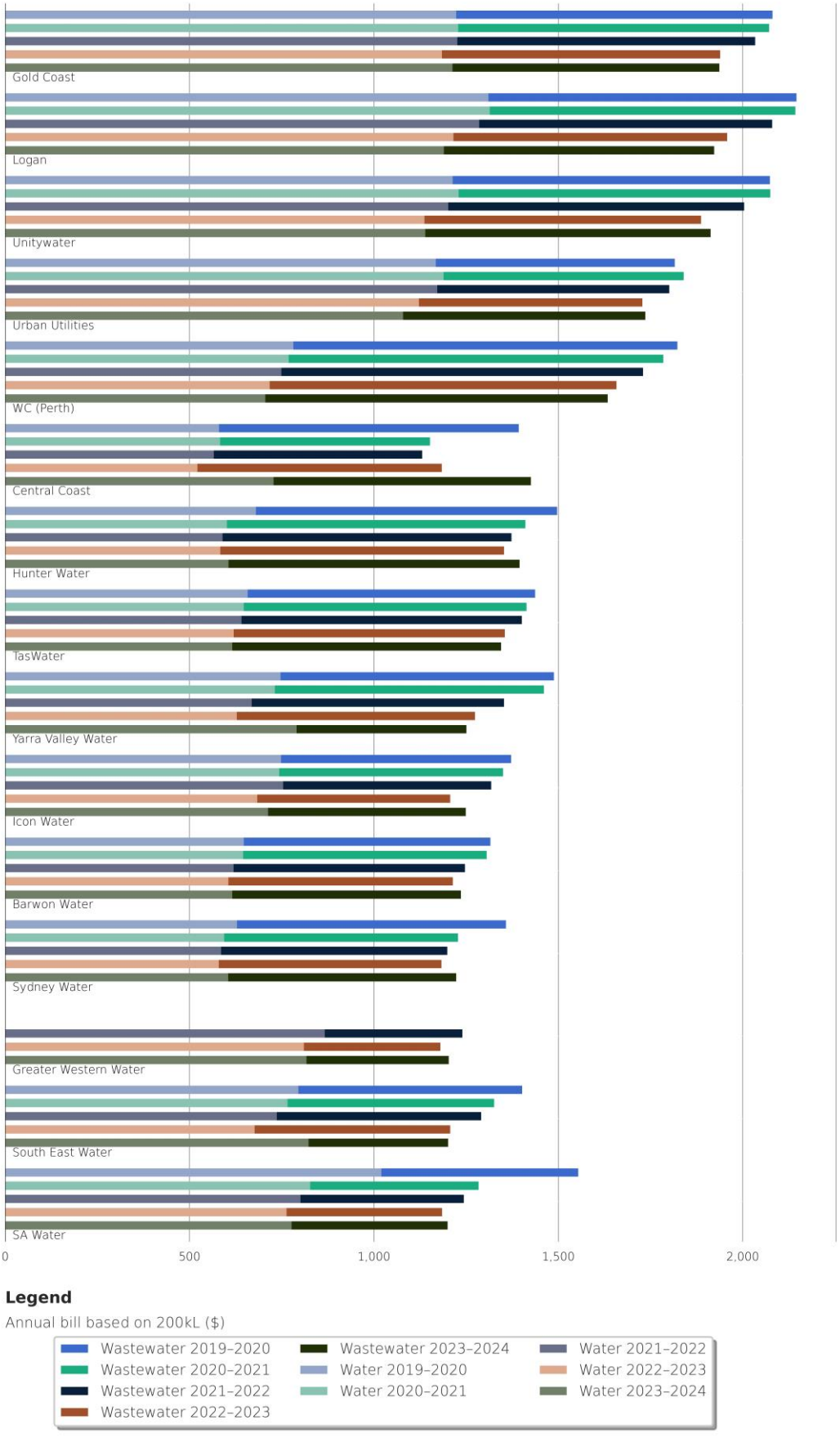


Figure 4.3 Annual bill based on 200 kL: water supply and wastewater ($) – Major utility group.

5. Finance

5.1. Total capital expenditure: water supply and wastewater – F16

Total capital expenditure ($000s) on water supply and wastewater (F16) provides a measure of the total level of capital investment by each utility and the size of the utility and its capital responsibilities.

Capital expenditure programs often affect operational expenditure. They are influenced by several factors, including the:

- age of a utility's infrastructure
- stage of each asset's lifecycle
- time and duration of a project.

Capital expenditure data is indexed using the consumer price index (CPI) to facilitate comparison in real terms.

Total capital expenditure for water supply and wastewater data for all utilities reporting in 2023–24 is shown in Table A5, Appendix A.

5.1.1. Key findings

Table 5.1 shows a summary of total capital expenditure for water and wastewater by utility size group. In real terms, total capital expenditure rose by 19% to nearly $7.5 billion compared to 2022–23 which was higher than the 16% increase in 2022–23 compared to 2021–22. This reflects the progression and expansion of capital programs, particularly works on wastewater assets, as well as higher project costs. 45 utilities reported an increase in capital spending out of 70 reporting utilities. As in previous years, Sydney Water Corporation (New South Wales) had the highest total capital expenditure in 2023–24 (~$2.12 billion).

Table 5.1 Overview of results: Total capital expenditure: water and wastewater ($000s)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Total		Change in total from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	2,119,910	70,564	13	2	5,125,981	6,177,293	21
	Sydney Water	Icon Water					
Large	83,623	12,438	9	3	564,202	600,199	6
	Coliban Water	Redland City					
Medium	84,100	3,859	13	6	431,064	524,519	22
	Eurobodalla	Albury					
Small	29,685	1,280	10	10	188,393	193,349	3
	Cassowary Coast	Snowy Monaro					
All size groups (national)	**2,119,910**	**1,280**	**45**	**21**	6,309,640	7,495,360	19
	Sydney Water	**Snowy Monaro**					

Note: Total capital expenditure for water and wastewater services in each year is calculated using data from active utilities reporting against F14 and F15 in both years.

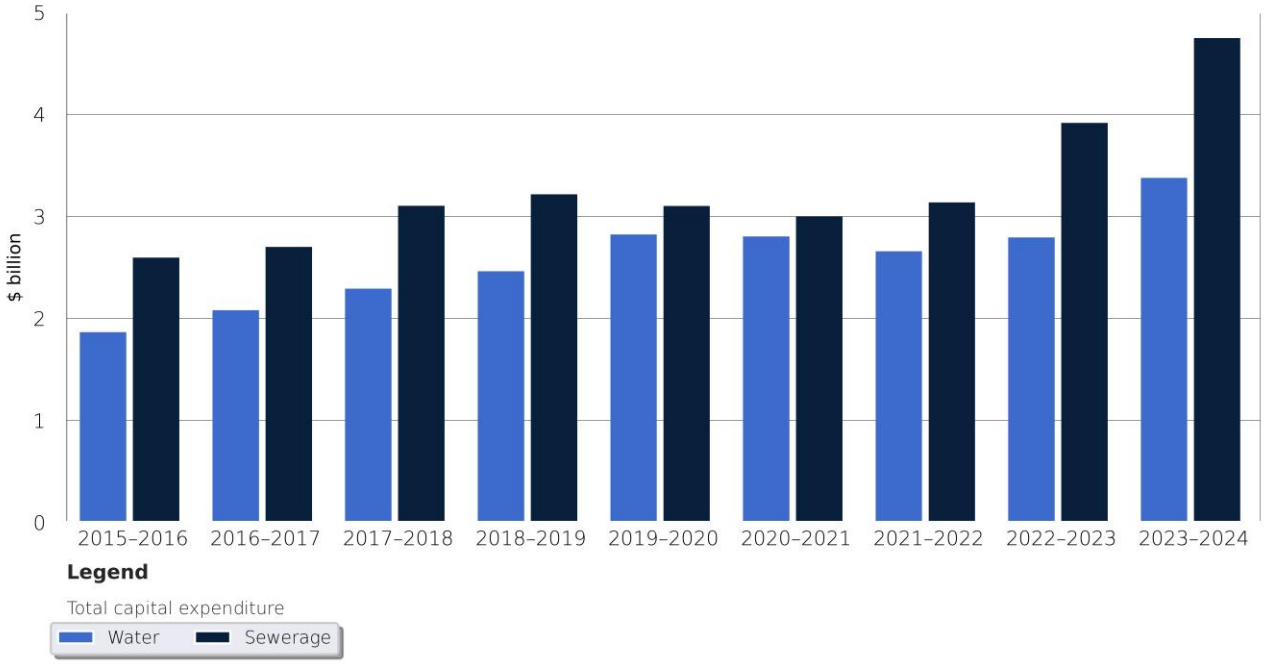


Figure 5.1 Total capital expenditure: water supply and wastewater ($ billion) for active utilities that reported all 9 years (excluding bulk water utilities)

5.1.2. Results and analysis – Major utility group

An increase in total capital expenditure across water and wastewater operations was reported by 13 out of the 15 reporting utilities in the Major utility group, resulting in a total increase of 21% (~$1.1 billion) in 2023–24 from the previous year. Only Urban Utilities (Queensland) and Icon Water Limited (Australian Capital Territory) reported small decreases of less than 7% in spending from 2022–23.

The utilities with the highest increases in total capital expenditure from the previous year were Water Corporation – Perth (Western Australia) with 55.6% increase, Unitywater (Queensland) with 41.1%, and TasWater (Tasmania) with 38.4%. Despite reporting a decrease in water supply project costs, TasWater (Tasmania) more than doubled spending on advancing works on wastewater assets.

5.2. Capital expenditure per property: water supply (F28) and wastewater (F29)

Capital expenditure ($/property) on water supply (F28) and wastewater (F29), on a per connected property basis, provides a measure of capital investment by each utility relative to its customer base. The normalisation on a per connected property basis facilitates a comparison between utilities.

Capital expenditure data is indexed using the consumer price index (CPI) to facilitate comparison in real terms. Capital expenditure data per connected property for water and wastewater services, for all utilities reporting in 2023–24, is shown in Table A6 and Table A7, Appendix A.

5.2.1. Key findings

Table 5.2 and Table 5.3 show a summary of the median capital expenditure of utilities providing water and wastewater services, respectively, by utility size group.

Table 5.2 Overview of results: Capital expenditure per property: water supply ($/property)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	549	76	10	5	242	222	-8.3
	SA Water	Gold Coast					
Large	944	46	8	4	348	365	5.0
	P&W (Darwin)	Redland City					
Medium	1,231	89	11	8	325	290	10.8
	Fitzroy River Water	Albury					
Small	6,008	0	11	9	408	286	29.9
	Mount Barker	Byron					
All size groups (national)	**6,008**	**0**	**40**	**26**	**308**	**307**	**-0.3**
	Mount Barker	**Byron**					

Note: Median capital expenditure per property: water supply ($/property) for each year is calculated using data from utilities providing water and wastewater services that reported against F28 in that year.

Table 5.3 Overview of results: Capital expenditure per property: wastewater ($/property)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	897	167	11	4	354	377	6
	TasWater	Icon Water					
Large	484	131	7	5	295	301	2
	North East Water	Toowoomba					
Medium	1,709	62	11	8	315	354	12
	Wingecarribee	Albury					
Small	2,312	0	9	11	250	225	-10
	Cassowary Coast	Byron					
All size groups (national)	**2,312**	**0**	**38**	**28**	**315**	**331**	**5**
	Cassowary Coast	**Byron**					

Note: Median capital expenditure per property: wastewater ($/property) in each year is calculated using data from all active utilities providing water and wastewater services that reported against F29 in that year.

The median capital expenditure for water supply per property remained almost unchanged nationally but was higher for 40 out of 69 reporting utilities compared to the previous year (Table 5.2). Mount Barker District Council (South Australia) in the Small size group had the highest increase (311.0%) compared to the previous year and the highest capital expenditure for water supply per property ($6,008/property) with the expansion of the municipality's recycled water network to supply non-residential customers.

The national median capital expenditure per property on wastewater services increased by 5% from $315/property in 2022–23 to $331/property in 2023–24 (Table 5.3). Increases in the median compared to the previous year were reported for all utility size groups except for the Small size groups. Cassowary Coast Regional Council (Queensland) had the highest capital expenditure per property on wastewater services at $2,312/property, nearly 4 times higher than the previous year, due to the acquisition and replacement of a regional sewerage treatment plant.

5.2.2. Results and analysis – Major utility group

Figure 5.2 shows a ranked breakdown of capital expenditure on a per connected property basis for the Major utility group. The figure shows the water supply (F28) and wastewater (F29) components of the total expenditure and reinforces the year-to-year variation.

There is typically volatility in capital expenditure due to the phasing of major projects. Out of the 15 utilities in the Major size group, 10 reported increases in capital expenditure per property for water supply and 11 for wastewater services. TasWater (Tasmania) had the highest combined water supply and wastewater capital expenditure per property in 2023–24. This was driven by a 150.7% increase in the wastewater component, a significantly higher increase than the other Major utilities. Unitywater (Queensland) reported the highest increase in water supply capital expenditure per property (70.1%), becoming the utility with the second highest combined capital expenditure per property for water supply and wastewater services in 2023–24.

Compared to the previous year, only Icon Water Limited (Australian Capital Territory) and Urban Utilities (Queensland) reported a decrease in the combined capital expenditure per property for water supply and wastewater services, with Icon Water Limited (Australian Capital Territory) reporting the lowest combined expenditure of all Major utilities in 2023–24.

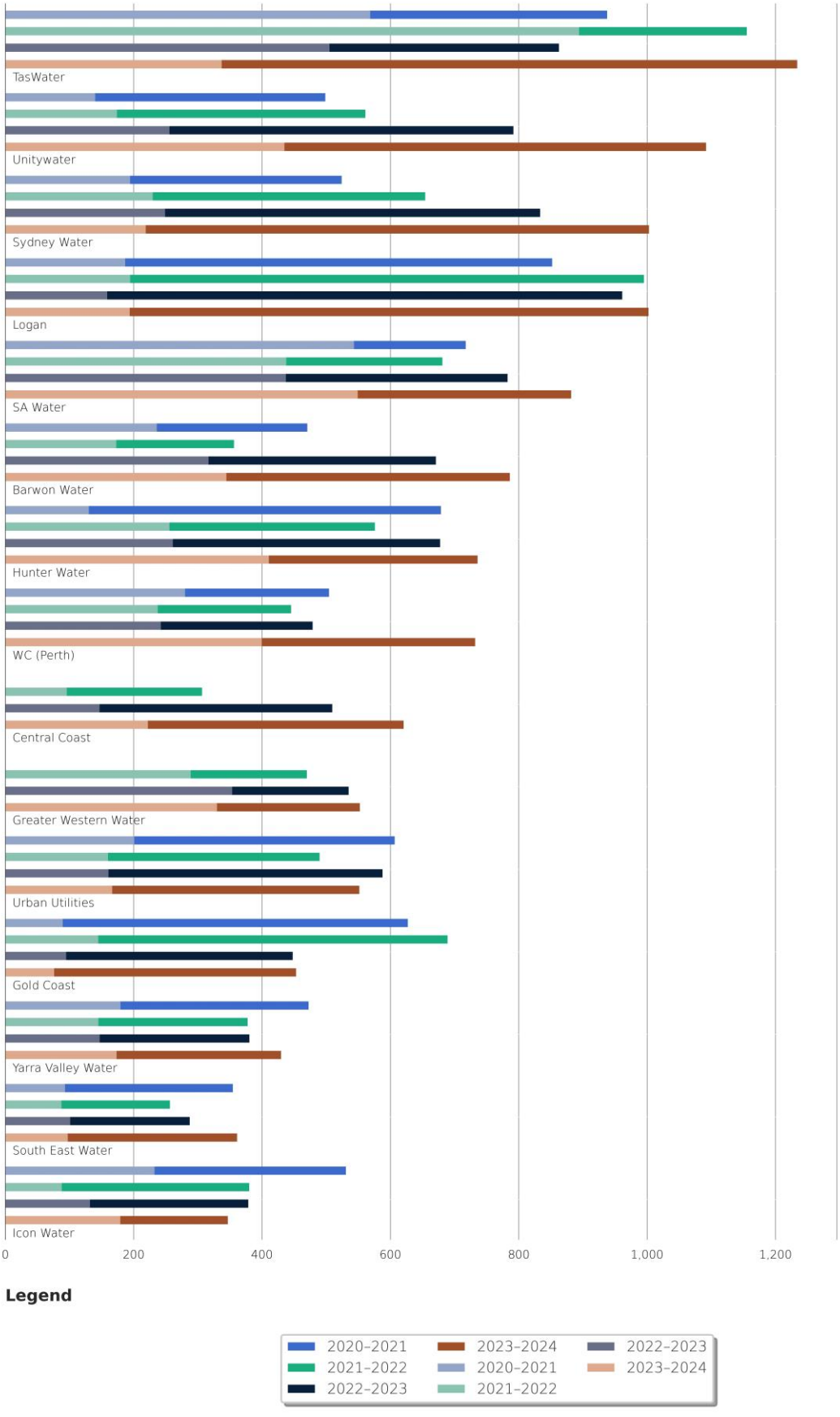


Figure 5.2 Capital expenditure: water supply and wastewater ($/property) – Major utility group

5.3. Community service obligations ratio – F8

Revenue from community service obligations (CSOs) as a percentage of a utility's total income (F8) is a measure of the extent to which activities undertaken by a utility are subsidised.

Payments for CSOs (F25) to a utility by a state or territory government are made when a utility is directed to undertake activities that they would not perform on a solely commercial basis. CSOs in the water sector may be provided to:

- allow reductions on bills to certain disadvantaged customer groups (for example, pensioners)
- allow utilities to charge common tariffs across all geographical regions despite cost differences
- ensure the delivery of government policy (for example, by administering rebates)
- allow utilities to provide services to high-cost areas where full cost recovery would otherwise result in unaffordable bills.

CSO data for all utilities reporting in 2023–24 is presented in Table A8, Appendix A.

5.3.1. Key findings

Table 5.4 shows a summary of the revenue from CSOs, by utility size group.

In 2023–24, 39 utilities reported decreases in the revenue received from CSOs relative to total income, while 19 utilities reported increases. Nationally, there was a 5% decrease in the median revenue from CSOs from 2022–23.

Table 5.4 Overview of results: Community service obligations ratio

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	0.1038	0	2	11	0.0242	0.0220	-9
	SA Water	Multiple utilities					
Large	0.0559	-0.0504	5	7	0.0169	0.0164	-2
	North East Water	WC (Mandurah)					
Medium	0.0645	0	6	13	0.0101	0.0089	-12
	GWMWater	Queanbeyan					
Small	0.1360	-0.4546	7	8	0.0071	0.0061	-14
	P&W (Alice Springs)	WC (Geraldton)					
All size groups (national)	**0.1360**	**-0.4546**	**20**	**39**	**0.0101**	**0.0096**	**-5**
	P&W (Alice Springs)	**WC (Geraldton)**					

Note: Median revenue from community service obligations (%) for each year is calculated using data from all utilities providing data in that year. In Western Australia, some regional schemes recover adequate revenue to cover the cost of service of the scheme, including the community service obligations, and these schemes partially offset the net loss of other regional services. When reported independently, these schemes will show a negative operating subsidy.

5.3.2. Results and analysis – Major utility group

In 2023–24, the Major size utility group reported a decrease of 9% in the median CSO payments relative to total utility income.

As in previous years, SA Water Corporation (South Australia) continued to report the highest proportion of revenue from CSOs (of 10.4% in 2023–24). For this utility, CSO payments are used to subsidise non-profitable water services, to provide water services in country areas at metropolitan water prices.

Only 2 Major utilities (Central Coast Council in New South Wales and Icon Water Limited in Australian Capital Territory) reported a small increase in CSO revenue relative to income. Barwon Water had the largest overall decrease in the absolute values of CSO ratio, from 4.46% in 2022–23 to 4.05% in 2023–34 (representing an absolute decrease of 0.41%). The Queensland utilities of City of Gold Coast, Logan City Council and Urban Utilities all reported no CSO payments this year.

6. Customer

6.1. Average duration of an unplanned interruption: water – C15

The average duration (minutes) of an unplanned interruption (C15), is the average time a customer is without water supply due to an unforeseen interruption that requires attention by the utility.

Unplanned interruptions include scheduled interruptions that exceed the time limit given in the original notification. The indicator is a measure of customer service, the condition of the water network and how effectively the network is managed.

The average duration is influenced by the:

- scale of the event causing the interruption
- location of the interruption (for example, the proximity to a repair crew and the depth of the burst pipe)
- utility's response policy for outlying areas
- number of maintenance and repair staff at the utility's disposal.

Note that a single event affecting a small number of properties for a long duration can cause large annual variations in this indicator, especially for smaller utilities.

Data on the average duration of an unplanned interruption (water supply) for all active utilities reporting in 2023–24 is presented in Table A9, Appendix A.

6.1.1. Key findings

Table 6.1 shows a summary of average duration of unplanned interruptions by utility size group.

Table 6.1 Overview of results: Average duration of an unplanned interruption: water (minutes)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	255.0	88.3	7	6	130.4	138.7	6
	SA Water	Barwon Water					
Large	153.0	35.9	1	9	96.0	102.5	7
	Shoalhaven	Cairns					
Medium	253.0	24.4	8	9	103.8	101.0	-3
	Tamworth	Gladstone					
Small	303.0	16.0	9	9	103.5	120.5	16
	Whitsunday	Armidale					
All size groups (national)	**303.0**	**16.0**	**25**	**33**	**111.5**	**117.5**	**5**
	Whitsunday	**Armidale**					

Note: Median average duration of an unplanned interruption: water (minutes) for each year is calculated for all active utilities that reported data for C15 in that year.

The median average duration of unplanned interruptions increased by 5% from 111.5 minutes in 2022–23 to 117.5 minutes in 2023–24 on the national scale. The longest and shortest duration of an unplanned interruption in water supply were in the Small size group, with Whitsunday Regional Council (Queensland) having the longest duration (303 minutes) and Armidale Regional Council

(New South Wales) having the shortest duration and also largest decrease from the previous year (90.4%, from 166 minutes in 2022–23 to 16 minutes in 2023–24).

Mackay Regional Council, part of the Medium size group in Queensland, reported the largest increase (312.2%, from 7.4 minutes in 2022–23 to 30.5 minutes 2023–24).

6.1.2. Results and analysis – Major utility group

Figure 6.1 shows a ranked breakdown of the average duration of an unplanned interruption for the Major utility group from 2019–20 to 2023–24. The figure highlights the large year-to-year variation in the indicator for all utilities in the Major size group that can result from a single major mains break.

In the Major utility size group, there were significant variations in the average duration of unplanned interruptions.

Logan City Council (Queensland) experienced a substantial increase of 76.5% from 2022–23, reaching 254 minutes. This increase was driven by a single large interruption recorded in May 2024 that impacted 100 customers for an extended period of time while repairs were completed. SA Water Corporation (South Australia) reported the highest average duration of unplanned interruption in water supply and a notable increase of 52.7%, with the duration increasing to 255 minutes in 2023–24.

Sydney Water Corporation (New South Wales) experienced a significant decrease of 22.5%, bringing the duration down to 179 minutes. City of Gold Coast (Queensland) and South East Water Corporation (Victoria) had minimal variance (less than 1%), while Hunter Water Corporation (New South Wales) remained unchanged.

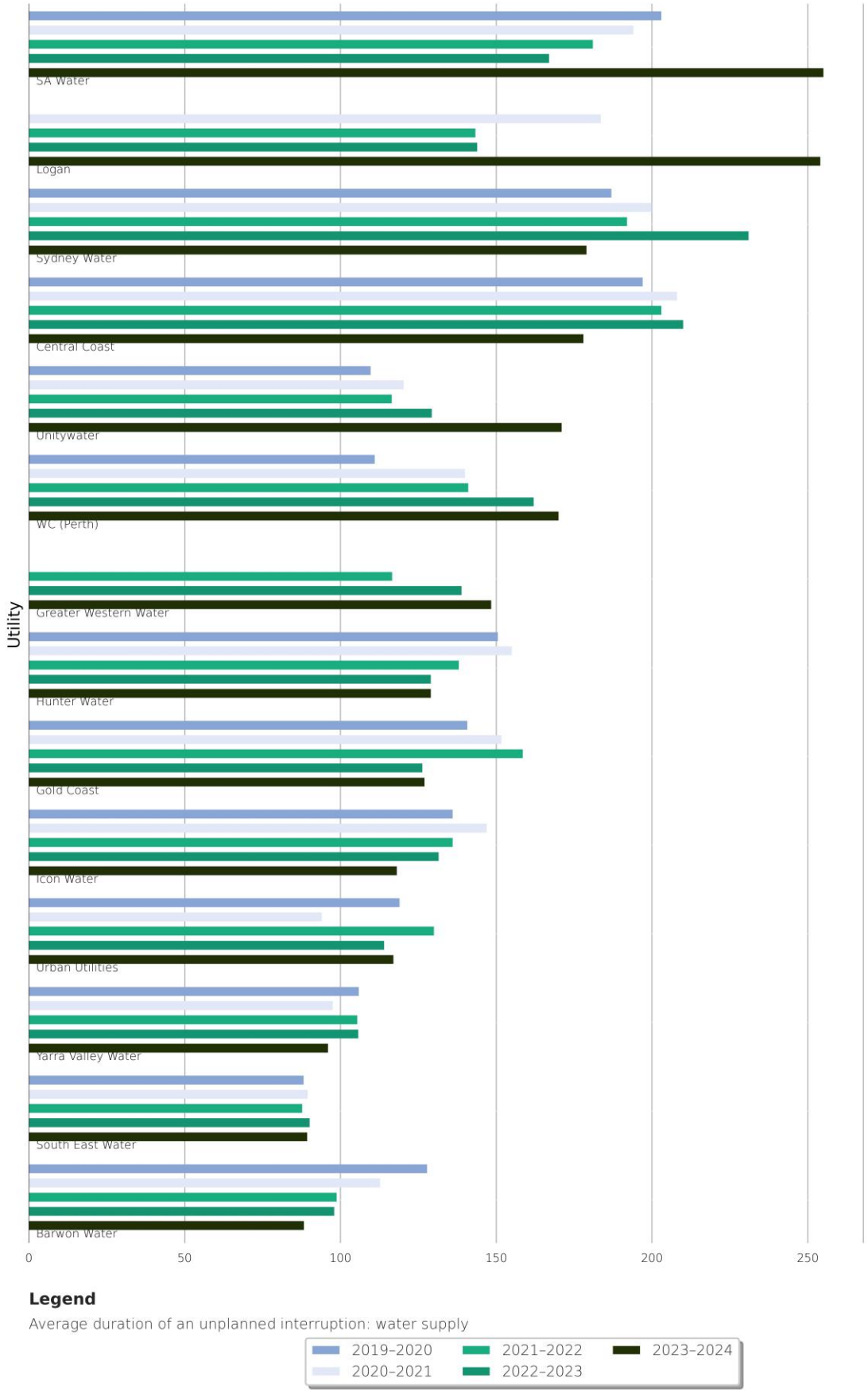


Figure 6.1 Average duration of an unplanned interruption: water (minutes) – Major utility group

7. Asset

7.1. Number of water main breaks, bursts and leaks per 100 km of water mains – A8

The number of water main breaks, bursts and leaks per 100 km of water mains (A8) is the total number of breaks, bursts and leaks in all distribution system mains[9], excluding breaks associated with headworks and transfer mains. It provides an indication of both customer service and the condition of the network. The number of main breaks is influenced by various factors, including:

- soil type
- rainfall
- pipe material
- age and condition of the network.

Data on the number of water main breaks, bursts and leaks per 100 km of water mains for all utilities reporting in 2023–24 are shown in Appendix A, Table A10.

7.1.1. Key findings

Figure 7.1 shows that the national median number of main breaks in 2023–24 remained unchanged from 2022–23, following the modest increases since 2021–22.

Table 7.1 shows a summary of the number of water main breaks per 100 km of water main by utility size group. The largest increase (90.2%) was reported in the Medium size group by Rockhampton Regional Council (Queensland), whereas the Small size group reported the largest decrease of 94.8% in Byron Shire Council in New South Wales. Additionally, TasWater (Tasmania) within the Major size group reported the highest number of water main breaks, bursts and leaks per 100 km of water (42.8), while Goulburn Mulwaree Council (New South Wales) in the Small size group reported the lowest number (1.1).

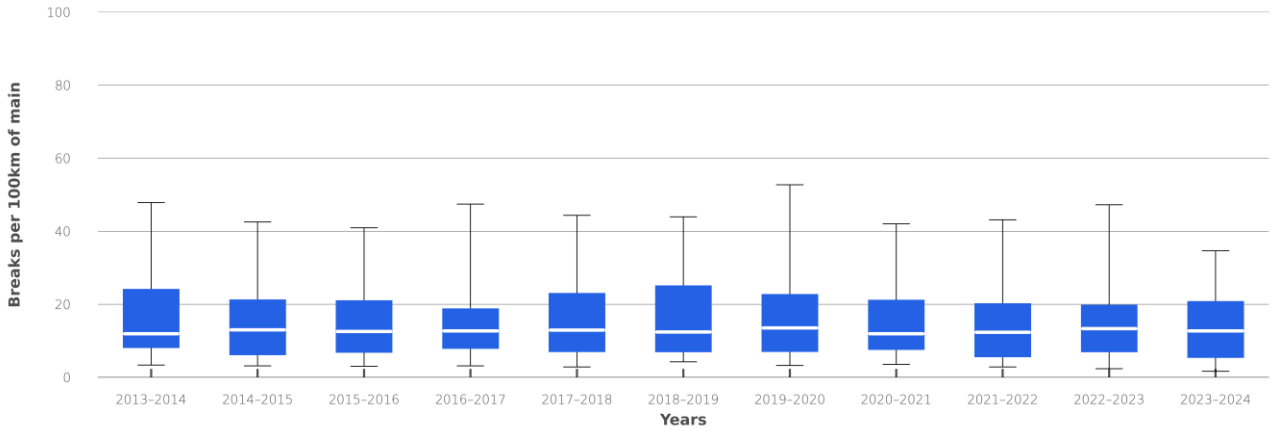


Figure 7.1 Number of water main breaks, bursts and leaks per 100 km of water mains

[9] The figure includes both potable and non-potable water mains.

Table 7.1 Overview of results: Number of water main breaks, burst and leaks per 100 km of water mains (mains breaks/100 km)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	42.8	3.5	7	7	16.8	17.9	7
	TasWater	Logan					
Large	25.0	2.2	7	5	16.4	15.0	-9
	Coliban Water	WC (Mandurah)					
Medium	39.8	1.5	10	11	10.0	11.6	17
	GWMWater	Clarence Valley					
Small	31.2	1.1	10	17	12.7	10.8	-15
	Cassowary Coast	Goulburn Mulwaree					
All size groups (national)	**42.8**	**1.1**	**34**	**40**	**13.3**	**12.7**	**-5**
	TasWater	**Goulburn Mulwaree**					

Note: The median for water main breaks, burst and leaks per 100 km of water mains in each year was calculated using data from all utilities (dual-service and single-service providers) reporting data against A8 in that year.

7.1.2. Results and analysis – Major utility group

Figure 7.2 shows a ranked breakdown of the water main breaks for each utility in the Major utility group from 2019–20 to 2023–24.

The Major utility group reported an increase (7%) in the number of breaks per 100 km of water mains, with 7 out of the 15 Major utilities reporting a decrease from 2022–23 (Unitywater in Queensland was the only utility that reported no change). Barwon Water (Victoria) reported the largest increase (23.7%), while the City of Gold Coast (Queensland) reported the largest decrease (18.1%) from 2022–23.

As in previous years, TasWater (Tasmania) reported the highest number of breaks per 100 km of water mains (42.8) followed by Yarra Valley Water Corporation in Victoria (34.6). However, both utilities saw decreases in their number of breaks per 100 km of water mains, with reductions of 9.3% and 2.3%, respectively from 2022–23.

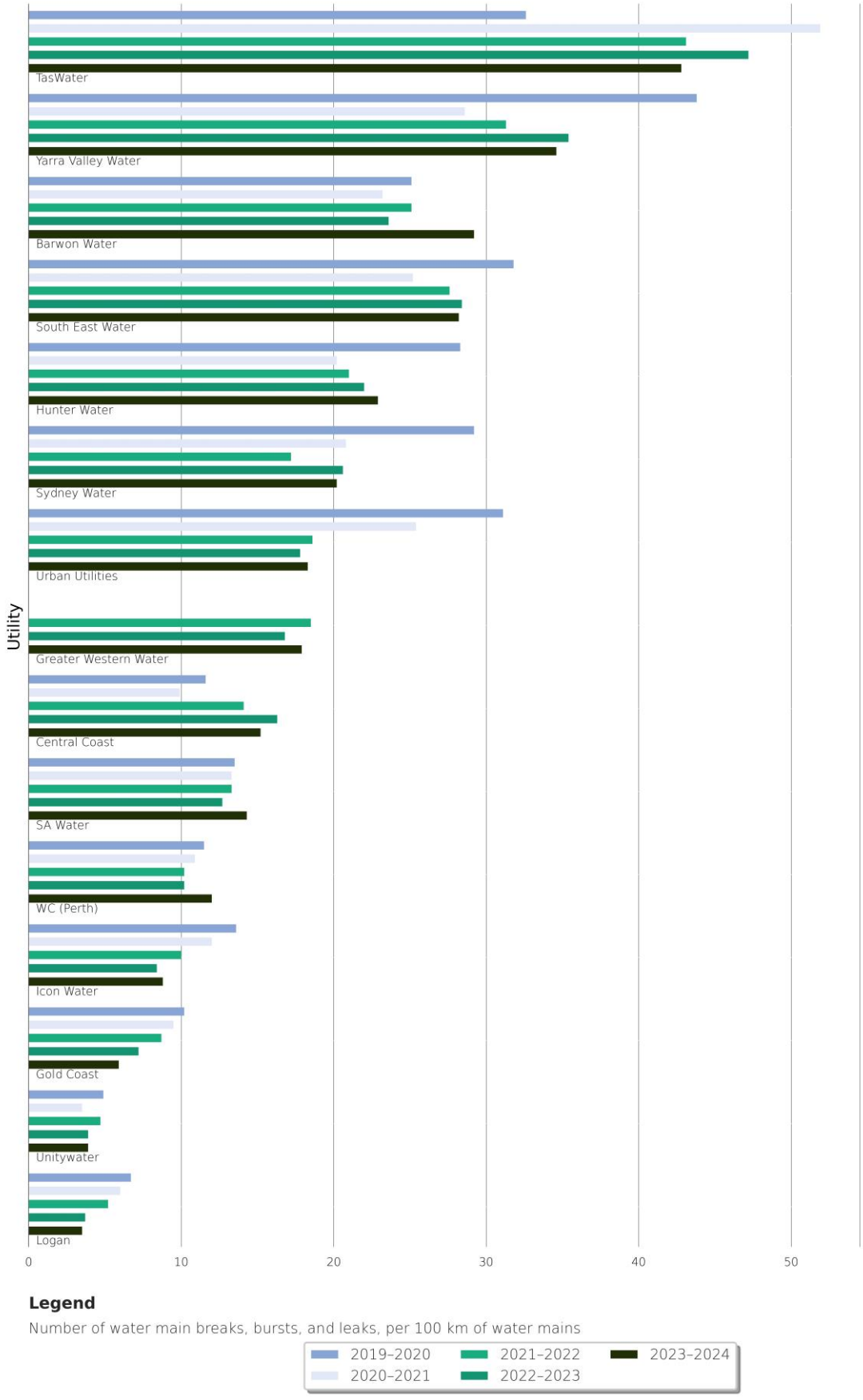


Figure 7.2 Water main breaks, bursts and leaks per 100 km of water mains – Major utility group

7.2. Number of sewer mains breaks and chokes per 100 km – A14 and number of property connection sewer breaks and chokes per 1,000 properties – A15

Indicator A14 reports the number of sewer breaks and chokes per 100 km of sewer mains, and A15 reports the number of property connection sewer breaks and chokes per 1,000 properties. The indicators are presented together to provide a complete picture of sewer system performance as utilities have sewer networks with various configurations.

- Some utilities have a very long property connection (for example, from the customer's sanitary drain to the middle of a road), while others have a very short or no property connection (that is, the sanitary drain may connect straight to the sewer main, which runs down an easement at the back of the property).
- Some utilities do not own[10] or maintain the property connections and therefore do not report on them in accordance with the definition of the indicator.
- Other utilities are responsible for only a portion of property sewer connections and so only report results on those for which they are responsible.

The performance of a sewerage system is influenced by:

- soil type
- pipe material
- sewerage configuration
- age
- tree root intrusion
- management of trade waste
- volume of sewage inflows
- rainfall.

Results reflect both the condition of the network and the level of customer service. For the reasons given above, care should be taken in comparing the performance of utilities against each other using these indicators.

Data on sewer mains breaks and chokes for all utilities reporting in 2023–24 are presented in Appendix A, Table A11. Property connection sewer breaks and chokes for all utilities reporting in 2023–24 are presented in Appendix A, Table A12.

7.2.1. Key findings

Table 7.2 shows a summary of the number of sewer mains breaks and chokes per 100 km of sewer main by utility group.

Table 7.3 shows a summary of the property connection sewer breaks and chokes per 1,000 properties by utility group.

[10] For such utilities, the property owner is responsible for the property's sewer connections.

Table 7.2 Overview of results: Number of sewer mains breaks and chokes per 100 km of sewer main (breaks and chokes/100 km)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	63.9	5.0	13	2	22.5	25.8	15
	TasWater	Gold Coast					
Large	42.8	4.9	4	8	14.6	13.2	-10
	Townsville	Gippsland Water					
Medium	94.0	1.1	9	10	11.5	10.6	-8
	Albury	Mackay					
Small	133.0	2.0	14	12	11.8	15.0	27
	Essential Energy	Bega Valley					
All size groups (national)	**133.0**	**1.1**	**40**	**32**	**14.3**	**14.8**	**3**
	Essential Energy	**Mackay**					

Note: The median number of sewer mains breaks (per 100 km of sewer main) in each year is calculated using data from all utilities (dual-service and single-service providers) reporting data against A14 in that year.

Table 7.3 Overview of results: Number of property connection sewer breaks and chokes per 1,000 properties (breaks and chokes/1,000 properties)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	33.00	0.17	14	0	3.19	3.81	19
	SA Water	Sydney Water					
Large	5.90	0.85	6	3	3.00	2.60	-13
	Townsville	Shoalhaven					
Medium	23.44	0.00	8	12	3.80	3.31	-13
	GWMWater	Albury					
Small	44.87	0.00	13	7	2.80	3.40	21
	Essential Energy	Mount Barker					
All size groups (national)	**44.87**	**0.00**	**41**	**22**	**3.39**	**3.40**	**0**
	Essential Energy	**Multiple utilities**					

Note: The median number of property connection sewer breaks and chokes per 1,000 properties in each year is calculated using data from all utilities (dual-service and single-service providers) reporting data against A15 in that year.

In 2023–24, there was a national median of 14.8 sewer main breaks and chokes per 100 km of sewer main, representing a 3% increase from 2022–23 (Table 7.2) with 40 utilities reporting an increase and 32 utilities reporting a decrease. The overall median of sewer breaks and chokes per 1,000 properties (Table 7.3) remained stable, with a small rise from 3.39 in 2022–23 to 3.40 in 2023–24. An increase was reported by 41 utilities, and 22 utilities reported a decrease.

The increase in the median number of sewer breaks and chokes was greatest in the Small size group followed by the Major size group.

Essential Energy in New South Wales from the Small size group, reported the highest number of property connection sewer breaks and chokes per 1,000 properties (44.8). In contrast, Albury City Council in New South Wales from the Medium size group reported no property connection sewer breaks and chokes, while Mount Barker District Council in South Australia from the Small size group reported just one incident during the reporting period.

7.2.2. Results and analysis – Major utility group

Figure 7.3 shows a ranked breakdown of the sewer mains breaks and chokes (per 100 km of sewer main) for each Major utility from 2019–20 to 2023–24 and Figure 7.4 shows a ranked breakdown of property connection sewer breaks and chokes per 1,000 properties.

Only 2 of the 15 Major utilities reported a decrease in sewer mains breaks and chokes per 100 km sewer main, while all utilities reported an increase in the number of property connection sewer breaks and chokes per 1,000 properties from 2022–23 to 2023–24.

Sydney Water Corporation (New South Wales) reported the largest increase in breaks and chokes per 100 km of sewer mains (46.4%, Figure 7.3) while it had the smallest property connection sewer breaks and chokes per 1,000 properties (0.2, Figure 7.4). SA Water Corporation (South Australia) reported the largest number of property connection sewer breaks and chokes per 1,000 properties (33, Figure 7.4) while TasWater (Tasmania) reported the largest number of sewer mains breaks and chokes per 100 km of sewer main (63.9, Figure 7.3) compared to 2022–23.

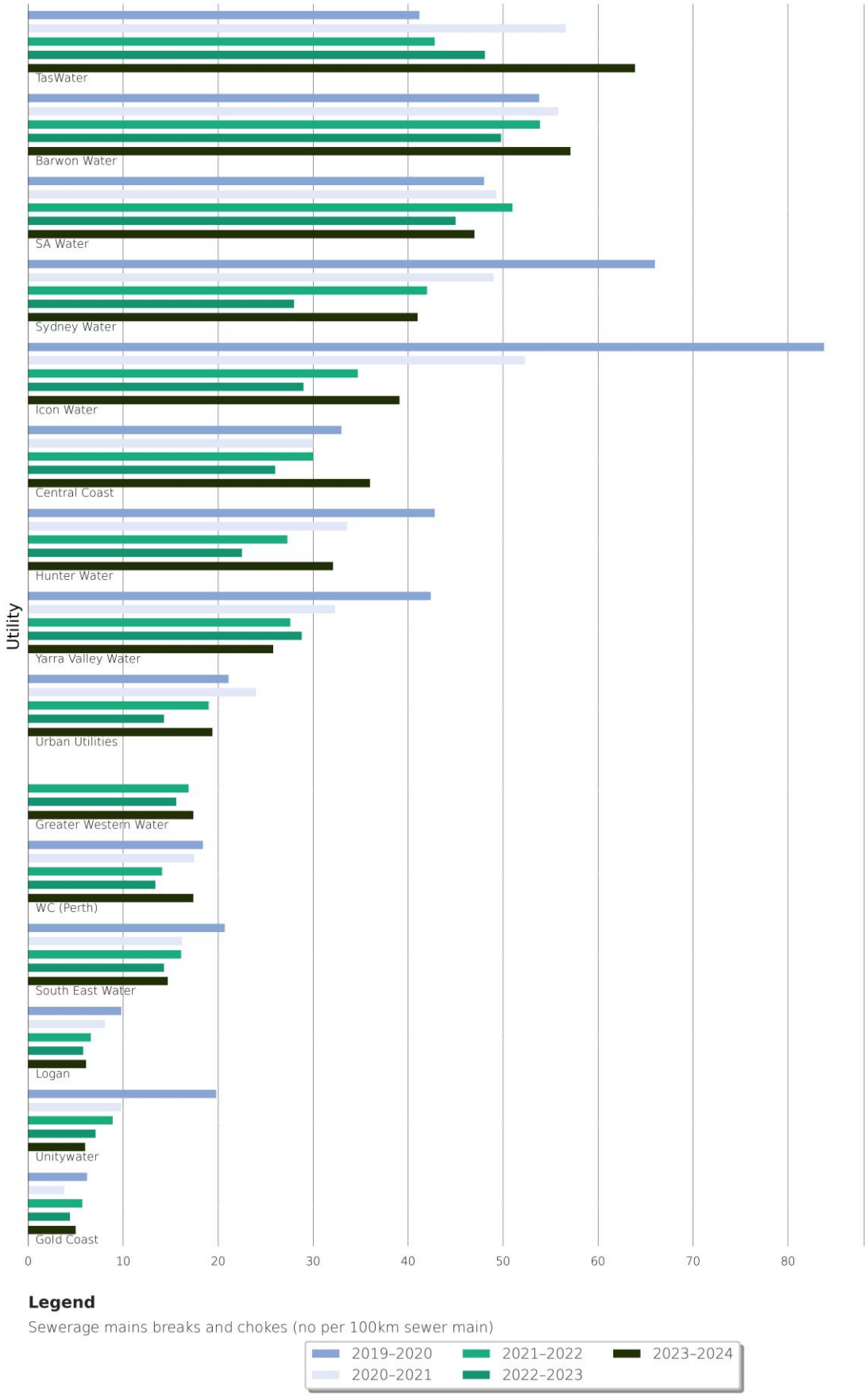


Figure 7.3 Sewer mains breaks and chokes per 100 km of sewer main – Major utility group

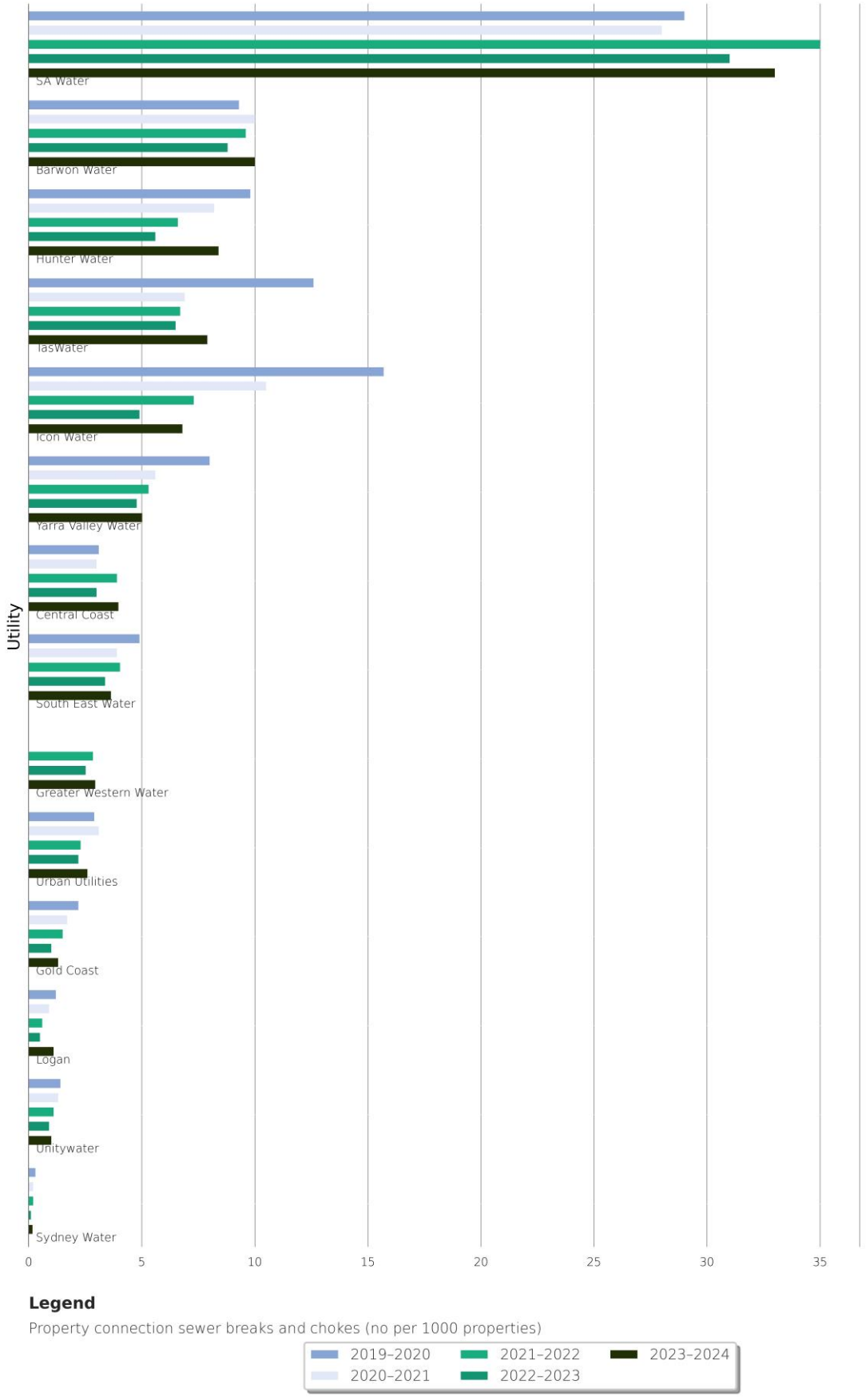


Figure 7.4 Property connection sewer breaks and chokes per 1,000 properties – Major utility group

7.3. Real losses: service connections – A10

'Real' losses (A10) are leakages and overflows from potable water mains, service reservoirs and service connections before the customer meter (L/service connection/day). This indicator excludes metering errors, unauthorised consumption (apparent losses), and unbilled authorised consumption (for example, water used for firefighting). Performance of this indicator may be influenced by the condition of mains, infrastructure and water pressure.

Real losses are estimated using a range of assumptions, including assumed errors in metered water deliveries, estimates of unmetered components, and metering of night flows, and may not be as accurate as other indicators (such as water main breaks) when comparing utilities.

Real loss data for all utilities reporting in 2023–24 is presented in Appendix A, Table A13.

7.3.1. Key findings

Table 7.4 shows a summary of the real losses by utility size group. From 2022–23 to 2023–24, the national median across all size groups increased by 3% to 76.0 L/service connection/day.

As in previous years, Cassowary Coast Regional Council (Queensland) in the Small size group reported the highest real losses among all utilities (457.7 L/service connection/day in 2023–24), with a 10.1% increase from the previous year. High losses for this service provider are due to aging infrastructure, long mains, customer meter under reads, long periods of wet weather creating difficult conditions for leak detection, and staff shortages. Westernport Water in Victoria, Queanbeyan–Palerang Regional Council and Goulburn Mulwaree Council both in New South Wales reported the lowest Real losses across the country, at 5, 9 and 9.5 L/service connection/day, respectively.

Table 7.4 Overview of results: Real losses: service connections (L/service connection/day)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	89.0	39.3	5	6	72.0	68.8	-4
	WC (Perth)	Logan					
Large	270.0	35.9	8	4	66.5	78.5	18
	P&W (Darwin)	Toowoomba					
Medium	146.1	9.0	9	10	69.0	75.7	10
	Fitzroy River Water	Queanbeyan					
Small	457.7	5.0	11	14	140.0	104.5	-25
	Cassowary Coast	Westernport Water					
All size groups (national)	**457.7**	**5.0**	**33**	**34**	**74.0**	**76.0**	**3**
	Cassowary Coast	**Westernport Water**					

Note: The median real losses (L/service connection/day) for each year are calculated using data from all utilities (dual-service and single-service providers) reporting data against A10 in that year.

Figure 7.5 shows a box-and-whisker plot of the real losses for all utilities reporting A10 for a given reporting year from 2013–14 to 2023–24. The figure highlights a smaller range of changes among utilities that have reported, compared to the previous year. The median for 2023–24 has increased

by almost 3% from the previous year, with the Large group reporting the highest increase percentage of 18% and Small group reporting the largest decrease percentage of 25% from 2022–23.

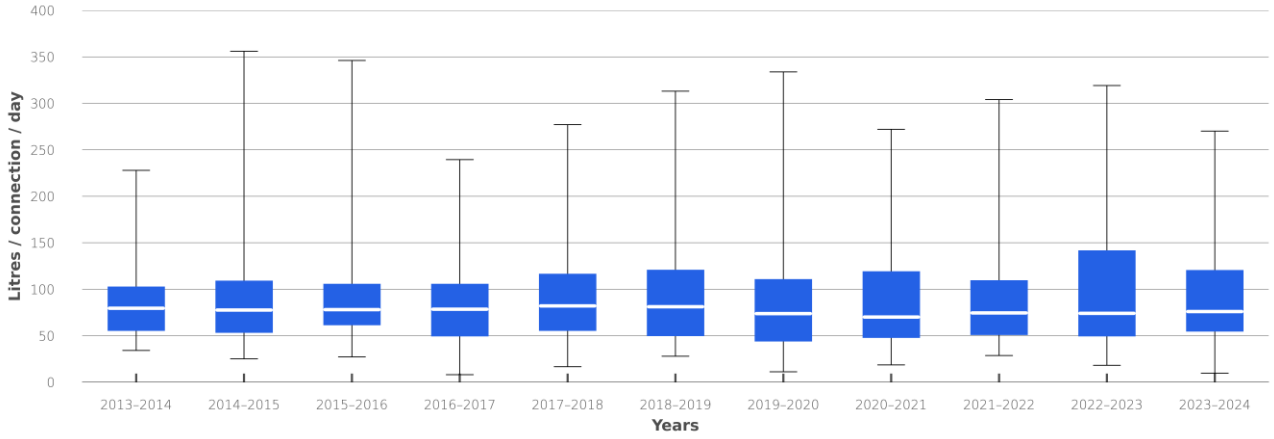


Figure 7.5 Real losses (L/service connection/day)

7.3.2. Results and analysis – Major utility group

Figure 7.6 shows a ranked breakdown of the real losses per annum for each Major utility from 2019–20 to 2023–24. Five utilities reported an increase in real losses from 2022–23, while Barwon Water (Victoria), Greater Western Water (Victoria), and SA Water Corporation (South Australia) remained unchanged. Unitywater (Queensland) reported the highest increase (20.1%), from 49.8 L/service connection/day in 2022–23 to 59.8 L/service connection/day in 2023–24. Conversely, Logan City Council (Queensland) reported the highest decrease (27.1%), from 53.9 L/service connection/day in 2022–23 to 39.3 L/service connection/day in 2023–24. TasWater (Tasmania) did not report to this indicator in 2023–24 due to a failure in auditing.

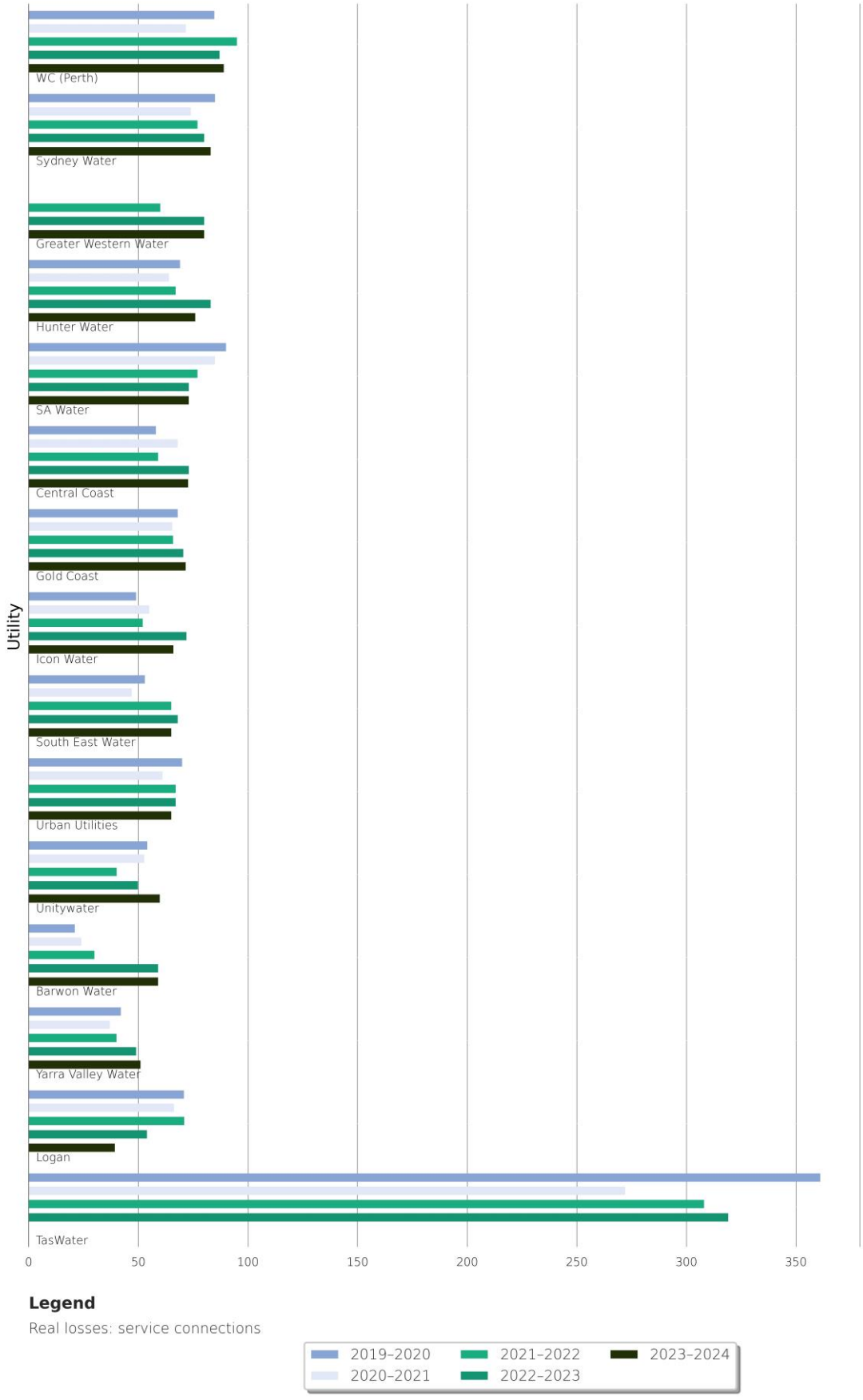


Figure 7.6 Real losses: service connections (L/service connection/day) – Major utility group

8. Environment

8.1. Total net greenhouse gas emissions per 1,000 properties – E12

The total net greenhouse gas emissions per 1,000 properties indicator (E12) reports the contribution of a utility's operations to greenhouse gas emissions (t CO_2 equivalent/1,000 properties). Utilities are required to update their calculations annually, and to refer to the National Greenhouse Accounts (NGA) Factors which are issued by the Commonwealth Department of Climate Change, Energy, the Environment and Water. Greenhouse gas emissions are reported in net terms – any quantity of carbon sequestered through activities such as the purchase of carbon offsets is deducted.

The NGA outlines 3 distinct types of emissions factors that may need to be calculated to estimate the full greenhouse impact of an organisation's activities:

- direct emission factors (Scope 1), which calculate the quantity of carbon dioxide equivalent (CO_2 equivalent) emitted per unit of activity at the point of emission release
- indirect emission factors (Scope 2), which calculate the greenhouse impact of purchasing and consuming electricity (that is, the impact of burning fuels – coal or gas – at the power station)
- various emission factors (Scope 3), including the impact of various activities – disposal of waste, employee business travel and the transportation of products.

Comparing different utilities' net greenhouse gas emissions is a difficult exercise and should be undertaken with caution due to the number of variables affecting emissions, including:

- sources of water
- gravity versus pumped networks
- geographical conditions (influencing the need for pumping)
- the number of large-volume customers
- the extent of industry within the customer base
- the prevailing greenhouse policy in the jurisdiction
- the method of calculation.

Total net greenhouse gas emissions data for 2023–24 is presented in Table A14, Appendix A.

8.1.1. Key findings

Table 8.1 shows a summary of the total net greenhouse gas emissions by utility size group.

The median total net greenhouse gas emissions decreased by 2% on the national scale, continuing the declining trend observed in previous years. Goulburn Mulwaree Council (New South Wales) in the Small size group reported the highest emissions of 850 t CO_2 equivalent/1,000 properties yet representing a slight decrease of 0.9% from the previous year. Logan City Council in Queensland reported the lowest emissions of 0 t CO_2 equivalent/1,000 properties, as a result of the purchased accredited carbon credits offsetting emissions for 2023–24.

Table 8.1 Overview of results: Total net greenhouse gas emissions per 1,000 properties (t CO_2 equivalent/1,000 properties)

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	503	0	5	8	166	156	-6
	Central Coast	Logan					
Large	465	157	4	8	337	309	-8
	Goulburn Valley Water	Redland City					
Medium	702	174	8	8	329	374	14
	MidCoast Council	Queanbeyan					
Small	850	71	10	11	386	384	-1
	Goulburn Mulwaree	Bathurst					
All size groups (national)	**850**	**0**	**27**	**35**	**329**	**322**	**-2**
	Goulburn Mulwaree	**Logan**					

Note: The median total net greenhouse gas emissions for each year is calculated using data from all utilities supplying both water and wastewater services reporting data for E12 for that year.

8.1.2. Results and analysis – Major utility group

The Major utility group reported a 6% decrease in the median net greenhouse gas emissions from 2022–23 to 2023–24. Yarra Valley Water Corporation (Victoria) had the highest percentage decrease of 34.8%. Similar to last year, Central Coast Council (New South Wales) was the highest total net greenhouse gas emitter with 503 t CO_2 equivalent per 1,000 properties, and Logan City Council (Queensland) was the lowest with 0 t CO_2. The City of Gold Coast (Queensland) had the highest increase in the median net greenhouse gas emissions, rising by 48.6% from 2022–23, bringing its emissions to 298 t CO_2 equivalent per 1,000 properties in 2023–24.

9. Health

9.1. Percentage of population for whom microbiological compliance was achieved – H3

This indicator reports the percentage of the population serviced by the utility for whom microbiological compliance was achieved.

Compliance is assessed against the Australian Drinking Water Guidelines 2011[11] or licence conditions imposed on the utility by their regulator. Typically, utilities record very high compliance. However, unforeseen events may deliver a lower compliance result, and the cause of non-compliance is not always traceable.

Microbiological compliance data for 2023–24 is shown in Appendix A, Table A15.

9.1.1. Key findings

Table 9.1 shows a summary of the percentage of population for which microbiological compliance was achieved by utility size group. In 2023–24, nationally and across all utility size groups, the median remained at 100% (no change from the previous year). The majority of utilities achieved 100% microbiological compliance, except for MidCoast Council (New South Wales) and South Gippsland Water (Victoria), both from the Medium size group, which achieved 97.0% and 94.5%[12] of the microbiological compliance, respectively.

Table 9.1 Overview of results: Percentage of population for which microbiological compliance was achieved

Utility group	Range		No. utilities with increase/decrease from 2022–23		Median		Change in median from 2022–23 (%)
	High	Low	Increase	Decrease	2022–23	2023–24	
Major	100.0	100.0	1	0	100.0	100.0	0
	Multiple utilities	Multiple utilities					
Large	100.0	100.0	2	0	100.0	100.0	0
	Multiple utilities	Multiple utilities					
Medium	100.0	94.5	1	2	100.0	100.0	0
	Multiple utilities	South Gippsland Water					
Small	100.0	100.0	0	0	100.0	100.0	0
	Multiple utilities	Multiple utilities					
All size groups (national)	**100.0**	**94.5**	**4**	**2**	**100.0**	**100.0**	**0**
	Albury	**South Gippsland Water**					

Note: The median percentage of population for which microbiological compliance was achieved for each year was calculated using data from all utilities supplying both water services and reporting data against H3 in that reporting year.

[11] www.nhmrc.gov.au/guidelines/publications/eh52, updated January 2022

[12] Due to impacts from torrential rain, now mitigated (refer to South Gippsland Water's 2023-24 annual report, available at https://www.sgwater.com.au/about-us/publications).

9.1.2. Results and analysis – Major utility group

Over the past 5 years, all utilities in the Major utility group achieved 100% microbiological compliance, except for South East Water Corporation (Victoria) which achieved 99.9% in the 2022–23 and 2020–21 reporting years.

Appendix A Individual utility group tables

- Tables A1 to A18 present a summary of key indicators by utility group for the period 2019–20 to 2023–24.
- Utilities are sorted in descending order based on their percentage changes in value from 2022–23 within each utility size group.

Table A1 W12 – Average annual residential water supplied (kL/property), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
SA Water	190	192	193	177	195	10.5
WC (Perth)	227	227	228	219	242	10.4
Barwon Water	161	159	160	147	162	9.9
TasWater	193	179	178	173	187	7.8
Icon Water	202	176	163	159	169	5.8
Hunter Water	156	151	151	152	159	4.4
Greater Western Water			140	137	143	4.4
Sydney Water	189	186	178	176	181	2.6
Gold Coast	175	167	163	166	170	2.4
South East Water	150	152	147	140	143	2.3
Central Coast	156	159	157	155	157	1.5
Yarra Valley Water	148	149	148	143	144	1.2
Urban Utilities	162	158	140	142	144	1.1
Unitywater	156	155	147	150	151	0.6
Logan	148	139	135	135	134	-0.6
Median	**162**	**159**	**157**	**152**	**159**	
Mean	**172**	**168**	**162**	**158**	**165**	
Large						
WC (Mandurah)	220	220	223	204	229	12.2
Goulburn Valley Water	261	254	240	217	243	12.0
Townsville	363	342	342	293	327	11.7
P&W (Darwin)	373	360	374	349	389	11.3
Toowoomba	162	150	135	130	140	8.2
North East Water	214	200	179	181	194	7.5
Coliban Water	195	194	192	175	184	5.4
Central Highlands Water (Vic)	151	147	147	142	147	4.1
Shoalhaven	155	155	142	143	142	-0.6
Gippsland Water	161	164	156	156	155	-0.7
Cairns	257	250	255	249	247	-0.9
Redland City	174	184	174	181	179	-1.4
Median	**205**	**197**	**185**	**181**	**189**	
Mean	**224**	**218**	**213**	**202**	**215**	
Medium						

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Dubbo	207	198	194	190	263	38.4
Bundaberg	261	284	236	188	245	30.8
Riverina Water (W)	338	264	257	248	309	24.5
Albury	246	218	192	195	235	20.3
Fitzroy River Water	383	365	317	279	328	17.5
Lower Murray Water	520	477	461	388	455	17.5
Gladstone	251	253	217	200	226	13.0
Wingecarribee	218	177	168	165	187	13.0
Clarence Valley	115	164	162	159	179	12.6
Tamworth	141	159	178	202	219	8.5
GWMWater	241	224	230	199	215	8.1
MidCoast Council	144	149	131	131	142	7.9
Fraser Coast	191	186	163	162	174	7.6
Wannon Water	141	139	144	131	140	7.3
Port Macquarie Hastings	144	154	152	177	177	-0.3
East Gippsland Water	153	145	126	134	131	-2.2
South Gippsland Water	115	118	115	115	111	-3.4
Queanbeyan	171	184	151	147	142	-3.6
Coffs Harbour	151	140	143	147	142	-3.9
Tweed	177	172	157	162	151	-6.8
Mackay	215	168	196	203	166	-18.1
Eurobodalla	121	117	113	108		
Median	**184**	**174**	**166**	**171**	**179**	
Mean	**211**	**202**	**191**	**183**	**206**	
Small						
Central Highlands	430	465	396	325	391	20.4
Essential Energy	277	250	236	218	251	15.1
WC (Geraldton)	286	291	288	276	313	13.5
Aqwest-Bunbury (W)	245	236	236	238	270	13.4
Busselton (W)	261	258	264	262	297	13.2
Bega Valley	163	148	135	128	145	13.1
Western Downs	201	193	168	169	190	12.6
Gympie	169	166	155	145	163	12.4
Livingstone	349	328	313	284	317	11.4
WC (Kal–Boulder) (W)	300	279	275	258	286	11.0
WC (Australind/Eaton)	289	283	281	275	304	10.6
Byron	200	199	199	193	212	10.2
Bathurst	144	156	157	168	184	10.0
Armidale	141	141	141	143	154	7.8
Goldenfields Water (W)	288	230	209	232	248	7.1
WC (Albany)	169	164	163	163	174	6.9
Goulburn Mulwaree	153	136	140	132	141	6.8
Whitsunday	304	276	277	257	274	6.5

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Mount Barker				96	102	6.3
Cassowary Coast	220	224	236	233	242	3.7
Westernport Water	89	93	92	88	91	3.5
Kempsey	146	175	125	127	130	2.1
P&W (Alice Springs)	451	432	420	390	397	1.6
Snowy Monaro		125		145	147	1.1
Lismore	127	123	126	115	116	1.0
Orange	163	160	158	148	148	0.0
Southern Downs	100	104	106	116	111	-4.0
Ballina	173	168	161	191	155	-19.1
Median	**201**	**193**	**183**	**180**	**187**	
Mean	**225**	**215**	**210**	**197**	**213**	

Table A2 W26 – Total recycled water supplied (ML), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Icon Water	75	27	24	17	31	82.4
Greater Western Water			6,233	5,961	8,549	43.4
SA Water	26,400	28,565	35,291	21,845	29,124	33.3
Logan	789	582	217	100	133	33.0
Unitywater	1,292	772	200	497	637	28.2
Central Coast	825	631	550	584	732	25.3
Hunter Water	4,651	3,206	3,132	3,215	3,668	14.1
South East Water	6,825	6,803	7,310	7,159	8,009	11.9
Gold Coast	7,306	7,466	6,146	7,036	7,360	4.6
Sydney Water	46,919	37,669	37,693	41,198	40,361	-2.0
WC (Perth)	20,681	22,579	21,759	23,337	22,660	-2.9
Urban Utilities	4,532	3,909	3,315	3,745	3,595	-4.0
Yarra Valley Water	2,427	2,851	3,069	3,097	2,925	-5.6
Barwon Water	4,683	3,951	3,642	3,145	2,734	-13.1
Median	**4,651**	**3,909**	**3,478**	**3,480**	**3,632**	
Mean	**9,800**	**9,155**	**9,184**	**8,638**	**9,323**	
Large						
Goulburn Valley Water	307	270	2,334	3,899	6,157	57.9
Toowoomba	2,105	1,809	713	840	1,296	54.3
Coliban Water	1,411	1,093	352	260	350	34.6
Shoalhaven	2,218	1,642	1,135	1,580	2,003	26.8
WC (Mandurah)	220	244	253	234	292	24.8
Central Highlands Water (Vic)	1,628	1,934	1,946	1,645	2,010	22.2
Cairns	533	1,731	1,847	1,982	2,184	10.2
Gippsland Water	1,969	2,128	2,043	2,137	2,232	4.4
Townsville	1,383	1,358	1,517	1,285	1,155	-10.1
North East Water	8,093	8,443	10,628	11,898	10,626	-10.7
Redland City	121	100	96	143	117	-18.2
P&W (Darwin)	0	0	0	0	0	
Median	**1,397**	**1,500**	**1,326**	**1,432**	**1,649**	
Mean	**1,666**	**1,729**	**1,905**	**2,159**	**2,368**	
Medium						
Wingecarribee	171	86	102	10	88	778.0
Mackay	4,034	446	446	446	3,656	719.7
Bundaberg	585	531	447	353	957	171.1
South Gippsland Water	122	194	189	116	245	111.2
Tamworth	3,622	1,818	2,188	2,414	3,648	51.1
MidCoast Council	1,398	685	593	659	903	37.1
Dubbo	2,159	1,483	1,732	1,329	1,630	22.7
Wannon Water	1,725	1,454	1,858	1,714	2,070	20.8
Coffs Harbour	1,281	774	474	608	666	9.6

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Tweed	874	777	532	618	643	4.0
Fitzroy River Water	735	748	760	818	849	3.8
East Gippsland Water	1,072	1,346	1,521	1,372	1,370	-0.1
Clarence Valley	386	332	216	310	306	-1.4
Wagga Wagga (S)	4,819	5,188	5,458	5,583	5,213	-6.6
Lower Murray Water	408	477	584	391	359	-8.2
Gladstone	2,494	3,812	3,603	3,845	3,322	-13.6
Albury	3,297	5,173	5,302	6,492	5,418	-16.5
GWMWater	0	2,698	2,084	4,344	3,592	-17.3
Fraser Coast	6,918	4,880	4,595	4,999	4,088	-18.2
Port Macquarie Hastings	610	384	291	343	277	-19.2
Queanbeyan	70	375	268	525	413	-21.4
Eurobodalla	433	190	111	71		
Median	**874**	**761**	**584**	**618**	**930**	
Mean	**1,618**	**1,539**	**1,450**	**1,624**	**1,805**	
Small						
Armidale	846	484	361	214	840	292.5
Lismore	17	21	332	7	21	195.7
Mount Barker			275	179	300	67.6
Kempsey	149	75	51	42	69	64.1
WC (Busselton) (S)	254	272	273	251	327	30.3
Orange	2,723	2,677	1,027	101	127	25.3
Byron	516	493	302	177	220	24.5
Whitsunday	317	386	365	314	366	16.6
Snowy Monaro		87	66	62	70	12.9
WC (Bunbury) (S)	127	115	118	95	104	9.5
WC (Geraldton)	210	224	216	245	268	9.4
Bega Valley	762	407	213	381	410	7.6
Essential Energy	617	499	444	377	402	6.6
P&W (Alice Springs)	927	1,019	1,148	1,107	1,178	6.4
Western Downs	1,105	754	695	669	711	6.3
Central Highlands	1,939	1,817	1,797	1,651	1,693	2.5
WC (Australind/Eaton)	721	740	742	775	785	1.3
Livingstone	1,047	836	852	730	739	1.2
WC (Albany)	2,085	2,357	2,351	2,298	2,290	-0.3
Gympie	97	93	68	88	83	-5.7
Kal-Boulder (S)	1,126	1,378	1,379	1,317	1,137	-13.7
Goulburn Mulwaree	237	338	260	264	225	-14.8
Westernport Water	76	83	106	119	90	-24.4
Southern Downs	894	863	2,013	1,831	1,365	-25.5
Ballina	688	748	573	960	393	-59.1
Bathurst	762	788	761	598	239	-60.1
Median	**688**	**446**	**332**	**248**	**284**	

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Mean	**730**	**627**	**579**	**495**	**482**	

Table A3 P8 – Typical annual bill ($), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Central Coast	1,228	1,052	1,029	1,001	1,235	23.4
SA Water	1,515	1,257	1,222	1,123	1,185	5.5
Hunter Water	1,374	1,270	1,235	1,220	1,276	4.6
Sydney Water	1,329	1,190	1,145	1,115	1,165	4.5
Urban Utilities	1,643	1,646	1,528	1,481	1,526	3.0
Icon Water	1,385	1,281	1,215	1,110	1,133	2.1
Greater Western Water			1,036	987	1,007	2.0
WC (Perth)	1,901	1,861	1,807	1,709	1,743	2.0
Barwon Water	1,222	1,202	1,147	1,087	1,106	1.7
Unitywater	1,766	1,752	1,663	1,563	1,589	1.7
Gold Coast	1,978	1,919	1,852	1,782	1,797	0.8
TasWater	1,429	1,387	1,374	1,323	1,330	0.5
Logan	1,896	1,849	1,752	1,647	1,642	-0.3
South East Water	1,171	1,183	1,074	999	990	-0.9
Yarra Valley Water	1,251	1,235	1,144	1,077	1,031	-4.3
Median	**1,407**	**1,276**	**1,222**	**1,123**	**1,235**	
Mean	**1,506**	**1,435**	**1,348**	**1,282**	**1,317**	
Large						
North East Water	1,113	1,066	967	951	1,023	7.6
Goulburn Valley Water	1,076	1,043	949	876	937	7.0
Coliban Water	1,595	1,564	1,477	1,380	1,432	3.8
WC (Mandurah)	2,081	2,037	2,002	1,875	1,929	2.9
P&W (Darwin)	2,198	2,133	2,121	1,982	2,036	2.7
Central Highlands Water (Vic)	1,457	1,418	1,331	1,264	1,291	2.1
Cairns	1,622	1,628	1,654	1,578	1,601	1.5
Shoalhaven	1,457	1,434	1,335	1,321	1,330	0.7
Gippsland Water	1,566	1,550	1,502	1,402	1,394	-0.6
Townsville	1,882	1,898	1,863	1,777	1,764	-0.7
Redland City	1,808	1,957	1,905	1,878	1,863	-0.8
Toowoomba	1,976	1,899	1,816	1,729	1,683	-2.7
Median	**1,608**	**1,596**	**1,578**	**1,490**	**1,516**	
Mean	**1,653**	**1,636**	**1,577**	**1,501**	**1,524**	
Medium						
Fitzroy River Water	1,628	1,621	1,522	1,525	1,677	10.0
Gladstone	2,035	2,092	1,966	1,972	2,131	8.1
Fraser Coast	1,845	1,865	1,776	1,738	1,855	6.7
Wingecarribee	1,674	1,603	1,548	1,498	1,591	6.2
Dubbo	1,748	1,741	1,699	1,616	1,707	5.6
South Gippsland Water	1,185	1,237	1,200	1,188	1,249	5.1
Clarence Valley	1,834	1,978	1,968	1,848	1,912	3.5
Lower Murray Water	1,196	1,144	1,092	1,061	1,096	3.3

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
East Gippsland Water	1,406	1,390	1,337	1,302	1,330	2.2
Wannon Water	1,314	1,281	1,218	1,149	1,173	2.1
Bundaberg	1,766	1,771	1,662	1,558	1,588	1.9
Albury	1,443	1,351	1,248	1,183	1,203	1.7
Tamworth	1,589	1,638	1,635	1,584	1,598	0.9
Mackay	1,802	1,794	1,723	1,527	1,533	0.4
Tweed	1,890	1,842	1,740	1,677	1,674	-0.2
Coffs Harbour	1,811	1,766	1,757	1,703	1,692	-0.6
Queanbeyan	1,707	1,950	1,763	1,687	1,672	-0.9
MidCoast Council	2,180	2,170	2,009	1,922	1,904	-0.9
GWMWater	1,618	1,568	1,471	1,377	1,339	-2.8
Port Macquarie Hastings	1,804	1,858	1,835	1,886	1,813	-3.9
Eurobodalla	2,061	2,062	1,996	1,913		
Median	**1,748**	**1,766**	**1,699**	**1,584**	**1,635**	
Mean	**1,692**	**1,701**	**1,627**	**1,567**	**1,587**	
Small						
Orange	1,410	1,412	1,372	1,283	1,548	20.7
Livingstone	2,040	2,042	2,076	1,977	2,200	11.3
Central Highlands	2,333	2,476	2,320	2,117	2,354	11.2
Kempsey	2,269	2,368	2,242	2,205	2,346	6.4
Essential Energy	1,616	1,566	1,487	1,443	1,533	6.2
Byron	2,195	2,334	2,302	2,223	2,354	5.9
Bega Valley	2,302	2,274	2,173	2,095	2,209	5.4
Bathurst	1,312	1,378	1,371	1,353	1,426	5.4
Gympie	1,477	1,514	1,452	1,392	1,460	4.9
Mount Barker			1,087	1,061	1,101	3.8
WC (Geraldton)	2,334	2,304	2,250	2,138	2,218	3.7
Westernport Water	1,351	1,375	1,323	1,274	1,321	3.7
Cassowary Coast	1,930	1,931	1,907	1,847	1,909	3.4
WC (Australind/Eaton)	2,279	2,212	2,151	2,048	2,099	2.5
Southern Downs	1,773	1,698	1,782	1,764	1,805	2.3
Armidale	1,250	1,517	1,540	1,497	1,525	1.9
Western Downs	1,688	1,681	1,575	1,548	1,573	1.6
Lismore	2,086	2,057	2,032	1,914	1,922	0.4
Snowy Monaro		1,822		1,775	1,774	-0.1
WC (Albany)	2,109	2,055	2,004	1,921	1,919	-0.1
Whitsunday	1,983	2,097	2,001	1,996	1,986	-0.5
P&W (Alice Springs)	2,381	2,298	2,224	2,095	2,053	-2.0
Goulburn Mulwaree	1,619	1,587	1,532	1,502	1,453	-3.3
Ballina	1,862	1,895	1,829	1,789	1,708	-4.5
Median	**1,956**	**1,931**	**1,907**	**1,818**	**1,857**	
Mean	**1,891**	**1,908**	**1,827**	**1,761**	**1,825**	

Table A4 P7 – Annual bill based on 200 kL ($), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Central Coast	1,393	1,152	1,131	1,184	1,426	20.4
Icon Water	1,372	1,350	1,318	1,207	1,249	3.5
Sydney Water	1,358	1,228	1,199	1,183	1,223	3.4
Hunter Water	1,497	1,411	1,373	1,353	1,395	3.1
Greater Western Water			1,240	1,180	1,203	1.9
Barwon Water	1,316	1,306	1,247	1,214	1,236	1.8
Unitywater	2,074	2,075	2,004	1,887	1,913	1.4
SA Water	1,554	1,284	1,244	1,194	1,200	0.5
Urban Utilities	1,816	1,840	1,801	1,728	1,736	0.5
Gold Coast	2,081	2,072	2,034	1,939	1,937	-0.1
South East Water	1,402	1,326	1,291	1,207	1,201	-0.5
TasWater	1,437	1,414	1,401	1,355	1,345	-0.7
WC (Perth)	1,823	1,785	1,730	1,658	1,634	-1.4
Logan	2,146	2,143	2,080	1,958	1,923	-1.8
Yarra Valley Water	1,488	1,461	1,353	1,274	1,251	-1.8
Median	**1,492**	**1,412**	**1,353**	**1,274**	**1,345**	
Mean	**1,626**	**1,560**	**1,496**	**1,435**	**1,458**	
Large						
Goulburn Valley Water	991	962	899	838	885	5.6
North East Water	1,071	1,068	1,025	1,001	1,040	3.9
Coliban Water	1,610	1,583	1,498	1,444	1,496	3.6
Cairns	1,558	1,549	1,558	1,503	1,528	1.7
Central Highlands Water (Vic)	1,580	1,552	1,460	1,403	1,419	1.1
Shoalhaven	1,550	1,525	1,494	1,432	1,444	0.8
Toowoomba	2,032	2,062	2,019	1,956	1,955	-0.1
Redland City	1,971	2,028	2,016	1,954	1,947	-0.4
Gippsland Water	1,662	1,640	1,610	1,504	1,498	-0.4
WC (Mandurah)	2,024	1,980	1,938	1,863	1,853	-0.5
Townsville	1,882	1,898	1,863	1,777	1,764	-0.7
P&W (Darwin)	1,797	1,768	1,732	1,661	1,637	-1.4
Median	**1,636**	**1,612**	**1,584**	**1,504**	**1,513**	
Mean	**1,644**	**1,635**	**1,593**	**1,528**	**1,539**	
Medium						
Gladstone	1,901	1,941	1,912	1,932	2,034	5.3
South Gippsland Water	1,359	1,421	1,393	1,381	1,453	5.2
Fraser Coast	1,913	1,894	1,876	1,819	1,910	5.0
Fitzroy River Water	1,410	1,428	1,414	1,455	1,519	4.4
Lower Murray Water	930	921	878	852	881	3.4
East Gippsland Water	1,544	1,531	1,473	1,435	1,480	3.1
Wingecarribee	1,636	1,651	1,614	1,575	1,619	2.8
Wannon Water	1,445	1,413	1,342	1,272	1,301	2.3

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Clarence Valley	2,086	2,084	2,078	1,963	1,971	0.4
Queanbeyan	2,751	2,308	2,250	2,175	2,183	0.4
Coffs Harbour	1,988	1,988	1,963	1,891	1,891	0.0
Tamworth	1,699	1,715	1,675	1,578	1,564	-0.9
Mackay	1,786	1,785	1,752	1,637	1,619	-1.1
Bundaberg	1,714	1,709	1,682	1,615	1,593	-1.4
MidCoast Council	2,418	2,422	2,325	2,239	2,185	-2.4
Albury	1,342	1,321	1,265	1,186	1,143	-3.6
Port Macquarie Hastings	2,006	2,034	2,024	1,980	1,900	-4.0
GWMWater	1,548	1,524	1,430	1,389	1,326	-4.5
Dubbo	1,731	1,746	1,713	1,639	1,564	-4.6
Tweed		3,537	3,441	3,298	2,460	-25.4
Eurobodalla	2,403	2,423	2,364	2,272		
Median	**1,722**	**1,746**	**1,713**	**1,637**	**1,606**	
Mean	**1,780**	**1,847**	**1,803**	**1,742**	**1,680**	
Small						
Orange	1,516	1,527	1,491	1,433	1,959	36.7
Livingstone	1,961	1,930	2,047	1,816	1,956	7.7
Kempsey	2,437	2,447	2,487	2,425	2,569	5.9
Mount Barker			1,087	1,061	1,101	3.8
Byron	2,311	2,338	2,305	2,229	2,313	3.8
Westernport Water	1,656	1,641	1,567	1,525	1,579	3.5
Central Highlands	1,832	1,838	1,857	1,875	1,941	3.5
Bega Valley	2,436	2,460	2,402	2,328	2,403	3.2
Cassowary Coast	1,909	1,906	1,867	1,811	1,863	2.9
Essential Energy	1,450	1,458	1,411	1,387	1,424	2.7
Gympie	1,507	1,540	1,512	1,473	1,509	2.4
Bathurst	1,460	1,498	1,488	1,447	1,471	1.7
Southern Downs	2,021	2,022	2,031	1,992	2,018	1.3
Lismore	2,434	2,424	2,397	2,318	2,323	0.2
Snowy Monaro		2,115	2,065	1,985	1,982	-0.2
Whitsunday	1,880	1,916	1,907	1,859	1,856	-0.2
Ballina	1,938	1,987	1,939	1,839	1,831	-0.4
Armidale	1,509	1,823	1,840	1,745	1,731	-0.8
WC (Geraldton)	2,086	2,049	2,006	1,937	1,917	-1.0
WC (Australind/Eaton)	2,022	1,980	1,928	1,850	1,827	-1.2
P&W (Alice Springs)	1,797	1,768	1,732	1,661	1,637	-1.4
WC (Albany)	2,198	2,157	2,107	2,018	1,986	-1.6
Goulburn Mulwaree	1,777	1,802	1,725	1,706	1,637	-4.0
Western Downs	1,656	1,672	1,604	1,606	1,477	-8.0
Median	**1,894**	**1,916**	**1,887**	**1,828**	**1,860**	
Mean	**1,900**	**1,926**	**1,867**	**1,805**	**1,846**	

Table A5 F16 – Total capital expenditure: water and wastewater ($000s), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
WC (Perth)	471,994	432,595	385,360	417,631	649,634	55.6
Unitywater	152,823	158,520	181,037	258,706	365,042	41.1
TasWater	151,814	191,737	245,034	180,809	250,243	38.4
South East Water	307,953	277,945	204,508	232,517	295,924	27.3
Central Coast	4,428	0	40,956	69,195	84,692	22.4
Sydney Water	1,063,003	1,074,092	1,356,666	1,745,595	2,119,910	21.4
Barwon Water	91,667	76,701	59,305	114,185	135,950	19.1
SA Water	668,682	558,456	516,860	588,994	679,824	15.4
Yarra Valley Water	396,872	387,737	314,508	320,951	368,805	14.9
Hunter Water	176,673	171,409	149,917	177,699	196,151	10.4
Logan	158,661	100,832	120,444	120,423	130,238	8.2
Greater Western Water			279,007	327,563	346,756	5.9
Gold Coast	132,519	162,419	180,479	118,632	121,039	2.0
Urban Utilities	365,160	384,834	315,698	377,631	362,521	-4.0
Icon Water	120,935	101,614	74,606	75,450	70,564	-6.5
Median	**167,667**	**181,573**	**204,508**	**232,517**	**295,924**	
Mean	**304,513**	**291,349**	**294,959**	**341,732**	**411,820**	
Large						
Redland City	4,364	11,389	3,872	6,970	12,438	78.5
P&W (Darwin)	24,231	23,959	27,941	38,063	67,251	76.7
Central Highlands Water (Vic)	22,074	24,118	22,917	29,755	48,065	61.5
Cairns	54,212	51,303	44,652	48,035	69,770	45.2
WC (Mandurah)	19,334	19,754	13,886	26,177	37,137	41.9
North East Water	28,380	29,686	35,208	29,425	38,640	31.3
Coliban Water	32,198	36,227	49,072	65,651	83,623	27.4
Goulburn Valley Water	44,993	44,380	42,491	28,568	31,379	9.8
Toowoomba	31,949	57,761	45,084	60,483	64,838	7.2
Gippsland Water	53,594	65,686	42,648	45,955	41,031	-10.7
Shoalhaven	28,667	29,056	29,764	51,449	41,223	-19.9
Townsville	70,564	59,125	97,746	133,671	64,804	-51.5
Median	**30,308**	**32,956**	**38,850**	**42,009**	**44,644**	
Mean	**34,547**	**37,704**	**37,940**	**47,017**	**50,017**	
Medium						
Queanbeyan	15,560	5,152	511	3,356	16,780	400.0
Wingecarribee	19,282	21,860	22,076	10,404	32,799	215.3
Coffs Harbour	7,319	3,622	3,400	3,713	9,139	146.1
MidCoast Council	6,145	18,980	29,463	29,141	42,640	46.3
Wannon Water	33,662	21,258	14,510	37,474	50,674	35.2
Bundaberg	20,256	26,722	15,306	15,985	21,271	33.1

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Fitzroy River Water	12,816	14,767	22,607	47,299	61,873	30.8
Eurobodalla	16,319	26,999	29,122	64,367	84,100	30.7
GWMWater	16,584	13,753	13,587	18,473	22,649	22.6
South Gippsland Water	25,448	24,040	17,316	20,328	24,726	21.6
Tweed	14,498	18,244	14,591	17,486	21,005	20.1
Fraser Coast	24,918	35,350	22,281	16,849	18,943	12.4
Lower Murray Water	11,344	17,065	28,666	17,629	18,168	3.1
Gladstone	16,137	22,336	17,035	11,907	12,194	2.4
Tamworth	25,799	24,562	11,313	19,452	19,245	-1.1
Albury	9,019	7,921	4,778	3,910	3,859	-1.3
East Gippsland Water	20,354	13,396	9,870	17,078	14,830	-13.2
Clarence Valley	8,068			17,987	14,187	-21.1
Mackay	29,415	20,394	24,745	28,206	19,648	-30.3
Port Macquarie Hastings	4,669	21,388	47,744	30,020	15,789	-47.4
Dubbo	0	82	3,568			
Median	**16,137**	**19,687**	**16,170**	**17,808**	**19,446**	
Mean	**16,077**	**17,895**	**17,624**	**21,553**	**26,226**	
Small						
Ballina	9,518	9,476	10,416	2,430	8,742	259.8
Cassowary Coast	17,561	8,368	10,201	11,519	29,685	157.7
WC (Australind/Eaton)	5,772	9,897	7,254	3,906	8,927	128.5
WC (Geraldton)	22,323	16,713	8,033	8,055	13,979	73.5
WC (Albany)	10,866	14,196	11,925	5,817	9,449	62.4
Whitsunday	30,409	13,883	8,012	4,332	6,778	56.5
Westernport Water	7,901	11,002	9,444	4,665	6,973	49.5
Central Highlands	4,974	11,382	13,726	14,335	17,019	18.7
Bathurst	13,440	9,304	6,425	13,678	13,906	1.7
Gympie	6,055	5,467	5,538	5,204	5,274	1.3
P&W (Alice Springs)	10,302	9,381	9,932	8,841	8,401	-5.0
Kempsey	4,492	9,271	4,633	9,980	8,192	-17.9
Livingstone	6,268	4,124	8,785	10,634	8,530	-19.8
Bega Valley	2,682	9,399		18,319	13,207	-27.9
Western Downs	30,409	6,538	5,041	10,321	6,600	-36.1
Mount Barker			7,286	14,676	8,574	-41.6
Southern Downs	9,361	11,214	8,520	14,092	7,908	-43.9
Armidale	0	4,304	2,390	12,913	5,715	-55.7
Goulburn Mulwaree	8,506	13,279	11,384	9,811	4,210	-57.1
Snowy Monaro		0	19,654	4,865	1,280	-73.7
Byron	3,531	5,726	0		0	
Essential Energy	9,391	11,255			2,471	
Lismore	9,584	9,013			6,500	
Orange	0	750	1,415	2,803		
Median	**8,934**	**9,381**	**8,033**	**9,811**	**8,192**	

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Mean	**10,152**	**8,867**	**8,096**	**9,105**	**8,797**	

Table A6 F28 – Capital expenditure: water ($/property), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Unitywater	159	140	174	256	435	70.1
WC (Perth)	319	280	238	242	400	64.8
Hunter Water	234	130	256	261	411	57.4
Central Coast	9	0	96	147	222	51.2
Icon Water	297	232	88	132	179	35.8
SA Water	616	543	438	437	549	25.6
Logan	208	187	195	159	194	22.0
Yarra Valley Water	190	179	145	147	173	18.0
Barwon Water	292	236	173	317	345	8.8
Urban Utilities	210	201	160	161	167	3.7
South East Water	91	93	87	101	97	-3.7
Greater Western Water			289	354	330	-6.8
Sydney Water	167	194	230	249	219	-12.0
Gold Coast	93	89	145	95	76	-19.7
TasWater	404	569	894	505	337	-33.2
Median	**209**	**191**	**174**	**242**	**222**	
Mean	**235**	**220**	**240**	**237**	**276**	
Large						
WC (Mandurah)	286	165	156	153	307	100.4
P&W (Darwin)	291	230	346	476	944	98.4
Cairns	335	344	287	355	653	84.1
Central Highlands Water (Vic)	143	115	67	172	280	62.5
Coliban Water	238	233	437	503	599	19.1
Toowoomba	311	683	506	755	870	15.1
North East Water	260	328	248	216	249	15.1
Shoalhaven	198	252	193	419	422	0.6
Goulburn Valley Water	379	379	357	193	191	-1.2
Gippsland Water	300	195	254	340	308	-9.4
Redland City	12	18	15	53	46	-12.6
Townsville	355	447	782	1,097	602	-45.1
Median	**289**	**242**	**270**	**348**	**365**	
Mean	**259**	**282**	**304**	**394**	**456**	
Medium						
Coffs Harbour	115	45	47	35	144	305.9
Fitzroy River Water	249	231	348	478	1,231	157.7
Gladstone	122	280	269	81	160	98.7
Bundaberg	491	697	317	182	323	77.2
MidCoast Council	53	226	447	387	643	66.1
Tweed	260	328	236	242	352	45.7
Fraser Coast	412	371	209	226	318	40.4
Wingecarribee	476	410	597	147	180	22.5

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
East Gippsland Water	630	449	191	196	236	20.3
Lower Murray Water	182	270	625	311	327	5.1
GWMWater	422	252	292	393	400	1.6
Queanbeyan	90	216	11	109	94	-13.1
Wannon Water	200	214	247	343	290	-15.6
Albury	204	174	138	117	89	-23.8
Tamworth	786	973	273	652	487	-25.3
South Gippsland Water	620	645	424	339	253	-25.4
Clarence Valley	210			641	404	-37.0
Mackay	166	196	361	396	237	-40.3
Port Macquarie Hastings	65	373	823	416	155	-62.8
Dubbo	0	0	85			
Eurobodalla	225	437	334	2,374		
Median	**210**	**275**	**283**	**325**	**290**	
Mean	**285**	**339**	**314**	**403**	**333**	
Small						
Mount Barker				1,462	6,008	311.0
Ballina	228	144	181	60	210	246.7
Whitsunday	862	763	532	83	266	222.1
WC (Australind/Eaton)	201	561	171	216	594	175.0
Cassowary Coast	893	401	475	570	1,093	91.6
WC (Geraldton)	627	776	321	326	606	85.8
Gympie	222	291	166	81	150	85.6
P&W (Alice Springs)	671	570	645	408	562	37.9
WC (Albany)	261	614	478	231	286	23.8
Central Highlands	258	691	768	969	1,073	10.7
Westernport Water	236	325	219	163	179	9.9
Goulburn Mulwaree	365	511	307	208	189	-8.9
Bathurst	631	436	282	707	497	-29.7
Southern Downs	554	518	366	713	485	-32.0
Western Downs	862	288	264	639	433	-32.2
Kempsey	178	277	212	414	277	-32.9
Bega Valley	63	410	733	918	604	-34.2
Livingstone	126	108	179	520	271	-47.8
Armidale	0	0	202	1,172	326	-72.2
Snowy Monaro		0	231	305	78	-74.4
Byron	166	130	0		0	
Essential Energy	782	889			202	
Lismore	348	310			263	
Orange	0	41	36	0		
Median	**260**	**401**	**264**	**408**	**286**	
Mean	**388**	**394**	**322**	**484**	**637**	

Table A7 F29 – Capital expenditure: wastewater ($/property), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
TasWater	357	369	261	358	897	150.7
South East Water	308	262	169	186	264	41.5
WC (Perth)	239	224	208	236	332	40.6
Sydney Water	360	330	424	585	784	34.1
Barwon Water	288	234	183	354	442	24.6
Greater Western Water			181	181	223	22.9
Unitywater	328	358	387	536	657	22.5
Central Coast	25	0	211	363	398	9.9
Yarra Valley Water	303	293	232	233	256	9.8
Gold Coast	425	538	544	353	377	6.8
Logan	1,187	665	801	802	808	0.7
SA Water	279	174	243	345	332	-3.8
Urban Utilities	378	406	330	427	385	-9.9
Hunter Water	471	549	320	416	325	-22.0
Icon Water	355	298	292	247	167	-32.1
Median	**342**	**314**	**261**	**354**	**377**	
Mean	**379**	**336**	**319**	**375**	**443**	
Large						
Redland City	65	188	53	64	167	162.6
Central Highlands Water (Vic)	189	243	269	239	371	55.0
North East Water	307	245	434	345	484	40.5
Coliban Water	194	247	193	339	465	37.5
Goulburn Valley Water	416	391	368	293	340	15.8
WC (Mandurah)	117	262	131	405	450	11.1
P&W (Darwin)	112	171	120	156	158	1.6
Cairns	395	346	310	281	262	-6.8
Gippsland Water	496	775	352	297	249	-16.1
Toowoomba	229	226	224	197	131	-33.4
Shoalhaven	444	382	419	680	438	-35.7
Townsville	533	264	379	473	145	-69.4
Median	**268**	**255**	**290**	**295**	**301**	
Mean	**291**	**312**	**271**	**314**	**305**	
Medium						
Queanbeyan	521	10	11	19	575	2883.1
Wingecarribee	578	799	567	435	1709	293.4
Tamworth	380	96	240	151	354	134.4
Wannon Water	668	314	93	573	967	68.8
GWMWater	117	212	158	216	364	68.0
Coffs Harbour	170	93	85	129	203	56.9
South Gippsland Water	673	545	419	655	960	46.7
Albury	166	150	52	45	62	38.2

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
MidCoast Council	120	273	300	355	409	15.2
Clarence Valley	152			180	198	10.0
Bundaberg	155	136	186	393	407	3.8
Lower Murray Water	171	254	218	216	212	-2.0
Gladstone	541	629	412	403	327	-19.0
Tweed	189	231	209	267	214	-19.6
Fraser Coast	264	599	394	208	147	-29.3
East Gippsland Water	245	108	242	571	403	-29.4
Fitzroy River Water	155	235	363	1024	669	-34.7
Port Macquarie Hastings	89	302	687	542	344	-36.5
Mackay	515	258	199	274	116	-57.7
Dubbo	0	5	91			
Eurobodalla	646	984	1229	836		
Median	**189**	**244**	**229**	**315**	**354**	
Mean	**310**	**312**	**308**	**375**	**455**	
Small						
Armidale	0	0	23	36	211	478.7
Ballina	419	503	515	72	349	384.7
Bathurst	160	106	89	63	293	369.2
Cassowary Coast	936	420	615	610	2,312	279.1
WC (Albany)	503	281	280	136	331	143.2
Westernport Water	242	324	328	99	211	112.3
Central Highlands	170	299	437	245	389	58.8
Livingstone	433	249	587	315	413	31.1
WC (Geraldton)	827	121	133	125	148	18.4
Kempsey	226	582	174	512	473	-7.4
Whitsunday	1,534	239	24	250	225	-10.3
Bega Valley	155	313	625	433	360	-17.0
Gympie	236	104	247	315	231	-26.7
WC (Australind/Eaton)	309	237	481	83	51	-38.5
Western Downs	1,534	369	234	346	177	-49.0
Mount Barker			519	1005	463	-53.9
Southern Downs	299	583	467	628	247	-60.6
P&W (Alice Springs)	171	201	152	322	98	-69.6
Goulburn Mulwaree	372	626	669	646	158	-75.5
Snowy Monaro		0	2,082	247	56	-77.1
Byron	179	422	0		0	
Essential Energy	130	212			38	
Lismore	361	356			192	
Orange	0	0	45	166		
Median	**271**	**281**	**304**	**250**	**225**	
Mean	**418**	**285**	**397**	**317**	**323**	

Table A8 F8 – Revenue from community service obligations (ratio), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Central Coast	0.0132		0.0150	0.0100	0.0105	5.0
Icon Water	0.0176	0.0251	0.0205	0.0162	0.0165	1.9
TasWater	0.0578	0.0245	0.0229	0.0221	0.0220	-0.5
WC (Perth)	0.0590	0.0630	0.0677	0.0623	0.0619	-0.6
Sydney Water	0.0511	0.0481	0.0487	0.0465	0.0456	-1.9
SA Water	0.1190	0.1103	0.1104	0.1070	0.1038	-3.0
Hunter Water	0.0522	0.0499	0.0473	0.0467	0.0447	-4.3
South East Water	0.0439	0.0488	0.0479	0.0488	0.0464	-4.9
Yarra Valley Water	0.0487	0.0539	0.0536	0.0504	0.0473	-6.2
Barwon Water	0.0435	0.0455	0.0425	0.0446	0.0405	-9.2
Unitywater	0.0080	0.0080	0.0077	0.0078	0.0068	-12.8
Greater Western Water			0.0346	0.0242	0.0206	-14.9
Gold Coast	0	0.0040	0.0043	0.0035	0	-100.0
Logan	0	0	0	0	0	
Urban Utilities	0	0	0	0	0	
Median	**0.0437**	**0.0455**	**0.0346**	**0.0242**	**0.0220**	
Mean	**0.0367**	**0.0370**	**0.0349**	**0.0327**	**0.0311**	
Large						
Redland City	0.0040	0.0030	0.0036	0.0036	0.0128	255.6
Toowoomba	0.0120	0.0080	0.0086	0.0111	0.0123	10.8
Cairns	0.0250	0.0220	0.0174	0.0062	0.0063	1.6
Central Highlands Water (Vic)	0.0513	0.0536	0.0473	0.0463	0.0470	1.5
P&W (Darwin)	0.0206	0.0346	0.0206	0.0189	0.0187	-1.1
Gippsland Water	0.0475	0.0427	0.0446	0.0440	0.0432	-1.8
North East Water	0.0630	0.0626	0.0639	0.0578	0.0559	-3.3
Townsville	0.0150	0.0160	0.0151	0.0148	0.0142	-4.1
Coliban Water	0.0413	0.0456	0.0449	0.0426	0.0400	-6.1
Shoalhaven	0.0120	0.0110	0.0110	0.0105	0.0092	-12.4
Goulburn Valley Water	0.0540	0.0601	0.0597	0.0528	0.0402	-23.9
WC (Mandurah)	-0.2344	-0.1405	-0.1341	-0.0787	-0.0504	-36.0
Median	**0.0228**	**0.0283**	**0.0190**	**0.0169**	**0.0164**	
Mean	**0.0093**	**0.0182**	**0.0169**	**0.0192**	**0.0208**	
Medium						
Eurobodalla	0.0090	0.0090	0.0094	0.0074	0.0217	193.2
Bundaberg	0.0310	0.0370	0.0177	0.0087	0.0179	105.7
Gladstone	0.0090	0.0100	0.0088	0.0069	0.0095	37.7
Tamworth	0.0090	0.0080	0.0070	0.0058	0.0068	17.2
Mackay	0.0020	0.0020	0.0020	0.0020	0.0021	5.0
South Gippsland Water	0.0575	0.0514	0.0534	0.0523	0.0548	4.8
Port Macquarie Hastings	0.0110	0.0100	0.0084	0.0083	0.0083	0.0

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
GWMWater	0.0678	0.0721	0.0673	0.0681	0.0645	-5.3
Coffs Harbour	0.0040	0.0040	0.0040	0.0074	0.0070	-5.4
Lower Murray Water	0.0591	0.0481	0.0490	0.0451	0.0424	-6.0
Clarence Valley	0.0100			0.0080	0.0074	-7.5
Wannon Water	0.0474	0.0522	0.0508	0.0528	0.0486	-8.0
Fitzroy River Water	0.0110	0.0110	0.0080	0.0116	0.0096	-17.2
East Gippsland Water	0.0630	0.0639	0.0666	0.0679	0.0561	-17.4
Tweed	0.0110	0.0100	0.0100	0.0104	0.0082	-21.2
MidCoast Council	0.0110	0.0110	0.0101	0.0101	0.0078	-22.8
Wingecarribee	0.0080	0.0080	0.0080	0.0110	0.0074	-32.7
Albury	0.0080		0.0088	0.0184	0.0108	-41.3
Fraser Coast	0.0070	0.0070	0.0071	0.0113	0.0052	-54.0
Queanbeyan	0.0040	0.0040	0.0026	0.0027	0	-100.0
Dubbo	0.0050	0.0050		0.0050		
Median	**0.0100**	**0.0100**	**0.0088**	**0.0101**	**0.0089**	
Mean	**0.0212**	**0.0223**	**0.0210**	**0.0201**	**0.0198**	
Small						
Gympie	0.0070	0.0240	0.0153	0.0111	0.0141	27.0
Western Downs	0.0930	0.0530	0.0684	0.0648	0.0793	22.4
Byron	0.0060	0.0050	0.0053	0.0071	0.0075	5.6
Lismore	0.0070	0.0080	0.0106	0.0076	0.0080	5.3
Ballina	0.0090	0.0070	0.0080	0.0077	0.0080	3.9
Westernport Water	0.0337	0.0357	0.0388	0.0382	0.0379	-0.8
P&W (Alice Springs)	0.1309	0.1581	0.1413	0.1401	0.1360	-2.9
Essential Energy	0.0090	0.0170	0.0170	0.0150	0.0145	-3.3
Goulburn Mulwaree	0.0070	0.0060	0.0071	0.0078	0.0068	-12.8
Kempsey	0.0090	0.0090	0.0077	0.0074	0.0064	-13.5
Bathurst	0.0070	0.0070	0.0070	0.0067	0.0054	-19.4
Bega Valley	0.0080	0.0080	0.0090	0.0080	0.0061	-23.7
WC (Geraldton)	-0.9320	-1.0678	-0.4408	-0.8339	-0.4546	-45.5
WC (Australind/Eaton)	-0.0315	-0.0059	-0.0388	0.0668	0.0298	-55.4
WC (Albany)	-0.0842	-0.0533	0.0718	-0.1269	-0.0446	-64.9
Armidale	0.0098	0.0070	0.0063		0.0025	
Cassowary Coast	0.0180	0	0	0	0	
Central Highlands	0	0	0	0	0	
Livingstone	0	0	0	0	0	
Mount Barker			0	0	0	
Orange	0.0090	0.0080	0.0080	0.0042		
Snowy Monaro			0	0	0	
Southern Downs	0	0	0	0	0	
Whitsunday	0.0930	0.0890	0.0498	0	0	
Median	**0.0075**	**0.0070**	**0.0074**	**0.0071**	**0.0061**	
Mean	**-0.0269**	**-0.0311**	**-0.0003**	**-0.0247**	**-0.0060**	

Table A9 C15 – Average duration of an unplanned interruption: water (minutes), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Logan		183.6	143.3	143.9	254.0	76.5
SA Water	203.0	194.0	181.0	167.0	255.0	52.7
Unitywater	109.7	120.3	116.5	129.3	171.0	32.3
Greater Western Water			116.6	138.9	148.4	6.8
WC (Perth)	111.0	140.0	141.0	162.0	170.0	4.9
Urban Utilities	119.0	94.0	130.0	114.0	117.0	2.6
Gold Coast	140.7	151.7	158.5	126.3	127.0	0.6
Hunter Water	150.5	155.0	138.0	129.0	129.0	0.0
South East Water	88.2	89.5	87.7	90.1	89.3	-0.9
Yarra Valley Water	105.9	97.5	105.4	105.7	96.0	-9.2
Barwon Water	127.8	112.7	98.8	98.0	88.3	-9.9
Icon Water	136.0	147.0	136.0	131.5	118.1	-10.2
Central Coast	197.0	208.0	203.0	210.0	178.0	-15.2
Sydney Water	187.0	200.0	192.0	231.0	179.0	-22.5
Median	**131.9**	**147.0**	**137.0**	**130.4**	**138.7**	
Mean	**139.7**	**145.6**	**139.1**	**141.2**	**151.4**	
Large						
Shoalhaven	153.0	74.0	113.0	80.0	153.0	91.3
North East Water	91.2	81.0	71.0	96.0	96.0	0.0
Central Highlands Water (Vic)	121.4	114.0	137.0	117.0	116.0	-0.9
P&W (Darwin)		139.0	102.1	77.0	75.8	-1.6
Goulburn Valley Water	106.3	93.7	117.5	84.2	78.6	-6.7
Townsville	103.1	98.8	800.1	134.5	118.0	-12.3
Gippsland Water	90.8	69.3	108.9	84.9	63.3	-25.4
WC (Mandurah)	64.0	62.0	64.0	114.0	82.0	-28.1
Cairns	76.9	45.0	36.7	61.3	35.9	-41.4
Coliban Water	112.5	141.5	105.0	195.0	109.0	-44.1
Redland City	110.0	135.1	160.0	273.0	130.0	-52.4
Toowoomba					144.5	
Median	**104.7**	**93.7**	**108.9**	**96.0**	**102.5**	
Mean	**102.9**	**95.8**	**165.0**	**119.7**	**100.2**	
Medium						
Mackay	19.6	23.3	11.9	7.4	30.5	312.2
Queanbeyan	38.0	220.0	76.0	56.6	130.0	129.7
Fitzroy River Water	55.3	33.8	33.9	35.8	75.5	110.9
GWMWater	102.9	83.9	80.0	102.6	134.4	31.0
Tweed	139.0	161.0	275.0	137.0	168.0	22.6
Lower Murray Water	66.6	59.0	52.4	58.8	63.7	8.3
Coffs Harbour	177.0	189.0	167.0	175.0	185.0	5.7
Riverina Water (W)	125.0	164.0	142.0	162.0	163.0	0.6

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Clarence Valley	180.0	180.0	95.0	96.0	96.0	0.0
Dubbo	123.0	125.0	125.0	120.0	120.0	0.0
Port Macquarie Hastings	90.0	90.0	90.0	90.0	90.0	0.0
South Gippsland Water	90.0	80.0	78.7	89.1	85.7	-3.8
Wingecarribee	245.0	185.0	204.0	141.0	135.0	-4.3
Fraser Coast	56.6	125.0	128.4	159.0	152.0	-4.4
East Gippsland Water	169.0	74.0	88.8	103.8	91.2	-12.1
Bundaberg	84.0	70.0	88.0	126.0	104.0	-17.5
Tamworth	86.0	210.0	2,289.0	422.0	253.0	-40.0
Wannon Water	158.0	138.5	150.0	172.0	98.0	-43.0
Gladstone	50.0	39.0	35.0	51.5	24.4	-52.6
Albury	95.0	87.0	213.0	148.0	62.0	-58.1
Eurobodalla	135.0	170.0	214.0	73.0		
Median	**95.0**	**125.0**	**95.0**	**103.8**	**101.0**	
Mean	**108.8**	**119.4**	**220.8**	**120.3**	**113.1**	
Small						
Busselton (W)	147.5	130.0	37.9	50.8	161.3	217.5
WC (Albany)	173.0	145.0	120.0	108.0	258.0	138.9
Southern Downs	90.0	246.5	247.5	36.8	86.4	134.8
P&W (Alice Springs)				78.0	144.7	85.5
Cassowary Coast	184.0	156.0	191.0	57.0	93.0	63.2
Westernport Water	58.4	71.8	83.9	93.7	139.0	48.3
Gympie	201.0	157.6	105.8	108.8	145.0	33.3
Goldenfields Water (W)		152.0	55.0	180.0	221.0	22.8
Western Downs	23.0	37.0	39.0	39.0	44.2	13.3
Orange	240.0	240.0	123.0	123.0	123.0	0.0
Central Highlands	30.0	30.0	30.0	30.0	30.0	0.0
Snowy Monaro		120.0	120.0	120.0	120.0	0.0
Livingstone	18.3	28.6	19.9	18.8	17.8	-5.3
WC (Australind/Eaton)	85.0	100.0	115.0	109.0	103.0	-5.5
Aqwest-Bunbury (W)	39.0	38.0	35.0	43.0	40.0	-7.0
WC (Geraldton)	89.0	102.0	109.0	99.0	90.0	-9.1
Whitsunday	278.3	235.0	231.0	352.5	303.0	-14.0
Lismore	90.0	120.0	162.0	162.0	121.0	-25.3
WC (Kal–Boulder) (W)	36.0	31.0	56.0	96.0	67.0	-30.2
Bega Valley	163.0	251.0	510.0	456.0	302.0	-33.8
Armidale	98.0	131.0	89.0	166.0	16.0	-90.4
Essential Energy					138.0	
Kempsey	238.0	195.0	104.0	112.0		
Median	**90.0**	**130.0**	**105.8**	**103.5**	**120.5**	
Mean	**120.1**	**129.4**	**123.0**	**120.0**	**125.6**	

Table A10 A8 – Water main breaks, bursts and leaks (mains breaks per 100 km), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Barwon Water	25.1	23.2	25.1	23.6	29.2	23.7
WC (Perth)	11.5	10.9	10.2	10.2	12.0	17.6
SA Water	13.5	13.3	13.3	12.7	14.3	12.6
Greater Western Water			18.5	16.8	17.9	6.5
Icon Water	13.6	12.0	10.0	8.4	8.8	4.8
Hunter Water	28.3	20.2	21.0	22.0	22.9	4.1
Urban Utilities	31.1	25.4	18.6	17.8	18.3	2.8
Unitywater	4.9	3.5	4.7	3.9	3.9	0.0
South East Water	31.8	25.2	27.6	28.4	28.2	-0.7
Sydney Water	29.2	20.8	17.2	20.6	20.2	-1.9
Yarra Valley Water	43.8	28.6	31.3	35.4	34.6	-2.3
Logan	6.7	6.0	5.2	3.7	3.5	-5.4
Central Coast	11.6	9.9	14.1	16.3	15.2	-6.7
TasWater	32.6	51.9	43.1	47.2	42.8	-9.3
Gold Coast	10.2	9.5	8.7	7.2	5.9	-18.1
Median	**19.4**	**16.8**	**17.2**	**16.8**	**17.9**	
Mean	**21.0**	**18.6**	**17.9**	**18.3**	**18.5**	
Large						
Redland City	3.0	3.9	5.9	6.6	8.1	22.7
North East Water	13.7	11.2	12.0	11.1	13.4	20.7
Townsville	27.6	21.8	19.1	17.7	20.7	16.9
Goulburn Valley Water	13.2	13.9	13.7	10.2	11.3	10.8
Coliban Water	26.4	21.5	23.7	23.1	25.0	8.2
Shoalhaven	11.2	6.5	5.8	19.2	20.6	7.3
Gippsland Water	20.5	19.7	18.6	16.5	16.7	1.2
Central Highlands Water (Vic)	19.7	16.8	17.7	16.3	16.2	-0.6
Cairns	20.1	20.6	19.8	25.6	24.5	-4.3
P&W (Darwin)	12.7	14.0	14.1	16.5	13.8	-16.4
WC (Mandurah)	2.3	2.7	2.0	2.9	2.2	-24.1
Toowoomba	9.1	9.1	12.3	14.4	8.3	-42.4
Median	**13.4**	**13.9**	**13.9**	**16.4**	**15.0**	
Mean	**15.0**	**13.5**	**13.7**	**15.0**	**15.1**	
Medium						
Fitzroy RiverA37:F54 Water	10.8	10.3	7.3	6.1	11.6	90.2
Wannon Water	5.1	7.3	8.0	6.9	12.6	82.6
Lower Murray Water	26.4	19.6	20.5	21.1	26.9	27.5
Albury	3.5	4.3	2.8	2.3	2.9	26.1
East Gippsland Water	15.4	13.4	10.6	11.7	14.2	21.4
Dubbo	5.4	12.9	15.1	12.6	15.0	19.0
MidCoast Council	13.7	15.1	15.2	10.2	11.5	12.7

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Fraser Coast	7.9	5.3	6.6	7.3	7.9	8.2
Coffs Harbour	11.0	9.1	8.2	5.0	5.4	8.0
Clarence Valley	86.1	7.3	3.9	1.4	1.5	7.1
Port Macquarie Hastings	7.9	8.5	1.1	2.8	2.7	-3.6
Riverina Water (W)	16.7	18.2	10.6	19.4	17.9	-7.7
Bundaberg	22.0	32.7	17.3	14.3	13.0	-9.1
South Gippsland Water	24.0	22.2	36.6	29.4	25.9	-11.9
GWMWater	45.2	41.7	39.9	46.2	39.8	-13.9
Gladstone	63.7	10.8	87.1	40.2	34.6	-13.9
Mackay	7.2	6.2	6.4	3.8	3.1	-18.4
Queanbeyan	21.4	23.2	23.5	17.6	13.4	-23.9
Tweed	9.3	11.8	11.1	9.7	5.7	-41.2
Wingecarribee	8.3	6.3	4.9	8.3	3.9	-53.0
Tamworth	3.7	4.7	2.9	3.9	1.6	-59.0
Eurobodalla	17.2	42.0	30.9	15.0		
Median	**12.3**	**11.3**	**10.6**	**9.9**	**11.6**	
Mean	**19.6**	**15.1**	**16.8**	**13.4**	**12.9**	
Small						
Cassowary Coast	20.8	21.9	27.3	21.5	31.2	45.1
WC (Australind/Eaton)	6.5	4.4	5.2	3.2	4.6	43.7
WC (Kal–Boulder) (W)	21.2	9.8	12.9	9.5	13.6	43.2
Westernport Water	11.0	10.5	19.8	13.3	15.5	16.5
Central Highlands	33.9	24.9	26.8	21.8	24.9	14.2
WC (Geraldton)	22.6	14.9	12.0	25.2	28.6	13.5
Whitsunday	16.2	19.5	23.2	25.9	28.5	10.0
Aqwest-Bunbury (W)	6.1	7.6	4.6	8.0	8.4	5.0
Livingstone	3.2	3.9	3.8	3.2	3.3	3.1
Lismore	52.7	20.9	12.3	11.4	11.7	2.6
Goldenfields Water (W)	18.5	15.1	8.6	11.2	10.8	-3.6
Orange	7.4	10.9	3.1	8.9	8.5	-4.5
Kempsey	5.5	5.9	19.5	17.4	16.6	-4.6
Gympie	9.8	10.6	16.0	14.0	12.7	-9.3
Busselton (W)	7.4	7.9	5.2	6.3	5.7	-9.5
Southern Downs	7.2	8.1	11.4	17.5	15.8	-9.7
Bathurst	4.2	3.5	3.9	13.5	11.7	-13.3
Essential Energy	22.3	9.9	4.5	5.2	4.5	-13.5
P&W (Alice Springs)	43.1	31.0	31.4	32.1	24.5	-23.7
WC (Albany)	11.5	13.5	11.9	12.1	8.8	-27.3
Ballina				5.5	3.7	-32.7
Mount Barker			6.0	11.1	7.3	-34.2
Bega Valley	4.7	11.6	3.0	8.1	5.3	-34.6
Western Downs	7.2	25.1	30.4	52.3	29.0	-44.6
Goulburn Mulwaree	14.4	4.7	2.8	2.0	1.1	-45.0

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Armidale	20.1	14.3	12.4	15.4	5.8	-62.3
Byron	13.1	48.2	61.9	55.4	2.9	-94.8
Snowy Monaro		8.9		16.1		
Median	**11.5**	**10.8**	**11.9**	**12.7**	**10.8**	
Mean	**15.6**	**14.1**	**14.6**	**16.0**	**12.8**	

Table A11 A14 – Number of sewer mains breaks and chokes per 100 km of sewer main (breaks and chokes/100 km), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Sydney Water	66.0	49.0	42.0	28.0	41.0	46.4
Hunter Water	42.8	33.6	27.3	22.5	32.1	42.7
Central Coast	33.0	30.0	30.0	26.0	36.0	38.5
Urban Utilities	21.1	24.0	19.0	14.3	19.4	35.7
Icon Water	83.8	52.3	34.7	29.0	39.1	34.8
TasWater	41.2	56.6	42.8	48.1	63.9	32.8
WC (Perth)	18.4	17.5	14.1	13.4	17.4	29.9
Barwon Water	53.8	55.8	53.9	49.8	57.1	14.7
Gold Coast	6.2	3.8	5.7	4.4	5.0	13.6
Greater Western Water			16.9	15.6	17.4	11.5
Logan	9.8	8.1	6.6	5.8	6.1	5.2
SA Water	48.0	49.3	51.0	45.0	47.0	4.4
South East Water	20.7	16.2	16.1	14.3	14.7	2.8
Yarra Valley Water	42.4	32.3	27.6	28.8	25.8	-10.4
Unitywater	19.8	9.8	8.9	7.1	6.0	-15.5
Median	**37.1**	**31.1**	**27.3**	**22.5**	**25.8**	
Mean	**36.2**	**31.3**	**26.4**	**23.5**	**28.5**	
Large						
Shoalhaven	3.0	3.0	17.0	13.0	20.0	53.8
Cairns	12.9	12.4	16.7	14.6	20.8	42.5
Gippsland Water	6.9	1.5	3.4	4.2	4.9	16.7
Toowoomba	13.5	20.1	18.7	14.6	14.7	0.7
Redland City	2.7	13.9	15.2	15.0	14.8	-1.3
Townsville	54.8	54.4	57.4	45.8	42.8	-6.6
Central Highlands Water (Vic)	19.2	19.7	16.7	16.5	15.1	-8.5
WC (Mandurah)	8.5	6.8	7.6	7.7	6.9	-10.4
Goulburn Valley Water	13.8	14.9	10.7	10.2	8.6	-15.7
North East Water	13.9	10.6	9.1	9.4	7.7	-18.1
Coliban Water	22.3	17.8	20.7	16.2	11.8	-27.2
P&W (Darwin)	12.7	8.6	6.9	17.0	7.9	-53.5
Median	**13.2**	**13.2**	**15.9**	**14.6**	**13.2**	
Mean	**15.4**	**15.3**	**16.7**	**15.3**	**14.7**	
Medium						
Coffs Harbour	88.0	82.0	31.0	10.0	30.0	200.0
Fraser Coast	9.4	6.5	3.6	2.6	4.5	73.1
Port Macquarie Hastings	6.0	15.0	30.0	6.0	9.0	50.0
South Gippsland Water	13.3	12.5	8.9	7.8	11.3	44.9
Albury	52.0	81.0	83.0	75.0	94.0	25.3
Queanbeyan	55.0	116.0	84.0	62.0	77.0	24.2
Dubbo	63.0	50.0	62.0	46.0	57.0	23.9

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
MidCoast Council	27.0	21.0	18.0	14.0	16.0	14.3
Wingecarribee	8.0	15.0	14.0	9.0	10.0	11.1
Clarence Valley	66.0	11.0	5.0	2.0	2.0	0.0
Tweed	0.0	1.0	6.0	6.0	6.0	0.0
Wagga Wagga (S)		57.0	55.0	57.0	56.0	-1.8
Wannon Water	9.2	8.1	5.2	7.0	6.7	-4.3
Lower Murray Water	16.0	10.4	17.0	13.9	13.0	-6.5
GWMWater	43.2	50.5	43.8	43.6	39.7	-8.9
Bundaberg	20.3	30.1	29.9	20.6	18.2	-11.7
East Gippsland Water	8.1	7.1	5.4	3.6	3.1	-13.9
Fitzroy River Water	18.3	16.5	16.3	16.3	10.6	-35.0
Tamworth	12.0	11.0	20.0	14.0	9.0	-35.7
Mackay	3.1	4.6	1.3	1.9	1.1	-42.1
Gladstone	8.4	2.0	9.4	6.9	2.3	-66.7
Eurobodalla	29.0	13.0	27.0	13.0		
Median	**16.0**	**14.0**	**17.5**	**11.5**	**10.6**	
Mean	**26.4**	**28.2**	**26.2**	**19.9**	**22.7**	
Small						
WC (Busselton) (S)	13.2	5.6	2.5	0.9	3.7	311.1
Ballina				5.0	15.0	200.0
Central Highlands	25.6	9.6	9.6	10.7	26.4	146.7
Orange	40.0	45.0	64.0	33.0	56.0	69.7
Mount Barker			1.0	2.0	3.0	50.0
Essential Energy	143.0	167.0	151.0	96.0	133.0	38.5
Armidale	58.0	47.0	41.0	29.0	40.0	37.9
WC (Bunbury) (S)	6.4	7.7	8.1	6.8	9.1	33.8
Lismore	24.0	28.0	24.0	16.0	20.0	25.0
Whitsunday	7.6	32.2	11.7	15.8	17.8	12.7
Gympie	14.2	15.6	19.0	15.5	17.3	11.6
Kal-Boulder (S)	34.0	24.3	13.7	11.8	13.1	11.0
Kempsey	12.0	33.0	29.0	32.0	35.0	9.4
Bathurst	36.0	44.0	23.0	22.0	23.0	4.5
WC (Australind/Eaton)	6.3	7.1	5.9	3.9	3.9	0.0
P&W (Alice Springs)	6.6	5.9	11.3	6.0	5.7	-5.0
Westernport Water	3.6	5.4	7.6	4.0	3.5	-12.5
Southern Downs	64.4	30.8	32.7	21.1	17.8	-15.6
WC (Albany)	21.1	22.9	16.5	17.3	13.5	-22.0
Goulburn Mulwaree	44.0	277.0	33.0	60.0	44.0	-26.7
Bega Valley	18.0	14.0	7.0	3.0	2.0	-33.3
Livingstone	3.9	4.2	4.8	4.1	2.4	-41.5
Cassowary Coast	3.0	6.9	3.5	8.2	4.6	-43.9
Snowy Monaro		6.0	13.0	219.0	116.0	-47.0
WC (Geraldton)	5.0	3.4	3.7	5.9	3.1	-47.5

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Byron	13.0	42.0	49.0	41.0	19.0	-53.7
Western Downs	4.7	4.0	3.6	7.5	2.9	-61.3
Median	**13.7**	**15.6**	**12.3**	**11.8**	**15.0**	
Mean	**25.3**	**35.5**	**22.7**	**25.8**	**24.1**	

Table A12 A15 – Number of property connection sewer breaks and chokes (breaks and chokes/1,000 properties), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Logan	1.20	0.90	0.60	0.50	1.10	120.0
Sydney Water	0.30	0.20	0.20	0.10	0.17	70.0
Hunter Water	9.80	8.20	6.60	5.60	8.40	50.0
Icon Water	15.70	10.50	7.30	4.90	6.80	38.8
Central Coast	3.10	3.00	3.90	3.00	3.97	32.3
Gold Coast	2.20	1.70	1.50	1.00	1.30	30.0
TasWater	12.60	6.90	6.70	6.50	7.90	21.5
Urban Utilities	2.90	3.10	2.30	2.20	2.60	18.2
Greater Western Water			2.84	2.52	2.94	16.7
Barwon Water	9.30	10.00	9.60	8.80	10.00	13.6
Unitywater	1.40	1.30	1.10	0.90	1.00	11.1
South East Water	4.90	3.90	4.04	3.38	3.64	7.7
SA Water	29.00	28.00	35.00	31.00	33.00	6.5
Yarra Valley Water	8.00	5.60	5.30	4.78	5.02	5.0
Median	**4.90**	**3.90**	**3.97**	**3.19**	**3.81**	
Mean	**7.72**	**6.41**	**6.21**	**5.37**	**6.27**	
Large						
Redland City	1.10	1.00	1.40	0.60	1.30	116.7
P&W (Darwin)	1.00	1.20	0.99	1.00	1.51	51.0
North East Water	2.70	3.20	3.50	3.00	3.97	32.3
Central Highlands Water (Vic)	4.60	5.60	5.15	4.04	5.23	29.5
Coliban Water	3.30	0.50	1.07	0.89	1.08	21.3
Toowoomba	2.30	1.60	1.10	3.10	3.40	9.7
Cairns	2.30	2.20	2.90	2.60	2.60	0.0
Gippsland Water	1.40	1.40	1.10	0.90	0.90	0.0
Goulburn Valley Water	7.50	5.80	5.70	4.40	4.20	-4.5
Townsville	5.40	7.20	4.60	6.20	5.90	-4.8
Shoalhaven	3.10	2.20	2.10	4.80	0.85	-82.3
Median	**2.70**	**2.20**	**2.10**	**3.00**	**2.60**	
Mean	**3.15**	**2.90**	**2.69**	**2.87**	**2.81**	
Medium						
Port Macquarie Hastings	0.20	0.03	0.40	0.10	2.42	2320.0
Gladstone	1.50	0.10	0.70	0.30	1.00	233.3
Bundaberg	7.60	11.00	7.80	3.80	6.30	65.8
Tweed	0.40	0.50	0.40	0.50	0.81	62.0
Wannon Water	2.00	1.90	2.60	1.90	2.80	47.4
Clarence Valley	4.20	3.40	6.90	3.60	5.30	47.2
GWMWater	33.10	35.50	23.73	21.07	23.44	11.2
Tamworth	2.90	6.30	2.40	4.70	4.92	4.7
Lower Murray Water	9.00	7.00	9.55	6.20	6.10	-1.6

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Coffs Harbour	3.00	3.10	11.20	13.70	12.69	-7.4
Mackay	1.20	1.70	1.40	1.20	1.10	-8.3
Fitzroy River Water	12.80	12.30	11.60	8.40	7.00	-16.7
Fraser Coast	1.50	3.40	3.80	4.00	3.20	-20.0
East Gippsland Water	1.30	1.70	0.67	0.90	0.70	-22.2
Queanbeyan		2.20	1.90	0.37	0.25	-32.4
Wagga Wagga (S)	8.50	6.90	7.00	6.20	4.12	-33.5
Wingecarribee	16.70	15.00	7.90	15.00	9.45	-37.0
South Gippsland Water	5.10	4.00	4.71	7.03	3.07	-56.3
Dubbo	8.60	8.90	9.80	10.00	3.41	-65.9
Albury	2.70	2.30	2.10	3.40	0.00	-100.0
Eurobodalla	4.20	6.00	4.60	1.80		
MidCoast Council					0.00	
Median	**3.60**	**3.40**	**4.60**	**3.80**	**3.20**	
Mean	**6.32**	**6.34**	**5.77**	**5.44**	**4.67**	
Small						
Southern Downs	7.10	5.70	5.40	2.80	6.40	128.6
Snowy Monaro				1.90	3.00	57.9
P&W (Alice Springs)	4.10	1.30	1.70	1.00	1.55	55.0
Whitsunday	0.70	3.00	2.70	1.00	1.50	50.0
Lismore	8.90	9.70	8.90	6.30	8.97	42.4
Bathurst	3.00	4.90	2.60	1.30	1.83	40.8
Cassowary Coast	0.30	0.70	0.80	1.40	1.90	35.7
Kempsey	23.50	12.40	12.20	9.70	13.09	34.9
Western Downs	4.60	8.10	6.90	5.10	6.50	27.5
Essential Energy	60.00	61.40	68.60	38.50	44.87	16.5
Gympie	1.90	2.30	1.10	0.80	0.90	12.5
Westernport Water	1.80	1.50	1.97	1.95	2.06	5.6
Livingstone	2.20	3.40	2.00	2.20	2.30	4.5
Mount Barker			1.20	0.00	0.00	0.0
Armidale	34.10	23.90	14.50	16.10	14.20	-11.8
Goulburn Mulwaree	12.60	10.40	9.10	8.40	6.37	-24.2
Byron	16.80	10.90	20.00	8.00	6.06	-24.3
Orange	10.00	13.40	28.30	13.80	9.74	-29.4
Ballina				5.40	3.40	-37.0
Central Highlands	8.60	10.90	13.20	11.50	6.60	-42.6
Bega Valley	0.80	1.90	0.50	1.30	0.34	-73.8
Kal-Boulder (S)	0.00	0.00				
Median	**4.60**	**5.70**	**5.40**	**2.80**	**3.40**	
Mean	**10.58**	**9.78**	**10.61**	**6.59**	**6.74**	

Table A13 A10 – Real losses: service connections (L/service connection/day), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Unitywater	54.1	52.7	40.1	49.8	59.8	20.1
Yarra Valley Water	42.0	37.0	40.0	49.0	51.0	4.1
Sydney Water	85.0	74.0	77.0	80.0	83.0	3.8
WC (Perth)	84.7	71.7	95.0	87.0	89.0	2.3
Gold Coast	68.0	65.5	65.8	70.5	71.6	1.6
Barwon Water	21.0	24.0	30.0	59.0	59.0	0.0
Greater Western Water			60.0	80.0	80.0	0.0
SA Water	90.0	85.0	77.0	73.0	73.0	0.0
Central Coast	58.0	68.0	59.0	73.0	72.7	-0.4
Urban Utilities	70.0	61.0	67.0	67.0	65.0	-3.0
South East Water	53.0	47.0	65.0	68.0	65.0	-4.4
Icon Water	49.0	55.0	52.0	72.0	66.0	-8.3
Hunter Water	69.0	64.0	67.0	83.0	76.0	-8.4
Logan	70.8	66.3	70.9	53.9	39.3	-27.1
TasWater	361.0	272.0	308.0	319.0		
Median	**68.5**	**64.8**	**65.8**	**72.0**	**68.8**	
Mean	**84.0**	**74.5**	**78.2**	**85.6**	**67.9**	
Large						
Redland City	25.6	18.4	28.5	16.2	55.7	243.8
Coliban Water	86.0	70.0	40.0	50.0	90.0	80.0
Cairns	31.3	34.8	64.5	77.7	121.0	55.7
WC (Mandurah)	47.8	45.3	58.3	44.3	65.5	48.0
Central Highlands Water (Vic)	42.0	50.0	50.0	32.0	46.0	43.8
North East Water	54.0	51.0	44.0	59.0	79.0	33.9
Goulburn Valley Water	80.0	107.0	78.0	93.0	102.0	9.7
Shoalhaven	74.0	52.0	90.0	74.0	79.2	7.0
Gippsland Water	71.0	70.0	72.0	84.0	78.0	-7.1
Toowoomba	73.4	96.3	98.6	42.7	35.9	-15.9
P&W (Darwin)	276.0	297.0	281.0	362.0	270.0	-25.4
Townsville	333.7	152.9	86.8	190.8	66.4	-65.2
Median	**72.2**	**61.0**	**68.2**	**66.5**	**78.5**	
Mean	**99.6**	**87.1**	**82.6**	**93.8**	**90.7**	
Medium						
Albury	19.0	23.0	17.0	20.0	33.2	65.9
Mackay	111.0	228.1	116.4	99.1	143.9	45.2
Wannon Water	70.0	70.0	70.0	70.0	90.0	28.6
GWMWater	40.0	70.0	90.0	110.0	138.0	25.5
Port Macquarie Hastings	87.0	64.0	70.0	13.0	16.1	23.8
Tamworth	87.0	82.0	61.0	48.0	56.4	17.5
Wingecarribee	77.0	76.0	88.0	69.0	80.5	16.7

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
South Gippsland Water	102.6	118.9	119.5	111.0	129.3	16.5
Gladstone	96.0	60.3	90.3	67.2	70.8	5.4
East Gippsland Water	43.1	56.0	47.0	50.0	50.0	0.0
Coffs Harbour	44.0	38.0	45.0	45.0	42.1	-6.5
Bundaberg	163.6	135.4	59.2	133.3	118.0	-11.5
Lower Murray Water	101.4	47.6	66.8	129.5	109.5	-15.5
Fraser Coast	62.1	53.1	59.0	48.6	40.6	-16.5
MidCoast Council	40.0	22.0	61.0	60.0	49.8	-17.0
Riverina Water (W)	2.0	63.0	101.0	100.0	70.3	-29.7
Clarence Valley	30.0	47.0	110.0	150.0	101.5	-32.3
Fitzroy River Water	194.1	170.0	224.0	220.8	146.1	-33.8
Tweed	96.0	109.0	146.0	174.0	114.5	-34.2
Queanbeyan	142.0	27.0	42.0	18.0	9.0	-49.8
Dubbo	121.0	106.0	77.0			
Eurobodalla	50.0	50.0	54.0	51.0		
Median	**82.0**	**63.5**	**70.0**	**69.0**	**75.7**	
Mean	**80.9**	**78.0**	**82.5**	**85.1**	**80.5**	
Small						
P&W (Alice Springs)	246.0	208.0	100.0	79.0	205.8	160.5
Central Highlands	261.9		304.0	246.8	372.1	50.8
Busselton (W)	162.0	127.0	177.0	145.0	177.0	22.1
WC (Geraldton)	162.6	234.4	226.8	212.9	238.9	12.2
Whitsunday	323.1	220.1	247.0	192.4	212.0	10.2
Cassowary Coast	472.4	372.5	358.4	415.7	457.7	10.1
Byron	49.0	44.0	26.0	69.0	75.8	9.9
Orange	39.0	29.0	35.0	43.0	47.0	9.3
Bega Valley	54.0	41.0	46.0	49.0	52.8	7.8
WC (Australind/Eaton)	87.4	168.4	104.9	137.4	147.2	7.1
Essential Energy	76.0	67.0	94.0	72.0	72.0	0.0
Aqwest-Bunbury (W)	95.0	103.0	107.0	140.0	137.0	-2.1
Livingstone	245.0	166.2	181.1	197.4	192.2	-2.6
Kempsey	45.0	45.0	35.0	57.0	55.0	-3.5
WC (Albany)	176.5	76.5	84.5	66.4	62.0	-6.6
Gympie	113.7	131.9	117.6	170.7	156.1	-8.6
Bathurst	47.0			144.0	126.5	-12.1
Goldenfields Water (W)	89.0	167.0	146.0	214.0	173.0	-19.2
Southern Downs	35.4	106.5	125.3	141.5	107.0	-24.4
Ballina	116.0	117.0	112.0	104.0	70.3	-32.4
WC (Kal–Boulder) (W)	120.6	79.6	100.2	141.4	92.9	-34.3
Lismore	78.0	75.0	108.0	161.0	102.0	-36.7
Western Downs	6.7	118.7	134.5	111.1	52.8	-52.5
Westernport Water	11.0	11.0	31.0	18.0	5.0	-72.2
Goulburn Mulwaree	40.0	12.0	38.0	40.0	9.5	-76.3

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Armidale	28.0	186.0	46.0		56.7	
Median	**88.2**	**111.8**	**107.0**	**140.0**	**104.5**	
Mean	**122.3**	**121.1**	**123.4**	**134.7**	**132.9**	

Table A14 E12 – Total net greenhouse gas emissions per 1,000 properties (t CO_2 equivalents/1,000 properties), by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
Gold Coast	250	236	239	200	298	48.6
SA Water	340	345	236	173	209	20.8
WC (Perth)	701	695	567	379	451	19.0
Unitywater	228	245	252	205	221	7.5
Central Coast	472	533	415	496	503	1.5
Logan	185	173	186	0	0	0.0
Urban Utilities	178	180	166	160	158	-1.3
Greater Western Water			78	61	58	-5.1
Hunter Water	344	309	294	271	255	-6.0
Icon Water	331	196	177	175	154	-12.1
Sydney Water	175	169	168	158	139	-12.4
Barwon Water	267	176	161	132	105	-20.5
South East Water	42	45	35	34	25	-25.7
Yarra Valley Water	35	26	20	18	12	-34.8
Median	**250**	**196**	**181**	**166**	**156**	
Mean	**273**	**256**	**214**	**176**	**185**	
Large						
WC (Mandurah)	377	387	323	193	254	32.0
Toowoomba	617	635	585	415	461	11.0
P&W (Darwin)	213	199	223	233	248	6.5
Central Highlands Water (Vic)	200	175	196	172	178	3.1
Cairns	228	247	224	214	211	-1.7
Goulburn Valley Water	1,290	1,170	555	490	465	-5.3
Townsville	433	165	482	483	424	-12.1
Redland City	184	190	180	180	157	-13.0
Coliban Water	385	341	338	258	219	-15.4
Gippsland Water	483	480	472	436	364	-16.7
North East Water	619	599	578	560	418	-25.4
Shoalhaven	739	763	831	676	399	-41.0
Median	**409**	**364**	**405**	**337**	**309**	
Mean	**481**	**446**	**416**	**359**	**316**	
Medium						
Clarence Valley	169	233	208	85	216	153.5
Dubbo	468	427	740	273	508	86.2
Tamworth	391	370	212	216	374	72.9
Wannon Water	611	528	578	502	566	12.7
Port Macquarie Hastings	527	489	489	539	593	10.0
Albury	367	271	339	374	391	4.5
Fitzroy River Water	562	559	535	500	519	3.9
Wingecarribee	733	913	1,064	643	667	3.7

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
GWMWater	473	436	492	377	377	0.0
Coffs Harbour	428	419	427	323	320	-0.8
South Gippsland Water	424	391	348	329	318	-3.3
Bundaberg	677	338	200	290	275	-5.2
Lower Murray Water	578	540	523	399	367	-8.1
East Gippsland Water	341	307	310	291	265	-8.9
MidCoast Council	416	454	844	803	702	-12.6
Tweed	413	391	380	377	324	-14.0
Queanbeyan	184	243	140	213	174	-18.3
Eurobodalla	495	404	177	260		
Mackay	0	0		0		
Median	**428**	**404**	**404**	**329**	**374**	
Mean	**435**	**406**	**445**	**358**	**409**	
Small						
Orange	668	672	672	229	283	23.6
Essential Energy	368	354	338	370	419	13.2
WC (Australind/Eaton)	321	310	287	223	245	9.8
WC (Albany)	364	362	372	292	315	8.2
Byron	386	417	416	379	403	6.4
Lismore	185	255	375	322	336	4.2
Cassowary Coast	290	319	257	329	342	3.7
WC (Geraldton)	343	322	325	279	289	3.7
P&W (Alice Springs)	878	882	757	712	730	2.5
Bega Valley	284	476	435	695	710	2.1
Snowy Monaro			1,096	734	729	-0.7
Goulburn Mulwaree	687	855	948	858	850	-0.9
Livingstone	391	370	439	393	384	-2.4
Armidale	326	719	124	491	449	-8.6
Southern Downs	168	599	646	571	518	-9.3
Kempsey	682	821	822	749	670	-10.6
Gympie	405	362	267	253	222	-12.3
Ballina	459	772	849	657	569	-13.4
Whitsunday	475	497	492	372	304	-18.2
Westernport Water	384	367	329	364	294	-19.2
Western Downs	475	509	600	562	437	-22.3
Bathurst	388	395	426	430	71	-83.4
Central Highlands	291	296	199			
Median	**385**	**406**	**426**	**386**	**394**	
Mean	**419**	**497**	**499**	**467**	**435**	

Table A15 H3 – Percentage of population where microbiological compliance was achieved, by utility size group, 2019–20 to 2023–24

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Major						
South East Water	100.0	99.9	100.0	99.9	100.0	0.1
Barwon Water	100.0	100.0	100.0	100.0	100.0	0.0
Central Coast	100.0	100.0	100.0	100.0	100.0	0.0
Gold Coast	100.0	100.0	100.0	100.0	100.0	0.0
Greater Western Water			100.0	100.0	100.0	0.0
Hunter Water	100.0	100.0	100.0	100.0	100.0	0.0
Icon Water	100.0	100.0	100.0	100.0	100.0	0.0
Logan	100.0	100.0	100.0	100.0	100.0	0.0
SA Water	100.0	100.0	100.0	100.0	100.0	0.0
Sydney Water	100.0	100.0	100.0	100.0	100.0	0.0
Unitywater	100.0	100.0	100.0	100.0	100.0	0.0
Urban Utilities	100.0	100.0	100.0	100.0	100.0	0.0
WC (Perth)	100.0	100.0	100.0	100.0	100.0	0.0
Yarra Valley Water	100.0	100.0	100.0	100.0	100.0	0.0
Median	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	
Mean	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	
Large						
Coliban Water	94.3	100.0	100.0	95.0	100.0	5.3
North East Water	100.0	100.0	100.0	99.9	100.0	0.1
Cairns	100.0	100.0	100.0	100.0	100.0	0.0
Gippsland Water	100.0	99.0	100.0	100.0	100.0	0.0
Central Highlands Water (Vic)	100.0	99.8	100.0	100.0	100.0	0.0
Goulburn Valley Water	100.0	99.5	100.0	100.0	100.0	0.0
P&W (Darwin)	100.0	100.0	100.0	100.0	100.0	0.0
Redland City	100.0	100.0	100.0	100.0	100.0	0.0
Shoalhaven	100.0	100.0	100.0	100.0	100.0	0.0
Toowoomba	100.0	100.0	100.0	100.0	100.0	0.0
Townsville	100.0	100.0	100.0	100.0	100.0	0.0
WC (Mandurah)	100.0	100.0	100.0	100.0	100.0	0.0
Median	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	
Mean	**99.5**	**99.9**	**100.0**	**99.6**	**100.0**	
Medium						
Bundaberg	100.0	100.0	100.0	99.9	100.0	0.1
Albury	100.0	100.0	100.0	100.0	100.0	0.0
Wannon Water	100.0	100.0	100.0	100.0	100.0	0.0
Tweed	100.0	100.0	100.0	100.0	100.0	0.0
Fitzroy River Water	100.0	100.0	100.0	100.0	100.0	0.0
Riverina Water (W)	100.0	100.0	100.0	100.0	100.0	0.0
Queanbeyan	100.0	100.0	100.0	100.0	100.0	0.0
Mackay	100.0	100.0	100.0	100.0	100.0	0.0

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Lower Murray Water	100.0	100.0	100.0	100.0	100.0	0.0
GWMWater	100.0	100.0	97.0	100.0	100.0	0.0
Gladstone	100.0	100.0	100.0	100.0	100.0	0.0
Fraser Coast	100.0	100.0	100.0	100.0	100.0	0.0
East Gippsland Water	100.0	100.0	100.0	100.0	100.0	0.0
Dubbo	100.0	100.0	100.0	100.0	100.0	0.0
Coffs Harbour	100.0	100.0	100.0	100.0	100.0	0.0
Clarence Valley	100.0	100.0	100.0	100.0	100.0	0.0
Wingecarribee	100.0	100.0	100.0	100.0	100.0	0.0
MidCoast Council	100.0	100.0	100.0	100.0	97.0	-3.0
South Gippsland Water	100.0	100.0	100.0	100.0	94.5	-5.5
Eurobodalla	100.0	100.0	100.0	100.0		
Port Macquarie Hastings	100.0	100.0	100.0		100.0	
Tamworth	100.0	100.0	100.0		100.0	
Median	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	
Mean	**100.0**	**100.0**	**99.9**	**100.0**	**99.6**	
Small						
Aqwest-Bunbury (W)	100.0	100.0	100.0	100.0	100.0	0.0
Lismore	100.0	100.0	100.0	100.0	100.0	0.0
Westernport Water	100.0	100.0	100.0	100.0	100.0	0.0
Western Downs	100.0	100.0	100.0	100.0	100.0	0.0
WC (Kal–Boulder) (W)	100.0	100.0	100.0	100.0	100.0	0.0
WC (Geraldton)	100.0	100.0	100.0	100.0	100.0	0.0
WC (Australind/Eaton)	100.0	100.0	100.0	100.0	100.0	0.0
WC (Albany)	100.0	100.0	100.0	100.0	100.0	0.0
Southern Downs	99.5	100.0	100.0	100.0	100.0	0.0
Snowy Monaro		100.0	100.0	100.0	100.0	0.0
P&W (Alice Springs)	100.0	100.0	100.0	100.0	100.0	0.0
Orange	100.0	100.0	100.0	100.0	100.0	0.0
Livingstone	100.0	100.0	100.0	100.0	100.0	0.0
Kempsey	100.0	100.0	100.0	100.0	100.0	0.0
Armidale	100.0	100.0	100.0	100.0	100.0	0.0
Gympie	98.6	100.0	100.0	100.0	100.0	0.0
Goulburn Mulwaree	100.0	100.0	100.0	100.0	100.0	0.0
Goldenfields Water (W)	100.0	100.0	100.0	100.0	100.0	0.0
Essential Energy	100.0	100.0	100.0	100.0	100.0	0.0
Central Highlands	100.0	100.0	100.0	100.0	100.0	0.0
Cassowary Coast	97.5	100.0	100.0	100.0	100.0	0.0
Byron	100.0	100.0	100.0	100.0	100.0	0.0
Busselton (W)	100.0	100.0	100.0	100.0	100.0	0.0
Bega Valley	100.0	100.0	100.0	100.0	100.0	0.0
Bathurst	100.0	100.0	100.0	100.0	100.0	0.0
Ballina	100.0	100.0	100.0	100.0	100.0	0.0

Utility	2019–20	2020–21	2021–22	2022–23	2023–24	Change from 2022–23 (%)
Whitsunday	100.0	100.0	100.0	100.0	100.0	0.0
Median	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	
Mean	**99.8**	**100.0**	**100.0**	**100.0**	**100.0**	

Appendix B Audit framework

Auditing is intended to provide enhanced confidence in the accuracy, completeness and reliability of reported information. Auditing promotes transparency and consistency in the process of collecting and reporting data, across all urban water utilities, in order to report performance results that are relevant and useful, and enable meaningful comparisons between utilities over time.

The National Water Commission, the Water Services Association of Australia and representative National Water Initiative (NWI) parties established the National Framework for Reporting on Performance of Urban Water Utilities Deed, which sets out how the parties will report on the performance of urban water utilities in accordance with the NWI. The deed requires parties to use all reasonable endeavours to ensure that a comprehensive audit of the data collected by each urban water utility under the National Performance Framework is undertaken at least once every 3 years.

The National Performance Framework 2013–14 auditing requirements and audit report template provide further detail about the requirements that a water utility must meet in order to report its results in the 2024 Urban NPR.

- Audits are to be conducted at a minimum of 3-year intervals.
- Indicators that have failed an audit will not be published (they need to be re-audited before they are published).
- Audits must be carried out by suitably qualified and independent auditors.
- The level of assurance to be provided is generally ‘reasonable’ assurance (although there are some instances in which ‘limited’ assurance is appropriate).
- Audits must be conducted under Australian Standard ASAE 3000: Assurance Engagements Other than Audits or Reviews of Historical Financial Information.
- Auditable indicators are those with the indicator codes W7, W8, W11, W11.3, W12, W18, W18.5, W19, W26, W27, A2, A3, A5, A6, A8 to A11, A14, A15, E1 to E3, E8, E12, E12.1, C2, C4, C8, C15, C17, C18, H3, H4, F1 to F8, F11 to F16, F20 to F30, P7 and P8.

Table B1 Status of indicator audits undertaken within each jurisdiction

Jurisdiction	Audit activities
Australian Capital Territory	The most recent audit conducted on the NPR data was for 2020–21, as it was an auditing year. Auditing is usually conducted every 3 years. The next audit will depend on auditing requirements following the NPR review.
New South Wales	In the New South Wales metropolitan area, the most recent audit conducted on the NPR data was in late 2024. The next audit is expected to be conducted in late 2025. IPART currently audits approximately one third of NPR indicators each year as part of its annual operational audit process. In the New South Wales country area, all the reported utilities have had their 2021–22 reported data audited as per the requirements. All water utilities in regional NSW reporting to the NPR will be required to conduct their next audit in the 2024–25 reporting year.
Northern Territory	The Department of Treasury and Finance has not required the utilities' providers, Power and Water – Alice Springs and Power and Water – Darwin, to audit their NPR data. Some NPR indicators are audited at an aggregate level.
Queensland	The Department of Local Government, Water and Volunteers has not required service providers to audit their NPR data.
South Australia	The most recent audit conducted on the NPR data was in October 2021 for 2020–21 data as it was an auditing year. Auditing is usually conducted every 3 years on the full set of indicators. The next audit will depend on auditing requirements following the NPR review.
Tasmania	TasWater's performance indicators are audited in 3 tranches over a 3-year period. In 2023–24, selected TasWater performance indicators were audited. This audit concluded in September 2024. A new 3-year audit cycle will commence in July 2025.
Victoria	Most Victorian National Performance Reporting data is audited under the Essential Service Commission's annual audit framework (refer section F7.5). The Essential Service Commission conducted their last regular annual audit of 2023–24 data in late 2024.
Western Australia	Urban NPR data audits were undertaken in 2021–22 in Western Australia. Auditing is conducted every 3 years. The next round of audits is scheduled for 2024–25.

Appendix C Utilities reporting

Utility name	Display name in tables and graphs	Jurisdiction	Utility size group
Albury City Council	Albury	New South Wales	Medium
Aqwest–Bunbury Water Corporation (W)	Aqwest–Bunbury (W)	Western Australia	Small
Armidale Regional Council	Armidale	New South Wales	Small
Ballina Shire Council	Ballina	New South Wales	Small
Barwon Water	Barwon Water	Victoria	Major
Bathurst Regional Council	Bathurst	New South Wales	Small
Bega Valley Shire Council	Bega Valley	New South Wales	Small
Bundaberg Regional Council	Bundaberg	Queensland	Medium
Busselton Water (W)	Busselton (W)	Western Australia	Small
Byron Shire Council	Byron	New South Wales	Small
Cairns Regional Council	Cairns	Queensland	Large
Cassowary Coast Regional Council	Cassowary Coast	Queensland	Small
Central Coast Council	Central Coast	New South Wales	Major
Central Gippsland Water	Gippsland Water	Victoria	Large
Central Highlands Regional Council	Central Highlands	Queensland	Small
Central Highlands Water	Central Highlands Water (Vic)	Victoria	Large
City of Gold Coast	Gold Coast	Queensland	Major
City of Kalgoorlie–Boulder (S)	Kal–Boulder (S)	Western Australia	Small
Clarence Valley Council	Clarence Valley	New South Wales	Medium
Coffs Harbour City Council	Coffs Harbour	New South Wales	Medium
Coliban Water	Coliban Water	Victoria	Large
Dubbo Regional Council	Dubbo	New South Wales	Medium
East Gippsland Water	East Gippsland Water	Victoria	Medium
Essential Energy	Essential Energy	New South Wales	Small
Eurobodalla Shire Council	Eurobodalla	New South Wales	Medium
Fraser Coast Regional Council	Fraser Coast	Queensland	Medium
Greater Western Water	Greater Western Water	Victoria	Major
Gladstone Area Water Board	GAWB	Queensland	Bulk water
Gladstone Regional Council	Gladstone	Queensland	Medium
Goldenfields Water County Council	Goldenfields Water (W)	New South Wales	Small
Goulburn Mulwaree Council	Goulburn Mulwaree	New South Wales	Small
Goulburn Valley Water	Goulburn Valley Water	Victoria	Large
GWMWater	GWMWater	Victoria	Medium
Gympie Regional Council	Gympie	Queensland	Small
Hunter Water Corporation	Hunter Water	New South Wales	Major
Icon Water Limited	Icon Water	Australian Capital Territory	Major
Kempsey Shire Council	Kempsey	New South Wales	Small
Lismore City Council	Lismore	New South Wales	Small
Livingstone Shire Council	Livingstone	Queensland	Small
Logan City Council	Logan	Queensland	Major
Lower Murray Water	Lower Murray Water	Victoria	Medium

Utility name	Display name in tables and graphs	Jurisdiction	Utility size group
Mackay Regional Council	Mackay	Queensland	Medium
Melbourne Water	Melbourne Water	Victoria	Bulk water
MidCoast Council	MidCoast Council	New South Wales	Medium
Mount Barker District Council	Mount Barker	South Australia	Small
North East Water	North East Water	Victoria	Large
Orange City Council	Orange	New South Wales	Small
Port Macquarie Hastings Council	Port Macquarie Hastings	New South Wales	Medium
Power and Water – Darwin	P&W (Darwin)	Northern Territory	Large
Power and Water – Alice Springs	P&W (Alice Springs)	Northern Territory	Small
Queanbeyan–Palerang Regional Council	Queanbeyan	New South Wales	Medium
Queensland Bulk Water Supply Authority	Seqwater	Queensland	Bulk water
Redland City Council	Redland City	Queensland	Large
Riverina Water County Council	Riverina Water (W)	New South Wales	Medium
Rockhampton Regional Council	Fitzroy River Water	Queensland	Medium
Rous Water	Rous Water	New South Wales	Bulk water
SA Water Corporation	SA Water	South Australia	Major
Shoalhaven City Council	Shoalhaven	New South Wales	Large
Snowy Monaro Regional Council	Snowy Monaro	New South Wales	Small
South East Water Corporation	South East Water	Victoria	Major
South Gippsland Water	South Gippsland Water	Victoria	Medium
Southern Downs Regional Council	Southern Downs	Queensland	Small
Sydney Water Corporation	Sydney Water	New South Wales	Major
Tamworth Regional Council	Tamworth	New South Wales	Medium
TasWater	TasWater	Tasmania	Major
Toowoomba Regional Council	Toowoomba	Queensland	Large
Townsville City Council	Townsville	Queensland	Large
Tweed Shire Council	Tweed	New South Wales	Medium
Unitywater	Unitywater	Queensland	Major
Urban Utilities	Urban Utilities	Queensland	Major
Wagga Wagga Council	Wagga Wagga (S)	New South Wales	Medium
Wannon Water	Wannon Water	Victoria	Medium
Water Corporation – Perth	WC (Perth)	Western Australia	Major
Water Corporation – Albany	WC (Albany)	Western Australia	Small
Water Corporation – Australind/Eaton	WC (Australind/Eaton)	Western Australia	Small
Water Corporation – Bunbury (S)	WC (Bunbury) (S)	Western Australia	Small
Water Corporation – Busselton (S)	WC (Busselton) (S)	Western Australia	Small
Water Corporation – Geraldton	WC (Geraldton)	Western Australia	Small
Water Corporation – Kalgoorlie–Boulder (W)	WC (Kal–Boulder) (W)	Western Australia	Small
Water Corporation – Mandurah	WC (Mandurah)	Western Australia	Large
WaterNSW	WaterNSW	New South Wales	Bulk water
Western Downs Regional Council	Western Downs	Queensland	Small
Westernport Water	Westernport Water	Victoria	Small
Whitsunday Regional Council	Whitsunday	Queensland	Small

Utility name	Display name in tables and graphs	Jurisdiction	Utility size group
Wingecarribee Shire Council	Wingecarribee	New South Wales	Medium
Yarra Valley Water Corporation	Yarra Valley Water	Victoria	Major

Appendix D Urban performance indicators

Indicator category	Indicator subcategory	Indicator code	Indicator name
Assets	Water treatment plants	A1	Number of water treatment plants providing full treatment (plants)
Assets	Other water assets	A2	Length of water mains (km)
Assets	Other water assets	A3	Number of properties served per km of water main (properties/km)
Assets	Wastewater assets	A4	Number of wastewater treatment plants (plants)
Assets	Wastewater assets	A5	Length of sewer mains and channels (km)
Assets	Wastewater assets	A6	Number of properties served per km of sewer main (properties/km)
Assets	Water main breaks	A8	Number of water main breaks, bursts and leaks per 100 km of water mains (mains breaks/100 km)
Assets	Water losses	A9	Infrastructure leakage index (ILI)
Assets	Water losses	A10	Real losses: service connections (L/service connection/day)
Assets	Water losses	A11	Real losses: water mains (kL/km water main/day)
Assets	Wastewater breaks and chokes	A14	Number of sewer mains breaks and chokes per 100 km (breaks and chokes/100 km)
Assets	Wastewater breaks and chokes	A15	Number of property connection sewer breaks and chokes per 1,000 properties (breaks and chokes/1,000 properties)
Assets	Water main breaks	IA8	Number of water main breaks, bursts and leaks (mains breaks)
Customers	Connected properties and population	C1	Population receiving services: water supply (population 000s)
Customers	Connected properties and population	C2	Number of connected residential properties: water supply (properties 000s)
Customers	Connected properties and population	C3	Number of connected non-residential properties: water supply (properties 000s)
Customers	Connected properties and population	C4	Total number of connected properties: water supply (properties 000s)
Customers	Connected properties and population	C6	Number of connected residential properties: wastewater (properties 000s)
Customers	Connected properties and population	C7	Number of connected non-residential properties: wastewater (properties 000s)
Customers	Connected properties and population	C8	Total number of connected properties: wastewater (properties 000s)
Customers	Water quality complaints	C9	Number of water quality complaints per 1,000 properties: water supply (complaints/1,000 properties)
Customers	Water service complaints	C10	Number of water service complaints per 1,000 properties (complaints/1,000 properties)
Customers	Wastewater service complaints	C11	Number of sewerage service complaints per 1,000 properties (complaints /1,000 properties)
Customers	Billing and account complaints	C12	Number of billing and account complaints per 1,000 properties: water supply and sewerage (complaints/1,000 properties)

Customers	Average duration of unplanned water supply interruptions	C15	Average duration of an unplanned interruption: water supply (minutes)
Customers	Water interruption frequency	C17	Number of unplanned interruptions per 1,000 properties: water supply (interruptions/1,000 properties)
Customers	Restrictions or legal action for non-payment of water bill	C18	Number of restrictions for non-payment of water bills per 1,000 properties (restrictions/1,000 properties)
Customers	Water quality complaints	IC9	Number of water quality complaints: water supply (complaints)
Customers	Water service complaints	IC10	Number of water service complaints (complaints)
Customers	Wastewater service complaints	IC11	Number of sewerage service complaints (complaints)
Customers	Billing and account complaints	IC12	Number of billing and account complaints: water supply and sewerage (complaints)
Customers	Total water and wastewater complaints	IC13	Number of water and sewerage complaints (complaints)
Customers	Water interruption frequency	IC17	Number of unplanned interruptions: water supply (interruptions)
Customers	Restrictions or legal action for non-payment of water bill	IC18	Number of restrictions for non-payment of water bills (restrictions)
Customers	Restrictions or legal action for non-payment of water bill	IC19	Number of legal actions taken for non-payment of water bills (legal actions)
Environment	Comparative wastewater treatment levels	E1	Percentage of wastewater only treated to a primary level (%)
Environment	Comparative wastewater treatment levels	E2	Percentage of wastewater only treated to a secondary level (%)
Environment	Comparative wastewater treatment levels	E3	Percentage of wastewater treated to a tertiary level (%)
Environment	Net greenhouse gas emissions	E8	Percentage of biosolids reused (%)
Environment	Net greenhouse gas emissions	E9	Net greenhouse gas emissions per 1,000 properties: water supply (t CO_2 equivalents/1,000 properties)
Environment	Net greenhouse gas emissions	E9.1	Net greenhouse gas emissions per ML: water supply – bulk utility (t CO_2 equivalents/ML)
Environment	Net greenhouse gas emissions	E10	Net greenhouse gas emissions per 1,000 properties: wastewater (t CO_2 equivalents/1,000 properties)
Environment	Net greenhouse gas emissions	E10.1	Net greenhouse gas emissions per ML: wastewater – bulk utility (t CO_2 equivalents/ML)
Environment	Net greenhouse gas emissions	E11	Net greenhouse gas emissions per 1,000 properties: other (t CO_2 equivalents/1,000 properties)
Environment	Net greenhouse gas emissions	E11.1	Net greenhouse gas emissions per ML: other – bulk utility (t CO_2 equivalents/ML)
Environment	Net greenhouse gas emissions	E12	Total net greenhouse gas emissions per 1,000 properties (t CO_2 equivalents/1,000 properties)
Environment	Net greenhouse gas emissions	E12.1	Total net greenhouse gas emissions per ML: bulk utility (t CO_2 equivalents/ML)

Environment	Comparative wastewater treatment levels	IE1	Volume of wastewater only treated to a primary level (ML)
Environment	Comparative wastewater treatment levels	IE2	Volume of wastewater only treated to a secondary level (ML)
Environment	Comparative wastewater treatment levels	IE3	Volume of wastewater treated to a tertiary level (ML)
Environment	Net greenhouse gas emissions	IE9	Net greenhouse gas emissions: water supply (t CO_2 equivalents)
Environment	Net greenhouse gas emissions	IE10	Net greenhouse gas emissions: wastewater (t CO_2 equivalents)
Environment	Net greenhouse gas emissions	IE11	Net greenhouse gas emissions: other (t CO_2 equivalents)
Environment	Net greenhouse gas emissions	IE12	Total net greenhouse gas emissions (t CO_2 equivalents)
Finance	Revenue	F1	Total revenue: water supply ($000s)
Finance	Revenue	F2	Total revenue: wastewater ($000s)
Finance	Revenue	F3	Total income for the utility ($000s)
Finance	Revenue	F4	Percentage of residential revenue from usage charges: water supply (%)
Finance	Revenue	F5	Revenue per property: water supply ($/property)
Finance	Revenue	F5.1	Revenue per ML: water supply – bulk utility ($/ML)
Finance	Revenue	F6	Revenue per property: wastewater ($/property)
Finance	Revenue	F6.1	Revenue per ML: wastewater – bulk utility ($/ML)
Finance	Revenue	F7	Total income per property ($/property)
Finance	Revenue	F7.1	Total income per ML: bulk utility ($/ML)
Finance	Revenue from community service obligations (CSOs)	F8	Community service obligations ratio
Finance	Costs	F9	Written-down replacement cost of fixed water supply assets ($000s)
Finance	Costs	F10	Written-down replacement cost of fixed wastewater assets ($000s)
Finance	Costs	F11	Operating cost per property: water supply ($/property)
Finance	Costs	F12	Operating cost per property: wastewater ($/property)
Finance	Capital expenditure	F14	Capital expenditure: water supply ($000s)
Finance	Capital expenditure	F15	Capital expenditure: wastewater ($000s)
Finance	Capital expenditure	F16	Total capital expenditure: water supply and wastewater ($000s)
Finance	Economic real rate of return	F17	Economic real rate of return: water supply
Finance	Economic real rate of return	F18	Economic real rate of return: wastewater
Finance	Economic real rate of return	F19	Economic real rate of return: water supply and wastewater
Finance	Dividends	F20	Dividend ($000s)
Finance	Dividends	F21	Dividend payout ratio
Finance	Net debt to equity	F22	Net debt to equity ratio
Finance	Interest cover	F23	Interest cover ratio

Finance	Dividends	F24	Net profit after tax (NPAT) ($000s)
Finance	Community service obligations (CSOs)	F25	Community service obligation ($000s)
Finance	Capital works grants	F26	Capital works grants: water supply ($000s)
Finance	Capital works grants	F27	Capital works grants: wastewater ($000s)
Finance	Capital expenditure	F28	Capital expenditure per property: water supply ($/property)
Finance	Capital expenditure	F29	Capital expenditure per property: wastewater ($/property)
Finance	Capital works grants	F30	Net profit after tax (NPAT) ratio
Finance	Costs	IF11	Operating cost: water supply ($000s)
Finance	Costs	IF12	Operating cost: wastewater ($000s)
Public Health	Water quality compliance	H1	Water quality guidelines (provided as text)
Public Health	Water quality compliance	H3	Percentage of population where microbiological compliance was achieved (%)
Public Health	Water quality compliance	H4	Number of zones where chemical compliance was achieved (zones)
Public Health	Water quality compliance	H5	Risk-based drinking water management plan externally assessed (yes/no)
Pricing	Residential tariff structure	P1	Tariff structure: water supply (provided as text)
Pricing	Residential tariff structure	P1.2	Fixed charge: water supply ($)
Pricing	Residential tariff structure	P1.3	Usage charge: step 1 ($/kL)
Pricing	Residential tariff structure	P1.4	Usage charge: step 2 ($/kL)
Pricing	Residential tariff structure	P1.5	Usage charge: step 3 ($/kL)
Pricing	Residential tariff structure	P1.6	Usage charge: step 4 ($/kL)
Pricing	Residential tariff structure	P1.7	Usage charge: step 5 ($/kL)
Pricing	Residential tariff structure	P1.12	Special levies: water supply ($)
Pricing	Residential tariff structure	P1.13	Income from special levies retained by the utility: water supply (yes/no)
Pricing	Residential tariff structure	P1.3a	Upper bound of usage: step 1 (kL)
Pricing	Residential tariff structure	P1.4a	Upper bound of usage: step 2 (kL)
Pricing	Residential tariff structure	P1.5a	Upper bound of usage: step 3 (kL)
Pricing	Residential tariff structure	P1.6a	Upper bound of usage: step 4 (kL)
Pricing	Residential tariff structure	P1.7a	Upper bound of usage: step 5 (kL)
Pricing	Annual bill	P2	Annual residential bill based on 200 kL per annum: water supply ($)
Pricing	Annual bill	P3	Typical residential bill: water supply ($)
Pricing	Residential tariff structure	P4	Tariff structure: wastewater (provided as text)
Pricing	Residential tariff structure	P4.1	Fixed charge: wastewater ($)

Pricing	Residential tariff structure	P4.2	Usage charge: wastewater ($/kL)
Pricing	Residential tariff structure	P4.3	Special levies: wastewater ($)
Pricing	Residential tariff structure	P4.4	Income from special levies retained by the utility: wastewater (yes/no)
Pricing	Annual bill	P5	Annual residential bill based on 200 kL per annum: wastewater ($)
Pricing	Annual bill	P6	Typical residential bill: wastewater ($)
Pricing	Annual bill	P7	Total annual residential bill based on 200 kL per annum: water supply and wastewater ($)
Pricing	Annual bill	P8	Total typical residential bill: water supply and wastewater ($)
Water Resources	Sources	W1	Volume of water sourced from surface water (ML)
Water Resources	Sources	W2	Volume of water sourced from groundwater (ML)
Water Resources	Sources	W3.1	Volume of water sourced from desalinated marine water (ML)
Water Resources	Transfers	W5	Total volume of water received from other service providers or operational areas within the urban water system (ML)
Water Resources	Transfers	W5.3	Volume of water, excluding recycled water, received from other service providers or operational areas within the urban water supply system (ML)
Water Resources	Transfers	W6	Volume of recycled water received from other service providers or operational areas within the urban water supply system (ML)
Water Resources	Sources	W7	Total volume of water sourced (ML)
Water Resources	Use	W8	Total volume of water supplied to residential customers (ML)
Water Resources	Use	W8.3	Volume of water supplied to residential customers (ML)
Water Resources	Use	W9	Total volume of water supplied to non-residential customers (ML)
Water Resources	Use	W9.3	Volume of water supplied to non-residential customers (ML)
Water Resources	Use	W10.1	Volume of non-revenue water (ML)
Water Resources	Production	W11	Total volume of urban water supplied (ML)
Water Resources	Production	W11.3	Volume of potable water produced for supply into the urban water supply system (ML)
Water Resources	Use	W12	Average volume of residential water supplied per property (kL/property)
Water Resources	Use	W13	Volume of water returned as environmental flows from outside of the urban water supply system (ML)
Water Resources	Transfers	W14.3	Volume of water, excluding recycled water, exported to other service providers or operational areas within the urban water supply system (ML)
Water Resources	Transfers	W15	Volume of recycled water exported to other service providers or operational areas within the urban water supply system (ML)
Water Resources	Wastewater collected	W16	Volume of wastewater, excluding trade waste, collected (ML)
Water Resources	Wastewater collected	W17	Volume of trade waste collected (ML)

Water Resources	Wastewater collected	W18	Total volume of wastewater collected (ML)
Water Resources	Transfers	W18.1	Volume of wastewater exported to other service providers or operational areas within the urban wastewater system (ML)
Water Resources	Transfers	W18.2	Volume of wastewater received from other service providers or operational areas within the urban wastewater system (ML)
Water Resources	Extraction for sewer mining	W18.3	Volume of wastewater taken through sewer mining (ML)
Water Resources	Inflow to plant	W18.4	Volume of wastewater inflow to wastewater treatment plants (ML)
Water Resources	Outflow from plant	W18.5	Volume of treated effluent outflow from wastewater treatment plants (ML)
Water Resources	Wastewater collected	W19	Average volume of wastewater collected per property (kL/property)
Water Resources	Use	W21	Volume of recycled water supplied to non-residential customers (ML)
Water Resources	Use	W23	Volume of recycled water supplied as environmental flows (ML)
Water Resources	Use	W25.1	Volume of recycled water supplied to managed aquifer recharge (ML)
Water Resources	Use	W26	Total volume of recycled water supplied (ML)
Water Resources	Use	W27	Recycled water as a percentage of total wastewater collected
Water Resources	Use	W28.4	Volume of urban stormwater supplied to residential customers (ML)
Water Resources	Use	W28.5	Volume of urban stormwater supplied to non-residential customers (ML)
Water Resources	Use	W29	Volume of treated wastewater disposals (ML)
Water Resources	Outflow from plant	W30	Volume of wastewater losses and discharges (ML)
Water Resources	Use	W31	Volume of water returned to surface water or groundwater from the urban water supply system (ML)
Water Resources	Use	W20	Volume of recycled water supplied to residential customers (ML)

Appendix E CPI indexation

Period	CPI-weighted average	Change from previous period	Change applied to values
2023–24	137.0	4.2	No change
2022–23	131.4	7.0	1.043
2021–22	122.8	4.4	1.115
2020–21	117.5	1.6	1.166
2019–20	115.7	1.3	1.184
2018–19	114.1	1.6	1.200
2017–18	112.3	1.9	1.220
2016–17	110.2	1.8	1.243
2015–16	108.3	1.4	1.265
2014–15	106.8	1.7	1.283
2013–14	105.0	2.6	1.305
2012–13	102.3	2.3	1.339
2011–12	100.0	2.4	1.370
2010–11	97.7	3.1	1.402
2009–10	94.8	2.4	1.445
2008–09	92.6	3.1	1.479
2007–08	89.8	3.3	1.525
2006–07	86.9	3.0	1.576
2005–06	84.4	3.2	1.623
2004–05	81.8	2.4	1.675
2003–04	79.9	2.4	1.714

Appendix F Jurisdictional summaries

Jurisdictional summaries are provided to document the institutional arrangements within each state and territory for the planning and management of water supply and wastewater services. These summaries are written by the states and territories and updated annually.

F1 Australian Capital Territory

F1.1 Introduction

The ACT Government has several water management roles and responsibilities including monitoring, reporting, regulation and policy development. In 2022, the ACT Government established the Office of Water within the Environment, Planning and Sustainable Development Directorate to provide a single point of contact within government on water management issues. The Office of Water leads in water policy and planning functions, along with the implementation of a suite of governance reforms to improve coordination, accountabilities and capability of the ACT water sector. The Office of Water has worked closely with the Commonwealth Government on key water reform initiatives through the National Water Reform Committee and under the Murray–Darling Basin Plan.

Reporting and compliance obligations for the ACT water sector exist under the following national and Territory legislation: *Water Act 2007*, *Corporations Act 2001*, and *Privacy Act 1988*, the *Independent Competition and Regulatory Commission Act 1997*, *Territory-Owned Corporations Act 1990*, *Work Safety Act 2008*, *Utilities Act 2000*, *Water Resources Act 2007, Environment Protection Act 1997*, *Water and Sewerage Act 2000* (for plumbing and sanitation services), and *Public Health Act 1997*.

F1.2 Water utilities in the ACT

Icon Water is the only water and wastewater utility in the ACT and is owned by the ACT Government. It manages the region's network of dams, water treatment plants, sewage treatment plants, reservoirs, water and sewage pumping stations, pipes and other related infrastructure – an asset base valued at around $3.4 billion. Icon Water also manages an investment, valued at around $1.0 billion, in the ActewAGL joint venture operating in the energy sector. This investment is managed through 2 subsidiary companies, Icon Retail Investments Limited and Icon Distribution Investments Limited.

Icon Water provides water and wastewater services to approximately 200,000 connected properties, with over 3,400 km of water mains. Icon Water also provided water services to Queanbeyan–Palerang Regional Council under the Queanbeyan Water Supply Agreement 2008.

F1.3 Operation and regulation of water utilities

The ACT Health Directorate regulates water quality under the Territory's *Public Health Act 1997*. Water quality testing is undertaken in accordance with the Australian Drinking Water Guidelines 2011. Icon Water achieved 100 compliance with the Public Health (Drinking Water) Code of Practice (2007) in 2022–23.[13] Icon Water also published its Annual drinking water quality report 2022–23 in accordance with the code in 2023.

[13] Icon Water Limited 2023, *2022–23 Annual report to the ACT Government*, Canberra

The *Utilities Act 2000* provides for the Independent Competition and Regulatory Commission (ICRC) to issue licences and determine industry codes. The ICRC determines price directions for water utilities and regulates access agreements. A new price direction was issued in May 2023 for the next price path period up to 30 June 2028.[14] An annual fee adjustment is made for water and sewerage services, taking into account the forecast rate of inflation and the significant increase in Icon Water's capital expenditure program (particularly for sewerage services). Prices are updated annually for actual inflation, the cost of debt and any approved pass-through costs.

Icon Water's activity is also regulated by the Utilities Technical Regulator (UTR) under the *Utilities (Technical Regulation) Act 2014*, and the Environment Protection Authority under the *Water Resources Act 2007* and the *Environment Protection Act 1997*.

F1.4 Performance reporting

The 2023–24 to 2026–27 Business Strategy sets out Icon Water's strategic objectives, priorities and performance indicators. The statement was approved by the voting Shareholder in May 2023. Quarterly reports of progress on the priorities and performance indicators are provided to the voting Shareholders throughout the year.

The Icon Water annual report to the ACT Government 2022–23 was provided to the ACT Government in October 2023. The report provides information on the strategic priorities, performance indicators and financial performance.

Annual performance and compliance reports are also provided to ICRC and UTR.

F2 New South Wales

F2.1 Introduction

In New South Wales, urban water supply and sewerage services are provided by 3 state-owned water utilities, 93 local water utilities (LWUs)[15] and a number of privately owned utilities.

Various regulatory agencies have responsibility for the establishment and operation of the water utilities. The Independent Pricing and Regulatory Tribunal (IPART) is the licence-compliance and price regulator for the 3 major state-owned water utilities in New South Wales: Hunter Water Corporation, Sydney Water Corporation, and WaterNSW (bulk water services in metropolitan and regional New South Wales). IPART is also the licence-compliance regulator for private water utilities in NSW and the price regulator that determines the maximum prices that 2 regional LWUs – Central Coast Council and Essential Energy (Broken Hill) – and one private water utility (the Sydney Desalination Plant) can charge their customers for the provision of water services.

The NSW Department of Climate Change, Energy, the Environment and Water (NSW DCCEEW) oversees and monitors water utility performance and is the primary policy maker for all water utilities and regulator for the 93 regional LWUs, which provide services to a total urban population of 2.04 million (with coverage of 98 for water supply and 96 for sewerage). The infrastructure current replacement cost for regional LWUs is $27.5 billion and annual revenue is $1.5 billion as reported last financial year.

[14] ICRC 2023, *Regulated water and sewerage services 2023–28*, Final Report, Canberra

[15] In addition to 92 LWUs in regional New South Wales, Hawkesbury City Council provides sewerage services only within Sydney Water's area of operation.

A number of other agencies, including NSW Health, the NSW Environment Protection Authority (EPA), the Office of Local Government, SafeWork NSW, the Natural Resource Access Regulator and Dam Safety NSW, are each responsible for aspects of the regulation of New South Wales water utilities.

The state's water utilities have obligations under Australian and New South Wales legislation, including the *Australian Government's Corporations Act 2001*, *Privacy Act 1988*, and *Water Act 2007*, and the following New South Wales legislation: *Water Management Act 2000*, *Water Act 1912*, *Protection of the Environment Operations Act 1997*, *Independent Pricing and Regulatory Tribunal Act 1992*, *Environmental Planning and Assessment Act 1979*, *State Owned Corporations Act 1989*, *Dams Safety Act 2015*, *Local Government Act 1993*, *Fisheries Management Act 1994*, *Public Health Act 2010*, *Fluoridation of Public Water Supplies Act 1957*, *Work Health and Safety Act 2011*, *Public Finance and Audit Act 1983*, *Water Industry Competition Act 2006*, *Hunter Water Act 1991*, *Sydney Water Act 1994*, and the *Water NSW Act 2014*.

F2.2 Establishment of water utilities

The 3 New South Wales state-owned utilities, (Sydney Water Corporation, WaterNSW, and Hunter Water Corporation), are created by and derive their responsibilities and areas of operations from their respective Acts (the *Sydney Water Act*, the *Water NSW Act*, and the *Hunter Water Act*) and operate as major utilities under the *Water Management Act*.

The 90 LWUs derive their responsibilities from and operate mainly under the *Local Government Act*. Four LWUs (Central Coast Council, Essential Energy, WaterNSW [but only in relation to the Fish River water supply scheme] and Cobar Water Board) operate as water supply authorities under the *Water Management Act*.

F2.3 Operation of water utilities

The regulatory oversight of water utilities in New South Wales is shared between different agencies. IPART regulates operating licences that have been issued to Sydney Water Corporation (under Part 5 of the *Sydney Water Act*), Hunter Water Corporation (under Part 5 of the *Hunter Water Act*), and WaterNSW (under Part 2 of the *Water NSW Act*). The operating licences include obligations relating to water quality, water conservation, performance standards, environmental/catchment management, asset management, customer relations, compliance, and performance reporting. IPART also determines the maximum prices these utilities can charge their customers for water services.

IPART conducts operational audits each year. These identify any areas of non-compliance and make recommendations to improve performance. It also undertakes end-of-term reviews of operating licences and makes recommendations to the relevant minister on the terms for renewal of the licences.

The NSW DCCEEW is the primary regulator of LWUs, under sections 56ff and 409(6) of the *Local Government Act* and administers the New South Wales Government's comprehensive Regulatory and Assurance Framework for Local Water Utilities. The legislation and the framework are the key policy and regulatory framework for strategic service planning, pricing, performance reporting and continuing performance improvement of the LWUs. A LWU paying a dividend to council's general fund needs to comply with section 4 of the Framework.

Under section 5 of the Regulatory and Assurance Framework for LWUs the NSW DCCEEW establishes expectations on, and provides assurance of, the effectiveness of LWU strategic planning.

Proposed construction or modification of water or sewage treatment works or for the development of a water-recycling system by LWUs in New South Wales requires ministerial approval under section 60 of the *Local Government Act*. The approval ensures that the proposed infrastructure is fit for purpose and can manage relevant risks. Under section 61 of the *Local Government Act*, the NSW DCCEEW conducts regular inspections of LWU treatment works and provides feedback and mentoring to the LWU operators.

The NSW DCCEEW conducts operator-training courses for LWU water and sewage treatment works operators.

The annual performance of each of the LWUs activities and outcomes is publicly reported via the interactive performance monitoring data dashboard and the NSW Water supply and sewerage benchmarking reports.

NSW Health regulates water quality in New South Wales and administers functions relating to water suppliers (Sydney Water Corporation, Hunter Water Corporation, and the regional LWUs) under the *Public Health Act*. NSW Health also enters into memoranda of understanding with the metropolitan water utilities (including WaterNSW) to facilitate interaction between the agencies and to establish the scope of drinking water management plans and procedures for communicating the results of water quality programs. NSW Health also conducts the NSW Drinking Water Quality Program, which tests and monitors the water quality of samples collected by the LWUs in accordance with the Australian Drinking Water Guidelines 2011.

Under the *Public Health Act*, each water supplier needs to prepare and implement a risk-based drinking water management system in accordance with the Australian Drinking Water Guidelines 2011. The water quality management system is regulated by NSW Health.

F2.4 Water utilities in New South Wales

Sydney Water Corporation, a statutory corporation wholly owned by the New South Wales Government, is Australia's largest water utility, with an area of operations covering almost 13,000 km^2. It provides drinking water, recycled water, wastewater services, and some stormwater services to more than 5 million people in Sydney, the Illawarra, and the Blue Mountains. Drinking water is sourced from a network of dams managed by WaterNSW, from the Hawkesbury River, and from the desalination plant at Kurnell before it is treated and delivered to customers.

WaterNSW is a state-owned corporation established in 2015 by the *Water NSW Act* through the merging of State Water Corporation and the Sydney Catchment Authority. WaterNSW supplies raw water in bulk. The urban component of WaterNSW reporting is based on the former Sydney Catchment Authority area of operations as defined in its operating licence and includes catchments in the Blue Mountains, Shoalhaven, Warragamba, upper Nepean, and Woronora areas. WaterNSW also provides bulk water services in regional New South Wales and operates the Fish River water supply scheme.

Hunter Water Corporation is a wholly state-owned corporation providing drinking water, recycled water, wastewater, and some stormwater services to more than 600,000 people in the lower Hunter region. The Hunter Water area of operations covers the local government areas of

Cessnock, Lake Macquarie, Maitland, Newcastle, Port Stephens and Dungog and parts of Singleton.

Regional New South Wales currently has 93 LWUs. Ninety of these LWUs are either general purpose local government councils or county councils, which are separate local government entities for the specific purpose of providing urban water services. Other LWUs operate as water supply authorities under the *Water Management Act*, including the Cobar Water Board, Essential Energy and WaterNSW for the Fish River Water Supply.[16] The 93 regional LWUs in New South Wales range in area from 285 km^2 (Orange City Council) to over 50,000 km^2 (Central Darling Shire Council), while the population served ranges from 1,000 (Central Darling Shire Council) to over 340,000 (Central Coast Council). There are 28 LWUs that serve 10,000 or more connected properties.

Performance monitoring and reporting are considered important for public accountability and have been strongly endorsed by the New South Wales Government, IPART and the Productivity Commission.[17]

The state-owned water utilities are required to report on the performance indicators in their operating licences and this reporting is audited through the operating licence audit. The operational audit report is presented to the responsible minister. These utilities also report the National Water Initiative (NWI) performance indicators required for the Urban NPR.

IPART audits roughly one-third of the auditable NWI indicators each year. The audit is conducted concurrently with the annual operating licence audits.

LWUs are required to annually report the fair value and the current replacement cost depreciation of their water supply and sewerage assets in their audited annual financial statements.

NSW DCCEEW annually reports the performance of all the New South Wales utilities by way of its NSW Performance Monitoring and Reporting System.

- The audit of the 30 NWI financial performance indicators is affected by the annual independent audit of the financial statements of each of the 93 regional LWUs.
- All the auditable non-financial performance indicators are independently audited every 3 years for each of the 28 regional NSW utilities that are required to report nationally.

The remainder of the information reported in the NSW Performance Monitoring and Reporting System is not independently audited. However, to assure data accuracy and reliability, the data are subject to a comprehensive quality assurance process.

The NSW Performance Monitoring and Reporting System functions as a ‘one stop shop’ to minimise red tape and to avoid duplication in reporting. The NSW Performance Monitoring and Reporting System provides LWU performance data to the Bureau of Meteorology annually (for the Urban NPR) and the Australian Bureau of Statistics, as well as for the NSW State of the Environment Report.

Private water utilities

[16] Councils exercising water supply and/or sewerage functions do so under the *Local Government Act*, including under a division generally administered by the minister responsible for water. Central Coast Council exercises its functions under both the *Local Government Act* and as a water supply authority under the *Water Management Act*. The Cobar Water Board, Essential Energy and WaterNSW (for the Fish River Water Supply) carry out their functions under the *Water Management Act*.

[17] Productivity Commission 2011, *Australia's urban water sector*, Report No. 55, Final Inquiry Report, Canberra.

In New South Wales, private utilities can also provide water and sewerage services if licensed under the *Water Industry Competition Act (WIC Act)*. The *WIC Act* is designed to encourage competition in the supply of water and wastewater services and facilitate private sector delivery of recycled water infrastructure. The Act sets out a licensing and compliance framework and establishes strict rules to ensure drinking water meets Australian standards, recycled water is 'fit for purpose', and all services are delivered safely, reliably and with minimal environmental impacts.

The Minister for Water administers the *WIC Act* and is advised by the NSW DCCEEW. IPART administers the licensing regime, which includes assessing licence applications and making recommendations to the Minister on whether to grant a licence, as well as auditing and enforcing licences.

The *WIC Act* commenced operation on 8 August 2008. There are 20 private water schemes licensed to operate in NSW. As of 30 June 2024, these schemes were providing services to 8,347 drinking water customers, 10,930 recycled water customers and 10,700 sewage customers. Comprehensive reforms of the licensing framework were enacted in 2021 and commenced on 1 March 2024 along with the supporting regulations. These reforms include:

- narrowing the focus of regulation on higher-risk utility-like schemes such as large-scale drinking water production, recycling and sewerage schemes or those servicing 30 or more residential or small business customers
- separating the licensing of operators and retailers from the approval of individual schemes so operators don't need to apply for a new licence for each scheme they operate
- requiring more rigorous assessment of the financial viability of proposed schemes to check their longevity and affordability and minimise the risk of last-resort events
- strengthening customer protection by introducing a deemed standard customer contract and new operator of last-resort arrangements to protect customers if an operator should fail financially.

Existing schemes are required to be transitioned to the new licensing framework by 1 March 2025.

F3 Northern Territory

F3.1 Introduction

The Northern Territory's *Water Supply and Sewerage Services Act 2000* provides the regulatory framework for the territory's water and sewerage industry. The NT Department of Treasury and Finance and the Northern Territory Treasurer are responsible for administering this Act insofar as it relates to economic regulation; the Northern Territory Minister for Essential Services in terms of its relationship to licensed supply and service; and the Department of Health (NT) in terms of its relationship to water quality standards.

The objectives of the *Water Supply and Sewerage Services Act* are:

- to promote the safe and efficient provision of water supply and sewerage services
- to establish and enforce standards of service in water supply and sewerage services
- to facilitate the provision of financially viable water supply and sewerage services
- to protect the interests of customers.

Among other things, this Act provides for the following:

- that the supply of water and sewerage services be licensed, and that licences issued by the Utilities Commission are for defined, gazetted, geographical areas
- that the Minister be responsible for the declaration of water supply and sewerage service licence areas (by notice in a government gazette).

Power and Water Corporation (the licensed utility) is subject to water quality monitoring programs and emergency directions issued by the Chief Health Officer (Department of Health).

The Northern Territory Utilities Commission is the independent industry regulator. It has responsibility for the licensing functions conferred by the *Water Supply and Sewerage Services Act*.

Statutory conditions of water and sewerage licences issued under this Act include:

- that the licensee monitors and reports to the Utilities Commission on compliance with the licence
- that the licensee procures an audit, if required by the Utilities Commission, of its compliance with the terms of the licence.

The *NT Water Act 1992* is another major piece of legislation pertaining to the regulation of water in the territory. This Act provides for the investigation, allocation, use, control, protection, management and administration of water resources, and for related purposes. The *Water Act* also allows for the issue of waste discharge licences and water extraction licences by the Controller of Water Resources (Department of Lands, Planning and Environment [NT]).

F3.2 Operation of water utilities

Power and Water Corporation is responsible for monitoring the quality of drinking water in line with its Drinking Water Operational and Verification Monitoring Program and reports the results to the Chief Health Officer. The program is based on the Australian Drinking Water Guidelines 2011.

While Power and Water Corporation has primary responsibility for providing safe drinking water, several government agencies are also involved. The Department of Health applies the guidelines and monitors compliance with them in the interest of public health, and the Department of Lands, Planning and Environment and the Northern Territory Environment Protection Authority (NT EPA) also have roles in protecting water quality, including the regulation and management of water resources and the regulation of pollution control.

The NT Department of Lands, Planning and Environment has a major role in protecting water quality through land-use planning in the territory. In addition, NT legislation such as the *Water Act* and the *Land Acquisition Act 1978* contain provisions for infrastructure and land use relating to water supply.

A condition of the waste discharge licences issued to Power and Water Corporation is the submission to the NT EPA of annual audit and compliance reports related to environmental impacts that discharged water may cause, and the assessment of water-recycling schemes. The corporation also investigates and reports to the NT EPA on pollution incidents under the *NT Waste Management and Pollution Control Act 1998*.

Water and sewerage tariffs and charges for the Power and Water Corporation are regulated by the NT Government via a Water and Sewerage Pricing Order issued by the Treasurer as Regulatory Minister. The Utilities Commission monitors compliance with the Pricing Order and enforces it

under section 23 of the *NT Utilities Commission Act 2000*. The Commission is also required to investigate any complaints made by customers about non-compliance with the prices outlined in the order.

F3.3 Water utilities in the Northern Territory

In the Northern Territory, Power and Water Corporation's water and sewerage business is licensed and is responsible for the supply of water and sewerage services to the territory's 5 major centres (Darwin, Katherine, Tennant Creek, Alice Springs and Yulara) and 13 minor centres.

No significant distinction between urban and rural areas is made under the legislation or the licensing framework under which Power and Water Corporation operates. Geographical coordinates (latitude and longitude) define the declared water supply and sewerage service licence.

F3.4 Performance reporting

Urban NPR data is gathered within the Power and Water Corporation by a central coordinator, who collates the report with information supplied by other areas in the organisation. Some key NPR indicators are provided to Power and Water Corporation's executive management, board and shareholders on a regular basis. Performance data that is publicly available are reviewed and/or signed off at the senior management level. NPR data are signed off at the senior management level. Some NPR indicators are audited at an aggregate level.

F4 South Australia

F4.1 Introduction

The *South Australia Water Industry Act 2012* and *Water Industry Regulations 2012* establish the regulatory framework for the water and sewerage industry, covering licencing of retail services, economic regulation, technical regulation, water planning, and customer complaint handling. The Act commenced on 1 July 2012 and governs all water industry entities providing 'retail services' to SA customers.

The Essential Services Commission (the Commission) is the independent economic regulator of water and sewerage retail services in the state. The Commission's primary objective is to protect the long-term interests of SA consumers with respect to the price, quality and reliability of those services. The Commission's role includes industry licensing, consumer protection, retail pricing regulation and performance monitoring.

The Department for Environment and Water (DEW) is responsible for the management of the state's water resources through administering the *Landscape South Australia Act 2019*; it also has a role in policy development relevant to the *Water Industry Act 2012*. Regional landscape boards are responsible for developing water allocation plans for prescribed water resource areas as required by the *Landscape South Australia Act 2019*.

The Office of the Technical Regulator (OTR) and DEW share the role of jurisdictional coordinator for National Performance Reporting, with DEW taking the lead on policy and OTR responsible for all other operational matters.

The OTR, which sits within the Department for Energy and Mining, has the following main functions under the *Water Industry Act 2012*:

- development of technical standards in connection with the water industry
- monitoring and regulating technical standards with respect to water and sewerage infrastructure and associated equipment, products and materials (including on the customer's side of any connection point) and plumbing
- providing advice in relation to safety or technical standards in the water industry to the Commission at its request, and in the plumbing industry
- fulfilling any further function assigned to the Technical Regulator under the Act.

SA Health ensures that drinking water is delivered to consumers according to the requirements of the *Safe Drinking Water Act 2011* and the *Safe Drinking Water Regulations 2012*. Under the *Safe Drinking Water Act 2011*, all drinking water providers must:

- register as a drinking water provider
- implement a risk management plan
- report water quality incidents to SA Health
- provide water quality results to consumers
- undertake commission audits and inspections.

Under the *SA Public Health Act 2011* and *SA Public Health (Wastewater) Regulations 2013*, SA Health protects public health through the development and administration of prescribed codes, protocols, guidelines and public health policy for wastewater management, treatment, disposal, and recycled water use. The Australian Guidelines for Water Recycling are also applied to South Australian recycled water schemes.

SA Health assess and approve the design, installation and ongoing operation of:

- community wastewater management systems (CWMS)
- CWMS wastewater treatment plants
- recycled water supply and use from CWMS and SA Water networks
- on-site wastewater systems >40 equivalent persons.

Under the *Environment Protection Act 1993*, the SA Environment Protection Authority licences the following activity:

Schedule 1(3)(4): Resource recovery, waste disposal and related activities – Wastewater treatment.

The conduct of wastewater treatment works, being sewage treatment works, a CWMS, winery wastewater treatment works or any other wastewater treatment works with the capacity to treat, during a 12 month period –

- in the case of works located wholly or partly within a water protection area – more than 5 megalitres of wastewater; or
- in the case of works located wholly outside of a water protection area – more than 50 megalitres of wastewater.

F4.2 Water utilities in South Australia

Any person or entity providing 'water retail services' to SA customers is required to be licensed by the Commission. The Commission has determined separate regulatory obligations for major retailers (those providing retail services to 50,000 or more connections) and small-scale networks (also known as minor and intermediate water retailers, with less than 50,000 connections). SA

Water Corporation is the only major retailer in SA, and there are currently 68 other retailers (mainly council-run operations and some private businesses). SA Water Corporation is a government entity and, as the state's main supplier of urban water, is required to deliver, monitor and report on its primary functions concerning:

- supply of water by reticulated systems
- storage, treatment and supply of bulk water
- removal and treatment of wastewater.

SA Water Corporation provides drinking water to approximately 837,410 connections, servicing around 99% of the state's drinking water customers. SA Water Corporation also provides sewerage services to approximately 661,480 connections, servicing around 87% of the state's sewerage customers.

Mount Barker District Council operates the largest Council wastewater treatment plant and recycled water scheme in SA. As an intermediate retailer Mount Barker District Council provides a retail service to over 14,300 connections across 7 townships in the Adelaide Hills Region.

F4.3 Operation of water utilities

Section 35 of the *Water Industry Act 2012* empowers the Commission to make a determination under the *SA Essential Services Commission Act 2002*, regulating prices, conditions relating to prices and price-fixing factors for water retail services.

The Commission made its final revenue determination for the *South Australian Water Corporation Act 1994* in June 2024, setting maximum allowed revenues for drinking water and sewerage retail services for the 4-year period from 1 July 2024 to 30 June 2028. SA Water Corporation and the South Australian Government are responsible for setting specific prices (such as supply and usage charges for residential and non-residential customers); however, those prices must comply with the Commission's allowed revenues.

A different, proportional approach to price regulation has been applied to other water retailers through a combination of pricing principles and a price-monitoring framework.

Pursuant to Part 4 of the *Essential Services Commission Act 2002*, the Commission is empowered to make industry codes and rules regulating the conduct or operations of a regulated industry or regulated entities. The Commission has devised a Water Retail Code for major retailers that sets out the minimum requirements to be complied with by SA Water Corporation when dealing with its customers, and it includes obligations relating to customer connections and the quality, safety and reliability of water and sewerage supply. SA Water Corporation is required to meet several operational service standards relating to customer service, service interruptions and new connections.

A water retail code for minor and intermediate retailers has been devised which sets out the behavioural standards and minimum requirements to be complied with by small scale networks when engaging with their customers.

F4.4 Performance reporting

Under the Commission's Monitoring and Evaluating Performance Framework, the SA Water Corporation is required to publicly report quarterly on its performance in meeting the annual service standards and provide annual performance self-assessments, supported by a system of verified

trust and accountability around that reporting. The SA Water Corporation also includes the following measures in its public reports:

- compliance with the Australian Drinking Water Guidelines 2011
- the Water Quality Management Index
- compliance with water and sewerage services targets
- the Incident Response Index.

To supplement SA Water's reporting, the Commission publishes an annual report identifying longer-term trends and systemic issues, as well as reporting on specific events where material or major issues arise.

The Commission also publishes annually on the performance of small-scale water networks. The focus of reporting is on residential retail services, and it informs customers about the quality and reliability of the services they receive at an aggregate level. The Commission places a strong emphasis on licensees constantly monitoring performance, outcomes and trends. Small-scale water networks are expected to regularly examine, identify, report on and explain performance outcomes, to provide greater transparency to consumers and stakeholders.

F5 Tasmania

F5.1 Introduction

The key piece of legislation governing the water and sewerage industry is the *Tasmanian Water and Sewerage Industry Act 2008* (the *Industry Act*). The *Industry Act* requires any persons or entities owning and/or operating water and/or sewerage infrastructure, or supplying water and/or sewerage services to others, to be licensed, unless exempted.

The Tasmanian Water and Sewerage Corporation Pty Ltd (TasWater) is the only licensed water utility in the state.

Industry regulators for the sector are the: Tasmanian Economic Regulator (TER), responsible for licensing, price regulation and service standards; Director, Environment Protection Authority (EPA) Tasmania, responsible for regulating wastewater treatment plants; Director of Public Health, responsible for regulating water quality and fluoridation; and the Secretary of the Tasmanian Department of Natural Resources and Environment Tasmania[18], responsible for water licence allocations and regulating dam safety.

In addition to the *Industry Act*, regulatory requirements are imposed by Tasmanian legislation including the *Environmental Management and Pollution Control Act 1994*, the *Public Health Act 1997* and the *Water Management (Safety of Dams) Regulations 2015*.

F5.2 Water utilities in Tasmania

Since 1 July 2013, TasWater has owned, controlled and operated water supply and sewerage systems in Tasmania. As the only licensed water utility in the state, TasWater manages all aspects of the water supply chain, from dams and reservoirs to customer property connections, and from customer sewer connections to wastewater treatment and disposal.

[18] On 1 December 2021, the then Tasmanian Department of Primary Industries, Parks, Water and Environment became the Department of Natural Resources and Environment Tasmania.

The licensing requirements place several regulatory obligations on TasWater through reference to various regulatory instruments such as codes and guidelines, as well as requiring the preparation of management plans in relation to matters such as asset and emergency management and compliance.

TasWater's objectives, as prescribed by the *Water and Sewerage Corporation Act 2012*, are to:

- efficiently provide water and sewerage functions in Tasmania
- encourage water conservation, the demand management of water and the reuse of water on an economic and commercial basis
- be a successful business and, to this end:
  - operate its activities in accordance with good commercial practice
  - deliver sustainable returns to its members
  - deliver water and sewerage services to customers in the most cost-efficient manner.

TasWater is owned by Tasmania's 29 councils, with the Tasmanian Government also becoming a shareholder in early 2019.

F5.3 Operation of water utilities

TasWater is subject to various economic, environmental, public health and customer service regulatory requirements.

The economic regulatory framework, established under the *Industry Act*, is focused on ensuring competitive market outcomes from the sector in relation to both price and service, ensuring the financial sustainability of the water and sewerage industry, and providing sufficient funding for capital expenditure to improve compliance with regulatory obligations and ensure operational efficiencies.

EPA Tasmania administers and enforces the provisions of the *Environmental Management and Pollution Control Act*, which is principally concerned with the prevention, reduction and remediation of environmental harm. The Director of Public Health is responsible for drinking water quality and safety through the application of drinking water quality guidelines and for the fluoridation of drinking water through the application of a code of practice. The Dam Safety Regulator monitors TasWater's performance against its dam safety regulatory obligations and guideline requirements to ensure that the dams TasWater is responsible for do not pose an unacceptable level of risk to the public.

Independent regulation of water and sewerage prices in Tasmania commenced on 1 July 2012.

Price reform of the industry is designed to transition customers to a single set of tariffs across the whole state by the statutory due date of 1 July 2020 (that is, customers are required to be paying the same price for the same service by this date). As previously reported, at the end of 2021–22 a small number of customers were still to be transitioned to consistent pricing. All of these remaining customers were transitioned during 2022–23. Price reform has also introduced two-part pricing for water (a fixed charge based on the size of the connection and a variable charge reflecting metered water consumption) and sewerage charges based on the assessed equivalent tenements (that is, the estimated demand placed on the system) of each property.

F5.4 Performance reporting

One of the TER's functions is to monitor and report on the state of the Tasmanian water and sewerage industry.[19] The performance indicators reported on are based on the National Performance Reporting (NPR) Framework with some additional Tasmanian-based measures, as set out in the TER's Tasmanian Water and Sewerage Industry Performance and Information Reporting Guideline, Version 1.7.

TasWater is required to carry out regular independent audits (performance appraisals) to assess:

- its compliance with, and the adequacy of, its management and compliance plans; and
- the quality and accuracy of its regulatory information, including the procedure for reporting performance information.

The NPR Audit Handbook requires independent audits of TasWater's performance indicators to be conducted at least once every 3 years. As a result, TasWater's performance indicators are audited in 3 tranches over a 3-year period.

The third tranche of TasWater's performance indicator was audited in September 2024 (with regard to 2023–24 performance indicators). This concludes the currebt audit cycle. A new 3-year cycle will commence in 2025 (with regard to 2024–25 performance indicators).

The TER's approach to regulatory reporting is set out in its Regulatory Reporting Guideline Version 5. Its approach to managing non-compliance is outlined in its Compliance Enforcement Policy Version 3.

F6 Queensland

F6.1 Introduction

Regulation of the urban water sector is undertaken by a number of Queensland Government departments with the aim of providing communities with access to safe and reliable water and sewerage services and ensuring efficient business operations, efficient water use, water security, protection of the environment, competition and the prevention of monopoly pricing.

F6.2 Water utilities in Queensland

There are 211 registered service providers in Queensland. Most are private entities and water boards that supply non-potable water for irrigation, commercial or stock and domestic purposes.

Drinking water services are provided to communities by:

- 4 bulk water entities (Seqwater, SunWater, Gladstone and Mount Isa Water Board)
- 2 distributor-retailers in South East Queensland (SEQ) (Urban Utilities, Unitywater)
- 3 local governments in SEQ (Gold Coast, Logan and Redland City Council)
- 66 local governments outside SEQ
- 8 private providers.

Despite the large number of drinking water service providers, 95% of connected properties receive water and sewerage services from the 22 service providers that report for the National Performance Reporting Framework. The remaining 5% of connected properties mostly receive

[19] The TER's annual water and sewerage state of the industry reports are available from www.economicregulator.tas.gov.au.

services from 53 small and very small service providers. In general, the communities these providers supply services to are very small and often geographically isolated.

This variation in scale means that there are significant differences in capacity and capability across the urban water sector in Queensland.

In addition to drinking water and sewerage service provision, there are 174 registered recycled water schemes in Queensland. Most of these schemes are managed by a local government (although a small number of recycled water schemes are run by a private entity).

F6.3 Operation of water utilities

Water and sewerage service provision is regulated by the *Water Supply (Safety and Reliability) Act 2008* (*Water Supply Act*). The purpose of the *Water Supply Act* is to provide for the safety and reliability of water supply across Queensland. The Department of Local Government, Water and Volunteers is the water supply regulator under the *Water Supply Act*.

The powers and obligations for service provision, performance reporting, drinking water management, and recycled water management apply to registered service providers according to the services provided.

To protect public health, the Department of Local Government, Water and Volunteers closely monitors and regulates drinking water in Queensland.

Drinking water service providers are required to have an approved drinking water quality management plan and must comply with the plan and any conditions placed upon the plan. Water quality monitoring and reporting are integral components of the plan. Drinking water service providers are responsible for ensuring the safe supply of drinking water to the community and managing incidents that compromise drinking water quality.

The *Public Health Act 2005* and the *Public Health Regulation 2018* include provisions relating to drinking water quality. As the administrator of the *Public Health Act* and the *Public Health Regulation*, Queensland Health has:

- set specific standards for drinking water quality in the Public Health Regulation
- the power to respond when drinking water supplied by a provider may present a risk to public health or be considered unsafe.

In addition to these provisions, Queensland Bulk Water Supply Authority (Seqwater) also has specific powers and obligations under the *Water Act 2000*, and SEQ services providers (Urban Utilities, Unitywater, City of Gold Coast, Logan City Council and Redland City Council) have powers and obligations under the *South-East Queensland Water (Distribution and Retail Restructuring) Act 2009*.

Other Acts that regulate one or more aspects of service provision include the:

- *Water Act 2000*
- *Public Health Act 2005*
- *Water Fluoridation Act 2008*
- *Environmental Protection Act 1994*
- *Local Government Act 2009*
- *Planning Act 2016*
- *Plumbing and Drainage Act 2018*

- *Queensland Competition Authority Act 1997*.

F6.4 Performance reporting

Urban water service providers are required to report on their performance under the Queensland Government Key Performance Indicator (KPI) Framework in the *Water Supply Act*. The KPI Framework captures almost all drinking water and sewerage service providers in Queensland and requires them to monitor and report on key performance indicators and publish an annual performance report. Larger providers (those with over 10,000 connections) are also required to report for the National Performance Reporting framework.

F7 Victoria

F7.1 Introduction

The Victorian Department of Energy, Environment and Climate Action (DEECA) has overall governance oversight, on behalf of the Victorian Minister for Water, for the establishment of water utilities and their performance in this state. This responsibility pertains to certain aspects of water utility performance and is also shared with the Victorian departments of Treasury and Finance (DTF, regarding business financial risks), Department of Health (DoH, regarding water quality), the Victorian Environment Protection Authority (EPA, regarding environmental performance), and the Essential Services Commission (ESC, regarding price regulation and service standards).

Reporting and compliance obligations are imposed by *Victorian legislation including the Water Act 1989*, the *Water Industry Act 1994*, the *Financial Management Act 1994*, the *Safe Drinking Water Act 2003*, and the *Environment Protection Act 1970*. In addition, regulatory instruments such as the Statement of Obligations (2015), the Water Industry Regulatory Order 2014, and the State Environment Protection Policy (SEPP) (Waters of Victoria) also impose some compliance and reporting obligations.

F7.2 Establishment of water utilities

The Victorian water sector is made up of 18 water utilities constituted under the *Victorian Water Act*. The key aspects of the frameworks governing drinking water quality, environmental protection, price regulation and consumer protection are the same across all 18 water utilities.

Under section 4I of the *Water Industry Act*, water utilities are subject to statements of obligations, issued by the Minister for Water following consultation with the Treasurer and the ESC, that impose obligations in relation to the performance of their functions and the exercise of their powers.

F7.3 Operation of water utilities

Apart from DEECA, 4 other agencies jointly oversee the regulation of water utility operation in Victoria.

The DTF oversees governance of the water utilities' proposed strategic directions and business management activities in terms of their potential for financial risk to the utility and its implications for the Victorian Government, focusing on the state's budget, net debt position and credit rating.

The DoH oversees governance of water quality under the *Safe Drinking Water Act* and the *Safe Drinking Water Regulations 2005*. This provides a framework for drinking water quality that

includes risk management obligations, a set of standards for key water quality parameters, and information disclosure requirements for water utilities. The Regulations establish an auditing framework.[20] Under the legislation, the DoH is required to publish an annual water quality report that is tabled in parliament by the Victorian Minister for Health.

The EPA regulates the environmental performance of the water utilities, particularly as it relates to treated wastewater quality, through a corporate licence (previously, each wastewater treatment plant was licensed). The level of wastewater treatment required usually depends on the type of waterway into which the treated wastewater is discharged. Under the licence provisions, water utilities must regularly sample and monitor wastewater quality and advise the EPA if there are specific incidents of non-compliance. A corporate licence also includes a requirement to submit an annual performance statement to the EPA.

Most wastewater treatment plants operated by the water utilities are subject to the SEPP (Waters of Victoria) schedules, which are developed and administered by the EPA. The schedules require wastewater treatment plant operators to ensure that the sustainable reuse of treated effluent and biosolids is maximised wherever possible.

Water utilities are also subject to EPA works approval permits before construction of new treatment plants or major alterations can begin.

The ESC is responsible for price regulation and setting service standards for water services in Victoria under Part 1A of the *Water Industry Act*, the *Essential Services Commission Act 2001* and the Water Industry Regulatory Order. The legislative framework provides the ESC with powers and functions to:

- make price determinations
- regulate standards and conditions of service and supply
- require regulated businesses to provide information.

F7.4 Water utilities in Victoria

The Victorian Government owns all 18 water utilities in the state. There are 4 water utilities in metropolitan Melbourne: Melbourne Water, Greater Western Water, South East Water Ltd and Yarra Valley Water Corporation. The 3 retailers (Greater Western Water, South East Water Ltd and Yarra Valley Water Corporation) deliver retail water supply and sewerage services to customers in the Melbourne metropolitan area. The 3 retailers also provide some localised sewerage services to their customers not connected to the Melbourne sewerage network.

Melbourne Water provides bulk water and bulk sewerage services in the Melbourne metropolitan area and manages rivers, creeks and major drainage systems in the Port Phillip and Westernport regions. Melbourne Water also controls the catchment for most of its supply.

Outside Melbourne, 12 regional urban water utilities provide water and sewerage services (Barwon Water, Central Gippsland Water, Central Highlands Water, Coliban Water, East Gippsland Water, Goulburn Valley Water, GWMWater (Grampians Wimmera Mallee Water), Lower Murray Water, North East Water, South Gippsland Water, Wannon Water and Westernport Water).

[20] Details of the drinking water regulatory framework, the audit arrangements and the annual drinking water quality report are available at https://www.health.vic.gov.au/water/drinking-water-quality-annual-reports.

Lower Murray Water also provides rural water services such as irrigation and stock and domestic supplies. GWMWater and Coliban Water also provide a piped rural water service for stock and domestic use.

Additionally, 2 rural water utilities (Goulburn–Murray Water and Southern Rural Water) provide irrigation and rural water services.

Most water utilities in regional Victoria have their own bulk water supplies. Goulburn–Murray Water, Southern Rural Water and GWMWater also provide both bulk and retail services.

Although owned by the Victorian Government, all 18 water utilities act as stand-alone entities and are responsible for their own management and performance. Each water utility has a chairperson and a board of directors appointed by the Minister for Water. The board has a range of responsibilities, including:

- setting the entity's strategic direction and steering the entity
- setting objectives and performance targets
- ensuring compliance with legislation and government policy.

Public sector directors must comply with the statutory directors' duties in the *Victorian Public Administration Act 2004*, the Directors' Code of Conduct, and common law directors' duties. In addition, directors of water utilities must also comply with requirements as set out in the *Water Act*.

Each water utility's board appoints a managing director who is responsible for the day-to-day management of the water utility under delegation from the board.

Each managing director sits on the board and is the primary link between the board and the water utility's management and staff. The managing directors are responsible for communicating board priorities and policies to management and staff and for presenting reports, submissions and budgets to the board. The board of each water utility reports to the Minister for Water via DEECA. In turn, the Minister for Water is responsible for reporting to parliament on the performance of each water utility. To assist with the management of the water industry, the Minister for Water is supported by the Water and Catchments Group within DEECA.

The *Financial Management Act* is the principal legislation governing financial reporting by water utilities. The Victorian Minister for Finance (through DTF) issues financial reporting directions under the *Financial Management Act* for the preparation of annual reports. The Minister for Water issues ministerial reporting directions to water utilities for performance reporting and other specific reporting requirements as part of their annual reports. DEECA is responsible for reviewing the annual reports of the water utilities and advising the Minister for Water on tabling the reports in parliament.

The Victorian Auditor-General's Office is responsible for auditing the annual financial statements and performance reports of water utilities. Some data reported in the NPR for Victorian water utilities are either taken directly from the published annual reports or derived from the annual reports.

In accordance with the *Water Act*, each water utility must submit an annual corporate plan that provides a statement of corporate intent, lists expected activities, and provides a financial forecast for the following 5 years. The Minister for Water (through DEECA) issues guidelines to the water utilities for the preparation of the corporate plans. DEECA and DTF are responsible for reviewing

the corporate plans (and business cases for major capital projects above a threshold value) and for advising the Minister for Water and the Treasurer, respectively.

Each water utility will also make periodic price submissions to the ESC. The regulatory period of a price determination typically covers a period of between 3–5 years but can cover a longer regulatory period. Price submissions include details about proposed revenue requirements and tariffs and pricing structures and are assessed by the Essential Services Commission. The process requires extensive customer engagement by the water utilities and the ESC.

F7.5 Performance reporting

One of the ESC's regulatory functions is to monitor and report publicly on the performance of the Victorian water utilities. The ESC's annual water performance reports are available on its website (www.esc.vic.gov.au/Water/Performance-reports/).

Under the Water Industry Regulatory Order, the ESC has the function of auditing:

- the compliance of a regulated water utility with the standards and conditions of service and supply specified by the ESC in any code or set out in the utility's price determination, and the systems and processes established by the water utility to ensure such compliance
- the reliability and quality of information reported by a water utility to the ESC, and the conformity of that information with any specification issued by the ESC
- the compliance of a water utility with asset management obligations imposed in any statement of obligations issued to it.

The annual audits are an important element of the regulatory framework. They verify that the information collected and reported by water utilities is accurate and reliable and provide evidence to customers and other stakeholders that regulatory obligations are being complied with. Most Victorian data reported in the NPRs are audited under those arrangements.

The audit approach is set out in the ESC's guideline for approving, conducting and reporting audits, which is available from the ESC's website (www.esc.vic.gov.au/Water/Codes-and-Guidelines).

F8 Western Australia

F8.1 Introduction

The WA Department of Water and Environmental Regulation has prime responsibility for water resource policy, planning, management and regulation, as well as the administration of water entitlements and water rights within the state. The reporting of water utility performance is the responsibility of the Economic Regulation Authority (ERA); however, the WA Department of Health and the Western Australian Environmental Protection Authority also have some reporting responsibilities.

Reporting and compliance obligations are imposed by Australian Government legislation including the *Corporations Act 2001* and the *Privacy Act 1988*, and by Western Australian legislation including the *Water Services Act 2012*, the *Metropolitan Water Supply, Sewerage and Drainage Act 1909*, the *Health (Miscellaneous Provisions) Act 1911* (which is being replaced by the new *Public Health Act 2016* over the next few years), the *Rights in Water and Irrigation Act 1914*, the

Country Areas Water Supply Act 1997, the *Environmental Protection Act 1986* and the *Planning and Development Act 2005*.

F8.2 Establishment of utilities

In Western Australia's legislative framework, water utilities are referred to as 'water service providers'.

Under the *Water Services Act*, the ERA is the independent regulator responsible for administering the licensing scheme for water services and for reporting on industry performance. To obtain a licence, a water service provider has to demonstrate that it has the financial and technical capacity to provide the service or services that are to be covered by a licence and that the grant of the licence is not contrary to the public interest.

F8.3 Operation of water utilities

The ERA and other agencies jointly oversee the operation of water service providers in Western Australia.

The Department of Health sets standards for drinking water quality and regulates the public health aspects of water supply (both potable and non-potable), pursuant to the *Health (Miscellaneous Provisions) Act*. The department also supports the Advisory Committee for the Purity of Water, which advises the WA ministers for Health and Water on issues associated with protecting public drinking water. The department has recently introduced new standards that regulate the public health aspects of sewerage services pursuant to the *Health (Miscellaneous Provisions) Act*.

The licence terms and conditions for service providers that supply drinking water require them to enter into a memorandum of understanding with the Department of Health for compliance with the health-related criteria in the Australian Drinking Water Guidelines.

The licence terms and conditions for service providers that provide sewerage services to more than 400 connections require them to enter into a memorandum of understanding with the Department of Health for compliance with the required public health standards.

Each memorandum of understanding is reviewed every 3 years, unless agreed otherwise.

The Department of Water and Environmental Regulation's responsibilities include the collection and analysis of water resources information, the protection of water quality and water resources, and water industry planning and policy, management, and regulation. The department also regulates the environmental impacts of water service providers through the *Environmental Protection Act*. The Act prescribes an environmental registration and licensing scheme, which sets limits on the type and volume of waste that can be discharged from a site. In some circumstances, the water service providers may be required to arrange for audits of their compliance with the conditions attached to their registration and provide a copy of the audit report to the department. The water service providers must notify the department if there is an unauthorised discharge of waste from registered premises.

The Environmental Protection Authority is an independent adviser to the WA Government on a broad range of environmental matters. The functions of the authority include conducting environmental impact assessments, preparing statutory policies for environmental protection, publishing guidelines for managing environmental impacts, and providing strategic advice to the WA Minister for Environment.

The Western Australian Planning Commission, a statutory authority that operates with the support of the Department of Planning, Lands and Heritage, oversees the land-use planning implications of the operations of the water service providers, according to requirements of the *Planning and Development Act*.

Prices for drinking water and sewerage services provided by the Water Corporation, Bunbury Water Corporation (trading as Aqwest), and Busselton Water Corporation (trading as Busselton Water) are set by the Minister for Water. The WA Government may request the ERA to undertake an independent review of pricing for the water corporations to provide advice to the government (the ERA has carried out 14 inquiries related to water pricing to date). Charges for sewerage services provided by local government authorities are set for each local government area and applied using a formula that depends on the type of property. Prices charged by private water and sewerage service providers are unregulated.

The *Water Services Act* requires licensees to arrange for an operational audit and a review of asset management system effectiveness at least once every 2 years (or longer, at the ERA's discretion). Independent auditors appointed by the ERA conduct the audit and review. The ERA approves the final audit and review reports, arranges for their publication on its website, and provides a copy of each report to the WA Minister for Water.

The Water Services Code of Conduct (Customer Service Standards) 2024 prescribes the customer service standards applicable to water and sewerage licensees. The Code is administered by the ERA in consultation with the Water Code Consultative Committee comprising representatives from industry, consumer representative organisations and government.

The *Water Services Code of Practice (Family Violence) 2020* came into effect on 8 December 2020 and prescribes minimum requirements for water service providers to support their residential customers who have been affected by family violence.

F8.4 Water utilities in Western Australia

Several water service providers are involved in delivering water supply and sewerage services in Western Australia. This report covers the larger service providers (the Water Corporation, Aqwest, Busselton Water and the City of Kalgoorlie–Boulder). There are other water supply and sewerage services that are delivered by privately owned operators and local government authorities.

The Water Corporation is a government trading enterprise operating under the *WA Water Corporations Act 1995* and the *WA Government Trading Enterprises Act 2023* that provides potable and non-potable water, bulk water, sewerage services, and drainage services to most areas of Western Australia. It also undertakes catchment management activities under delegation from the Department of Water and Environmental Regulation according to an operational agreement for catchment management between the 2 organisations. The Water Corporation is the principal supplier of water, sewerage and drainage services to hundreds of thousands of homes, businesses and farms, and provides bulk water to farms and growers' cooperatives for irrigation. Its services, projects and activities span more than 2.5 million km^2. It has regional offices in Perth, Bunbury, Albany, Karratha, Geraldton, Northam and Kalgoorlie.

Aqwest and Busselton Water are government trading enterprises operating under the *WA Water Corporations Act* and the *WA Government Trading Enterprises Act*.

The Aqwest licence permits the supply of potable water to the greater Bunbury region.

The Busselton Water licence permits the supply of potable water to the regional centre of Busselton, approximately 200 km south of Perth, and the Busselton–Capel and Blackwater groundwater areas. Busselton Water also supplies bulk water to the Water Corporation in Dunsborough.

The state-owned corporations (Aqwest, Busselton Water and the Water Corporation) are subject to performance reporting requirements under the *WA Financial Management Act 2006*. The annual reports prepared by Aqwest, Busselton Water and the Water Corporation include non-financial performance indicators that are independently audited by the WA Office of the Auditor General.

The City of Kalgoorlie–Boulder provides sewerage and non-potable water services to Kalgoorlie–Boulder, located 550 km east of Perth in the Goldfields district. The non-potable water supplied to customers is sourced from recycled effluent.

There are also several small licensed or exempted water service providers in the state. The licensed service providers include Aquasol Aqua Ferre (Muchea) (trading as Muchea Water), Athena Water Solutions, BHP Nickel West, BHP Iron Ore, Country Heights, Hamersley Iron, Lancelin South, Moore River Water Services, the Rottnest Island Authority, Robe River Mining Company, Peel Water, TMC Witchcliffe, WA Sewage, and Water West North Dandalup.

F8.5 Performance reporting

Licensees are required to provide the Economic Regulation Authority with data for performance monitoring purposes, as set out in the licence and the ERA's Performance indicators and definitions handbook – water service providers. Licensees are required to submit completed performance reports to the ERA for each year ending 30 June. Where possible, the performance indicators for licensees who are not required to report under the National Water Initiative Agreement have been aligned with the NPR indicator set for consistency.

The ERA's Water compliance reporting manual requires licensees to report to the ERA on their compliance with the terms and conditions of their licence for each year ending 30 June. The ERA uses the compliance reports to monitor the overall level of compliance by licensees. The content of each report is confidential to the licensee and the ERA.

The ERA publishes performance data provided by licensed urban service providers that do not report under the Urban NPR and 3 of WA's largest rural water service providers[21]. Most of the urban performance indicators are consistent with those of the NPR. Except for the licensees that report under the Urban NPR, licensees are not subject to the data audit requirements of the NPRs. For those licensees not reporting under the NPR, confirmation of the accuracy of the performance data they report to the ERA is assessed in the operational audits.

[21] Water, sewerage and irrigation performance data (2019 – onwards) - Economic Regulation Authority Western Australia

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